Introduction to Software Engineering

COSC 336

ReSource (Phases IV and V)

Design Document

Architecture Design and Detailed Design

Intelligent AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System

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Introduction

Prepared by: Mohamed Almarzooqi

Purpose and Scope

Prepared by: Mohamed Almarzooqi

This document is the design of ReSource, the university system for exchanging surplus campus resources and managing assets through their life cycle. Sections 1 to 7 give the architecture design requested for Phase 4 [1], [2]: the context and data-flow diagrams, the architecture and its design decisions, the AI services, the human approval points, security, and mockups of the eight screens named in the project description. Sections 8 to 19 give the detailed design requested for Phase 5 [1], [3]: the use case, class, sequence, activity, state, component and deployment diagrams, the AI recommendation workflow, the entity relationship diagram (ERD), the interfaces between modules and AI services, the recommendation score and explanation design, the sustainability calculations and the model-monitoring records.

The design covers the whole system specified in the Phase 3 Requirements Document [4]: the 13 use cases UC-1 to UC-13, the feature groups FR-1 to FR-13, the nonfunctional requirements and the business rules. Essential and Desirable requirements are designed. The three Future requirements (FR-1.14 bulk registration, FR-5.33 registration of usable components and FR-7.R8 scheduled reports) are left out of the design, and nothing in the design prevents adding them. The design keeps the Django and MySQL stack, the team, the student IDs and the scope approved in Phase 2 [5].

Document Conventions

Prepared by: Mohamed Almarzooqi

Requirement IDs (FR, AI, PER, SAF, SEC, QUA, BR, LEG and ORG) and use case IDs (UC) are those of the Phase 3 document [4]. Each design element cites the requirements it meets, and Appendix A traces every requirement group to its design. Design elements have their own IDs: DR-n for decision records, H-n for human approval points and D-n for data stores; the data-flow processes are numbered 1.0 to 7.0 on Level 1 and 3.1 to 3.6 on Level 2. Amounts are in UAE dirhams (AED) and dates are written DD/MM/YYYY.

All sample data in the figures and mockups is synthetic, as ORG-2 requires. It follows one example: the Physics department needs two oscilloscopes for a teaching lab on SAN Campus (request R-2026-0391), and Electrical Engineering has listed three surplus 100 MHz oscilloscopes on Main Campus (asset A-2026-0412). The other screens use other assets so that every role and module is shown.

Role names follow Phase 3. Table I-1 gives the abbreviations used in the diagrams, which save space in the figures.

Table I-1: Roles and abbreviations

Abbreviation	Role (Phase 3)	Main use cases
DR	Department Representative	UC-2, UC-3, UC-5, UC-6, UC-11

Abbreviation	Role (Phase 3)	Main use cases
AC	Asset Coordinator	UC-1, UC-4, UC-5, UC-6, UC-7, UC-10
AD	Administrator	UC-4, UC-5, UC-6, UC-7, UC-12
PT	Procurement Team	UC-4, UC-9, UC-12
MS	Maintenance Staff	UC-6
FO	Finance Officer	UC-12
SO	Sustainability Officer	UC-6, UC-7, UC-12
SA	System Administrator	UC-8, UC-13
UM	University Management	UC-12
None	Applicant (no active account yet)	UC-13

Design Summary

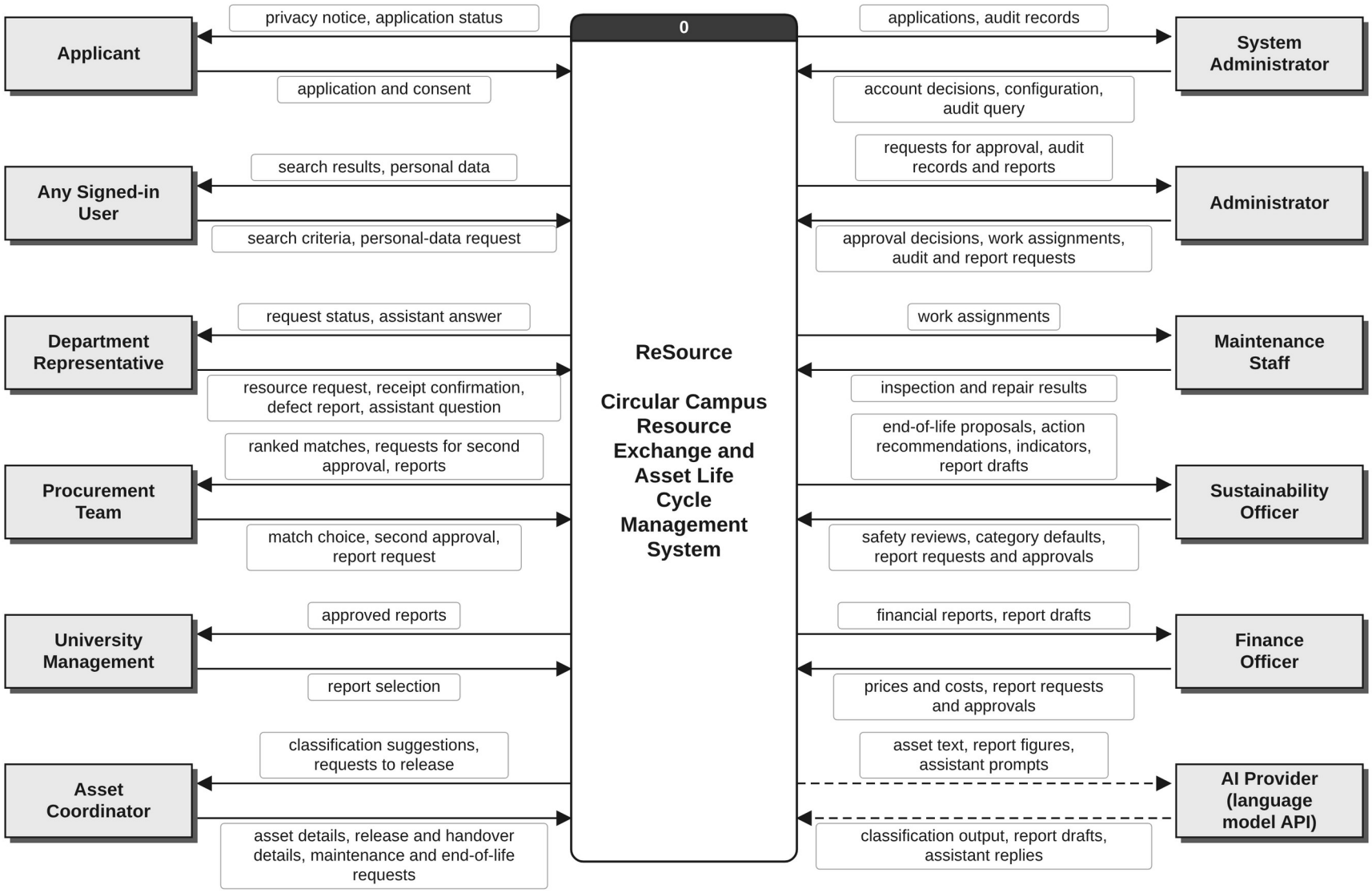
Prepared by: Mohamed Almarzooqi

ReSource is one Django web application with one MySQL database, organized as a modular monolith: nine business modules with clear borders, an AI gateway (AIGateway) that is the only way to reach the AI services, and a background worker for jobs that nobody waits for (Section 3). Semantic matching uses a small open embedding model that runs on the server, so request and asset text used for matching never leaves the university server. A hosted language model is used only for classification text, the assistant and report drafting, through the gateway, with personal data removed (SEC-11). Every AI output is a suggestion with a confidence level and a reason; a named person decides at each of the approval points in Section 5, and every module keeps working with forms, standard search and standard reports when the AI is unavailable (AI-6, PER-6).

1. Context Diagram

Prepared by: Mohamed Almarzooqi

Figure 1-1 shows ReSource as a single process, numbered 0, with every external entity that gives data to the system or receives data from it. The entities are the nine roles of Phase 3, the Applicant (a person without an active account, UC-13), "Any Signed-in User" for the functions that every role shares (searching the catalogue and viewing personal data, FR-2.2, FR-6.11), and the AI Provider, the hosted language model reached through AIGateway (DR-4). Flows are grouped and named with nouns; there are no data stores and no internal steps on this level.



Above each pair: data from ReSource to the entity. Below: data from the entity to ReSource.
Dashed: exchanged with the external language-model service through AIGateway, without personal data (SEC-11).

Figure 1-1: Context diagram

Some participants are deliberately not entities. Facilities Management, approved recycling vendors and donation recipients have no account; their evidence is entered by campus staff (Phase 3, Section 5.1), so their data reaches the system through the Sustainability Officer and the Asset Coordinator. The e-mail service and university single sign-on are not part of the prototype (TBD-6, ORG-3) and are shown as later integration points in Section 6.6. The embedding model runs inside ReSource and the backup copy holds ReSource's own data, so neither is an outside party in the logical data flow.

Table 1-1: External entities and their data flows

External entity	Data in (to ReSource)	Data out (from ReSource)	Source in Phase 3
Applicant	application and consent	privacy notice, application status	UC-13; FR-6.1, FR-6.2
Any Signed-in User	search criteria, personal-data request	search results, personal data	UC-2; FR-2.2, FR-6.11, SEC-9
Department Representative	resource request, receipt confirmation, defect report, assistant question	request status, assistant answer	UC-3, UC-5, UC-6, UC-11; FR-3, FR-4.10, FR-5.1, FR-11
Asset Coordinator	asset details, release and handover details, maintenance and end-of-life requests	classification suggestions, requests to release	UC-1, UC-4, UC-5, UC-7, UC-10; FR-1, FR-4.1, FR-4.7, FR-9
Procurement Team	match choice, second approval, report request	ranked matches, requests for second approval, reports	UC-4, UC-9, UC-12; FR-4.3, FR-8.7, FR-12.1
University Management	report selection	approved reports	UC-12; FR-7.R1, FR-12.9
System Administrator	account decisions, configuration, audit query	applications, audit records	UC-8, UC-13; FR-6.3, FR-6.8, FR-7.A3
Administrator	approval decisions, work assignments, audit and report requests	requests for approval, audit records and reports	UC-4 to UC-7, UC-12; FR-4.2, FR-4.12, FR-5.3, BR-8
Maintenance Staff	inspection and repair results	work assignments	UC-6; FR-5.3 to FR-5.8
Sustainability Officer	safety reviews, category defaults, report requests and approvals	end-of-life proposals, action recommendations, indicators, report drafts	UC-6, UC-7, UC-12; FR-5.23, FR-10, FR-13.4
Finance Officer	prices and costs, report requests and approvals	financial reports, report drafts	UC-12; FR-13.4, FR-12.9
AI Provider	classification output, report drafts, assistant replies	asset text, report figures, assistant prompts	UC-10 to UC-12; FR-9, FR-11, FR-12; SEC-11

Every entity sends and receives at least one flow, and the 24 grouped flows are the ones balanced against Level 1 in Table 2-3.

2. Data-Flow Diagrams

Prepared by: Mohamed Almarzooqi

2.1 Level 1

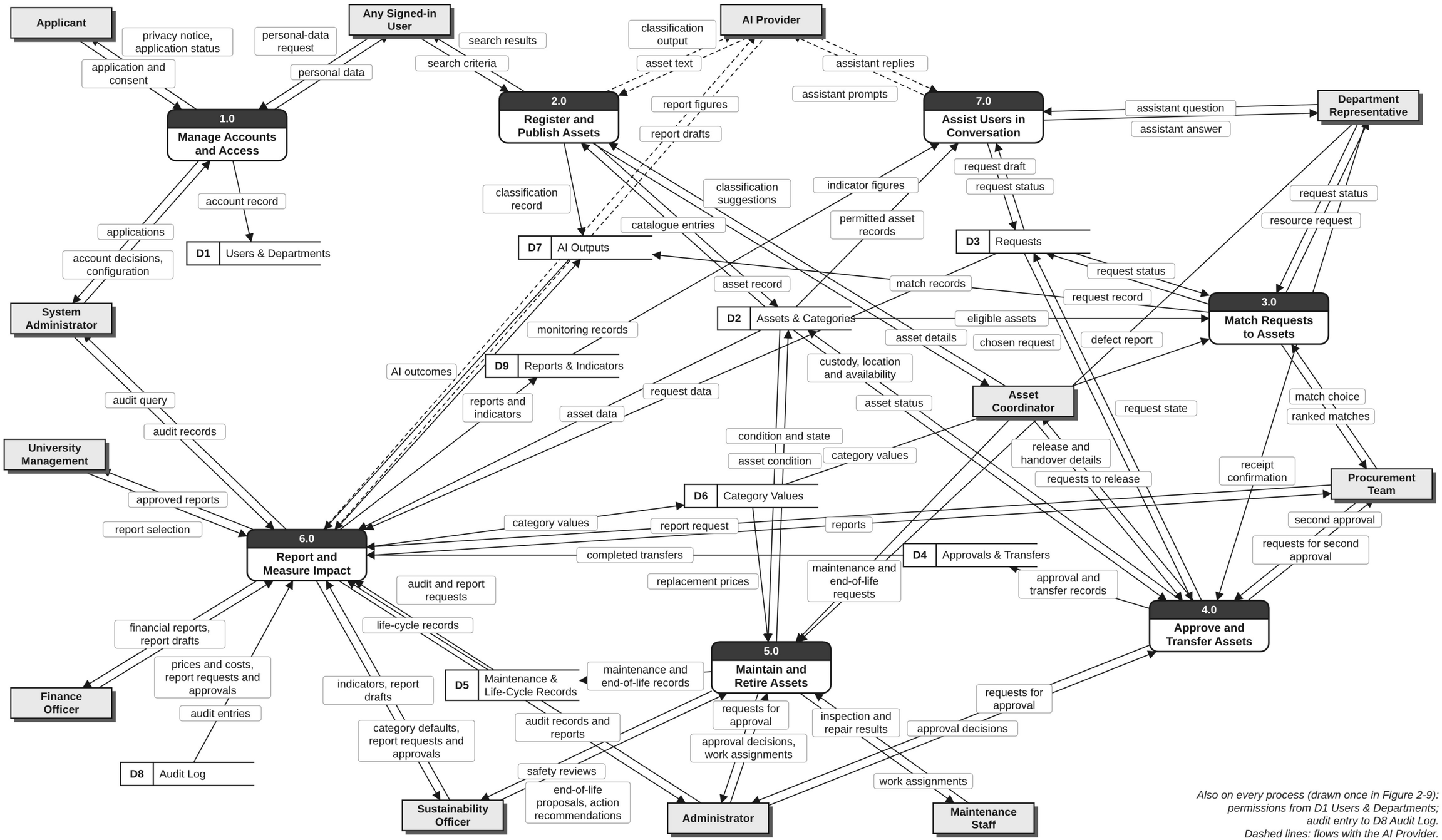
Prepared by: Mohamed Almarzooqi

Level 1 splits process 0 into seven processes, one for each main job of the system (Table 2-1). Each process is a verb and a noun, is numbered 1.0 to 7.0 and serves a group of Phase 3 use cases. The nine data stores D1 to D9 hold data that waits between processes (Table 2-2).

Table 2-1: Level 1 processes

ID	Process	Main job	AI used	Requirements and use cases
1.0	Manage Accounts and Access	Registers applicants with consent, activates accounts, assigns roles and departments, and returns personal data on request.	None	FR-6; SEC-1 to SEC-9; UC-8, UC-13
2.0	Register and Publish Assets	Records and publishes assets, gets classification suggestions and answers catalogue searches.	ClassificationService (AI Provider and local model)	FR-1, FR-2, FR-9; UC-1, UC-2, UC-10
3.0	Match Requests to Assets	Records requests and open needs, finds and ranks suitable assets and records the match decision.	EmbeddingService, MatchingEngine, RankingModel	FR-3, FR-8; UC-3, UC-9
4.0	Approve and Transfer Assets	Takes the release and approval decisions, reserves units, records the transfer and the receipt.	None	FR-4; BR-3 to BR-6; UC-4, UC-5
5.0	Maintain and Retire Assets	Records defects, inspections and repairs, and takes end-of-life proposals through safety review, approval and completion.	ActionRecommender (Desirable)	FR-5, FR-10; SAF-1 to SAF-9; UC-6, UC-7
6.0	Report and Measure Impact	Calculates indicators, produces standard and AI reports for review, keeps category values, answers audit queries and records weekly model-monitoring metrics.	ReportGenerator (AI Provider)	FR-7.R, FR-7.A, FR-12, FR-13; UC-12
7.0	Assist Users in Conversation	Answers questions and prepares request drafts in conversation (Desirable).	AssistantService (AI Provider)	FR-11; UC-11

Figure 2-1 is the complete Level 1 diagram: the 12 entities, the seven processes, the nine stores and every flow between them, on one large page. Figures 2-2 to 2-8 repeat the same diagram one process at a time at a larger size, so each flow label can be read easily; the same entity or store therefore appears on several of those figures. Two flows are used by every process and are drawn once (Figure 2-9): each process reads permissions from D1 and writes an audit entry to D8 (FR-6.7, FR-7.A1). Table 2-2 shows which process writes and reads each store.



Also on every process (drawn once in Figure 2-9): permissions from D1 Users & Departments; audit entry to D8 Audit Log.
Dashed lines: flows with the AI Provider.

Figure 2-1: Level 1 data-flow diagram

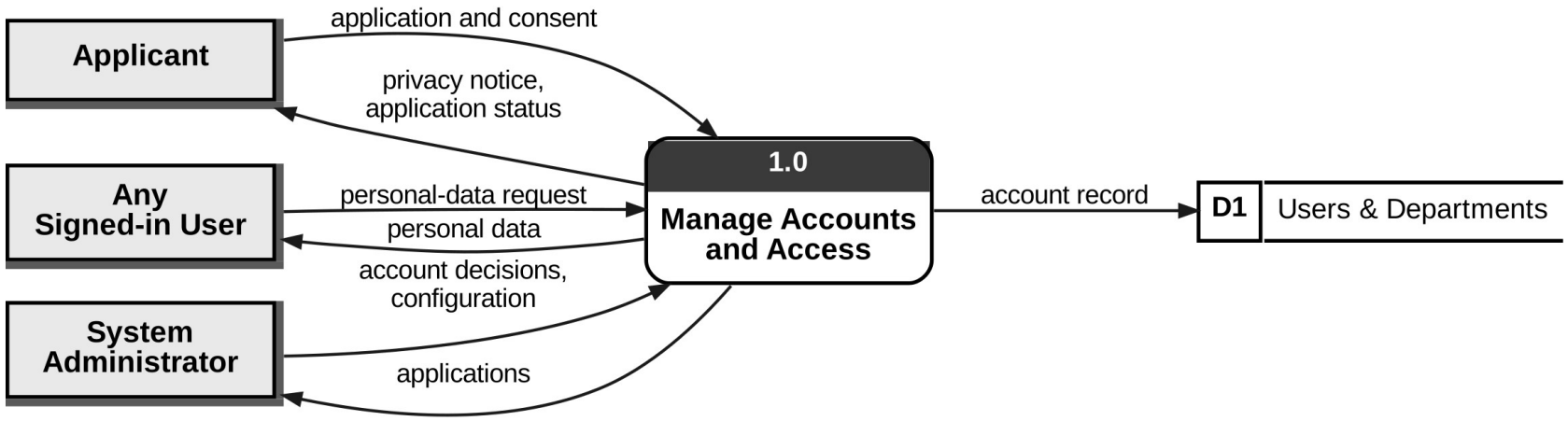


Figure 2-2: Level 1, process 1.0

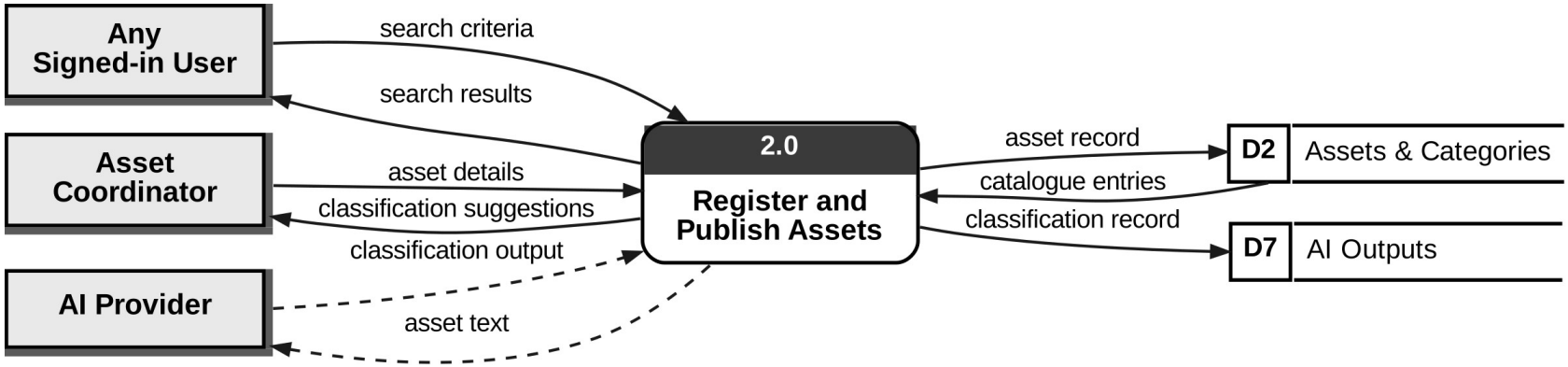


Figure 2-3: Level 1, process 2.0

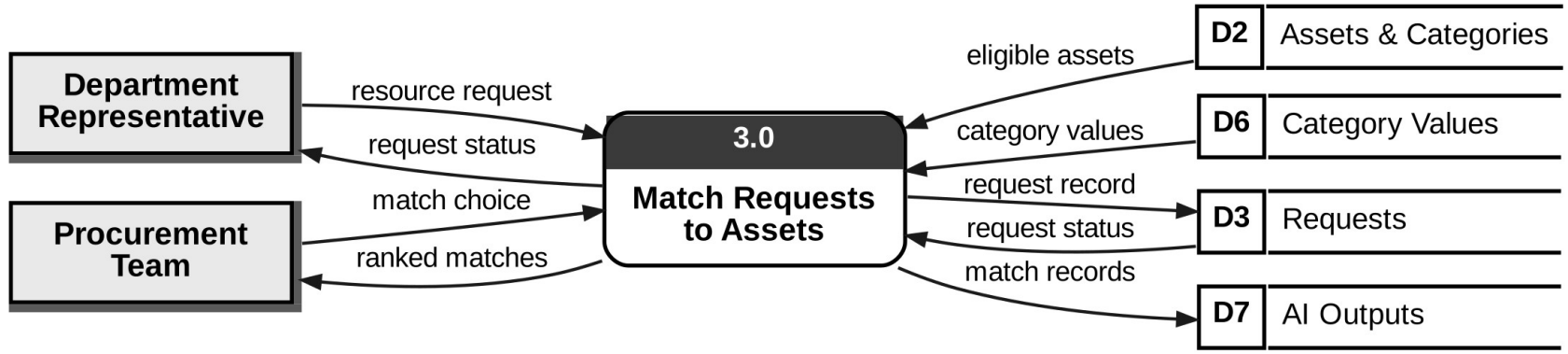


Figure 2-4: Level 1, process 3.0

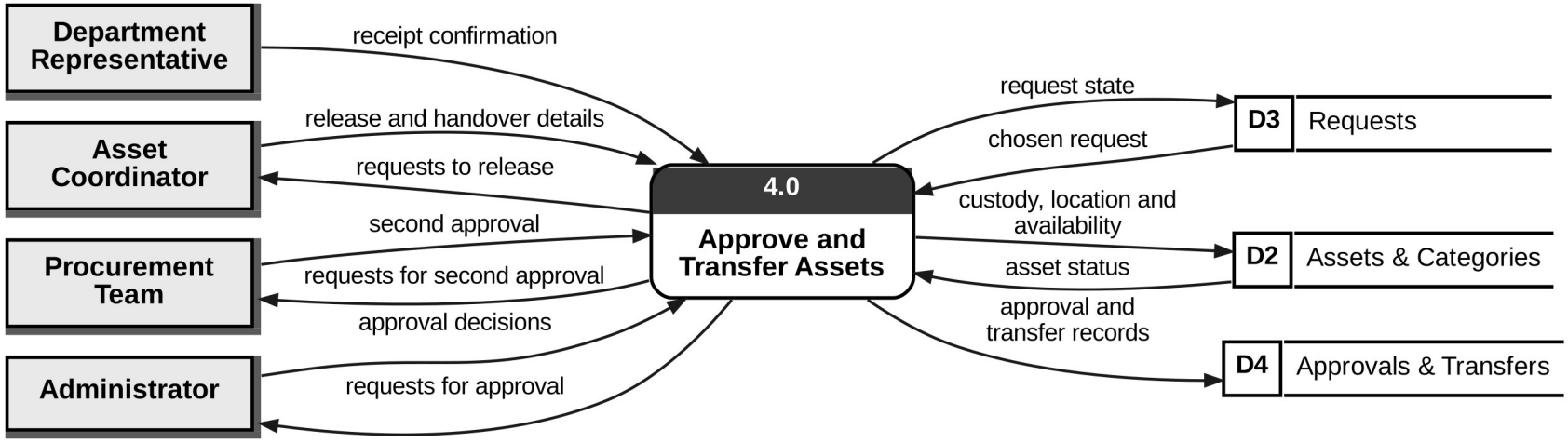


Figure 2-5: Level 1, process 4.0

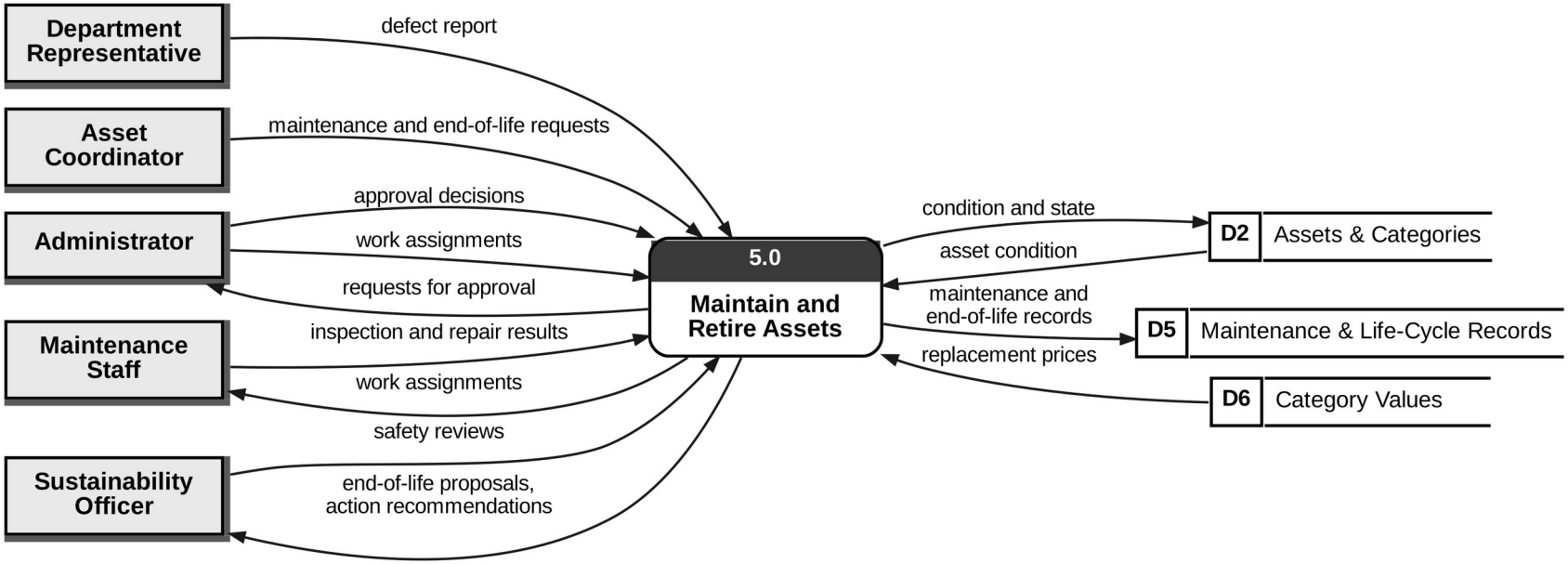


Figure 2-6: Level 1, process 5.0

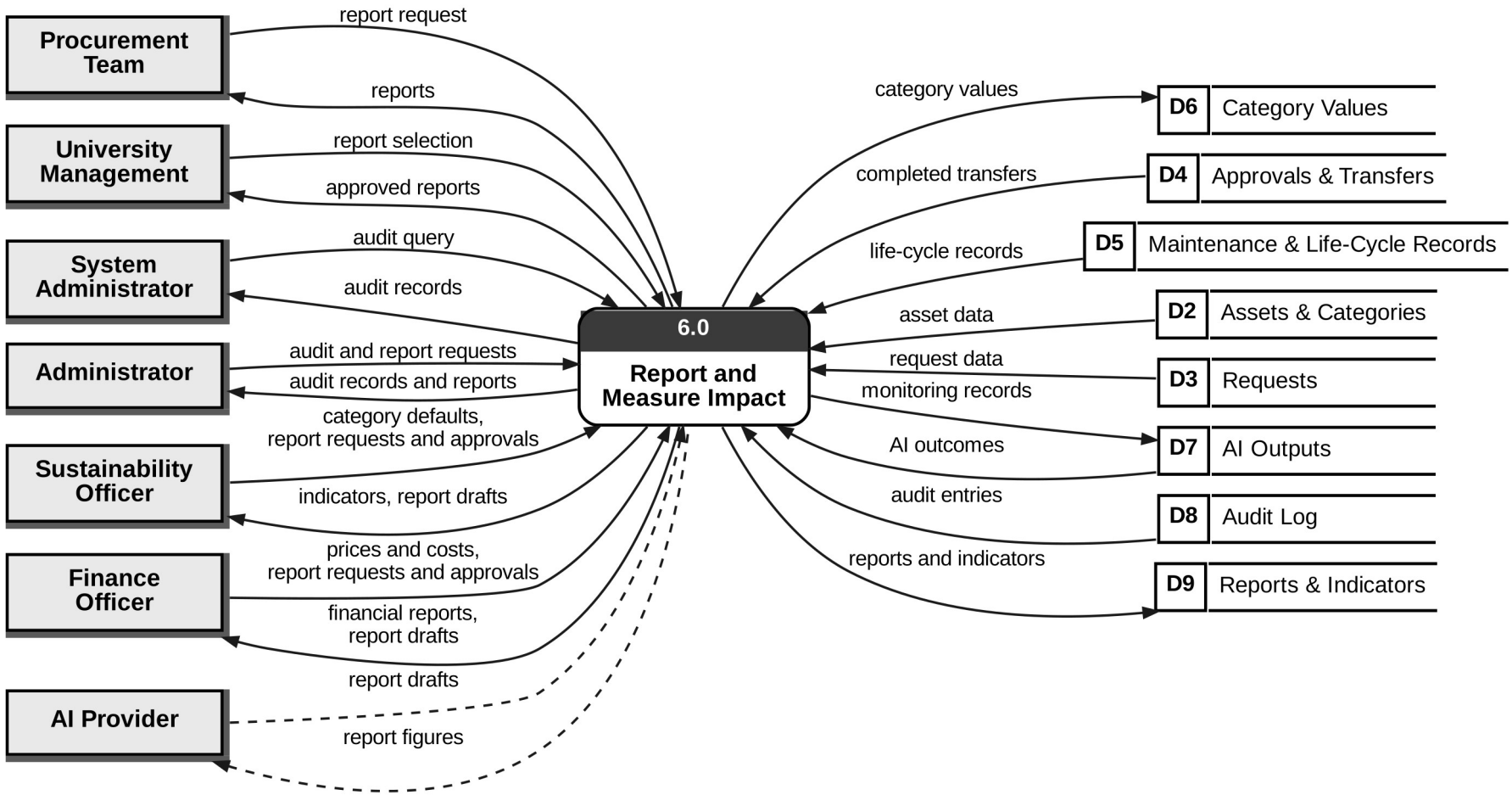


Figure 2-7: Level 1, process 6.0

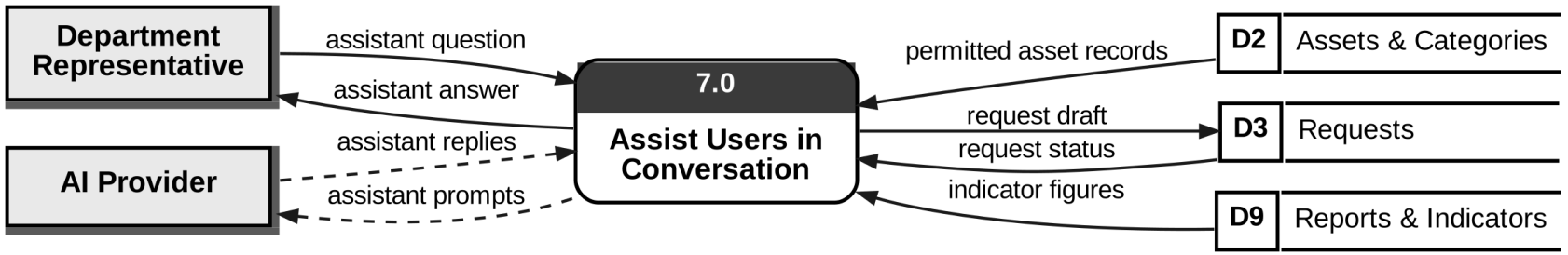


Figure 2-8: Level 1, process 7.0

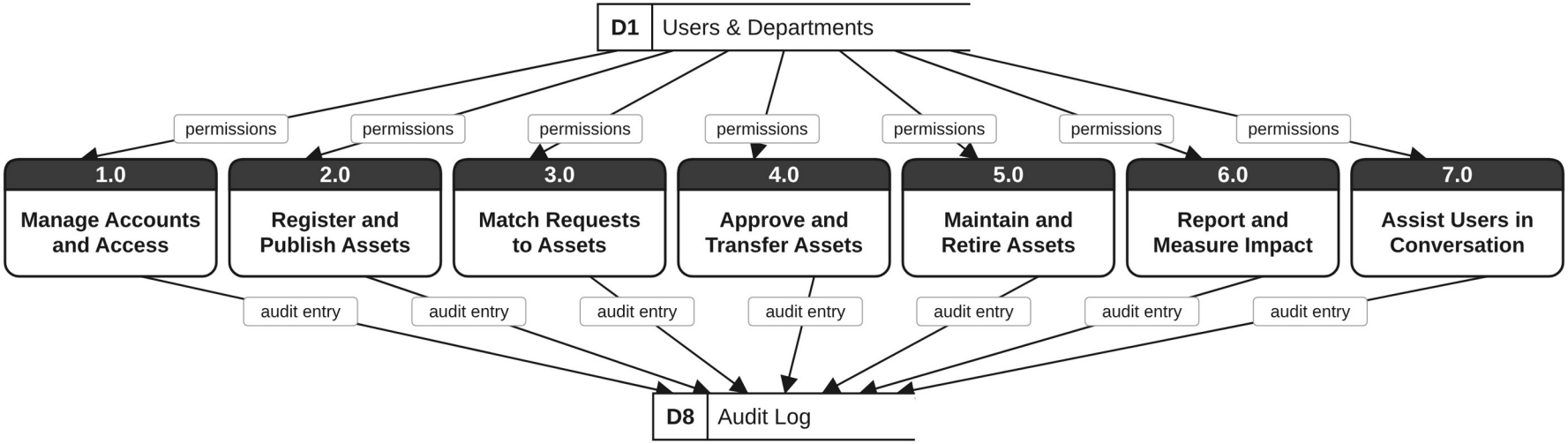


Figure 2-9: Level 1, common flows

2.2 Data Stores

Prepared by: Mohamed Almarzooqi

Table 2-2: Data stores

ID	Store	Contents	Written by	Read by	ERD tables (Section 15)	Requirements
D1	Users & Departments	Users, roles, departments, consent records	1.0	All processes	DEPARTMENT, USER, ROLE, USER_ROLE, CONSENT_RECORD	FR-6; SEC-7, SEC-8
D2	Assets & Categories	Assets, photos, categories, text embeddings, life-cycle state	2.0, 4.0, 5.0	2.0, 3.0, 4.0, 5.0, 6.0, 7.0	ASSET, ASSET_FILE, ASSET_EMBEDDING, CATEGORY	FR-1, FR-2, FR-9
D3	Requests	Requests, request drafts, request history	3.0, 4.0, 7.0	3.0, 4.0, 6.0, 7.0	REQUEST	FR-3, FR-11.2
D4	Approvals & Transfers	Approvals, reservations, transfers, receipt records	4.0	6.0	APPROVAL, RESERVATION, TRANSFER, TRANSFER_DOCUMENT	FR-4; BR-3 to BR-6
D5	Maintenance & Life-Cycle Records	Maintenance requests, inspections, end-of-life proposals, vendors, life-cycle events	5.0	6.0	MAINTENANCE_REQUEST, INSPECTION, MAINTENANCE_RECORD, END_OF_LIFE_PROPOSAL, VENDOR, LIFECYCLE_EVENT	FR-5
D6	Category Values	Versioned prices, costs, weights, years of use, emission factors	6.0	3.0, 5.0	CATEGORY_VALUE	FR-13.4, FR-13.5
D7	AI Outputs	AI output records, match recommendations, model versions, reviewer decisions	2.0, 3.0, 6.0	6.0	MODEL_VERSION, AI_OUTPUT, RECOMMENDATION, HUMAN_REVIEW, EVALUATION_RUN, MONITORING_RECORD	AI-5, FR-7.A4, FR-8.7, FR-9.9
D8	Audit Log	Append-only audit entries	All processes	6.0	AUDIT_ENTRY	FR-7.A1 to FR-7.A6
D9	Reports & Indicators	Calculated indicators, AI report drafts and approved reports	6.0	7.0	SUSTAINABILITY_IMPACT, IMPACT_VALUE_USED, AI_REPORT	FR-12, FR-13

Each store is written by at least one process and read by at least one other process or by the same process at a later time, so no store is a dead end. The data stores are the Phase 4 data model; Section 15 refines them into 32 tables. The NOTIFICATION table keeps a copy of outgoing flows already shown in Figure 2-1, such as requests to release and request status, so it is not a separate store.

2.3 Balancing Level 1 Against the Context Diagram

Prepared by: Mohamed Almarzooqi

Table 2-3 checks every grouped flow of the context diagram. A grouped flow is split on Level 1 into parts, each handled by one process; the parts keep the same entity and direction, and together they make up the grouped flow. Level 1 has no external flow that is not in the context diagram.

Table 2-3: Balancing Level 1 against the context diagram

#	Context flow	External entity	In/ out	Level 1 process (part)	Same name and direction
1	application and consent	Applicant	in	1.0	Yes
2	privacy notice, application status	Applicant	out	1.0	Yes
3	search criteria, personal-data request	Any Signed-in User	in	2.0 (search criteria), 1.0 (personal-data request)	Yes
4	search results, personal data	Any Signed-in User	out	2.0 (search results), 1.0 (personal data)	Yes
5	resource request, receipt confirmation, defect report, assistant question	Department Representative	in	3.0 (resource request), 4.0 (receipt confirmation), 5.0 (defect report), 7.0 (assistant question)	Yes
6	request status, assistant answer	Department Representative	out	3.0 (request status), 7.0 (assistant answer)	Yes
7	asset details, release and handover details, maintenance and end-of-life requests	Asset Coordinator	in	2.0 (asset details), 4.0 (release and handover details), 5.0 (maintenance and end-of-life requests)	Yes
8	classification suggestions, requests to release	Asset Coordinator	out	2.0 (classification suggestions), 4.0 (requests to release)	Yes
9	match choice, second approval, report request	Procurement Team	in	3.0 (match choice), 4.0 (second approval), 6.0 (report request)	Yes
10	ranked matches, requests for second approval, reports	Procurement Team	out	3.0 (ranked matches), 4.0 (requests for second approval), 6.0 (reports)	Yes
11	report selection	University Management	in	6.0	Yes
12	approved reports	University Management	out	6.0	Yes
13	account decisions, configuration, audit query	System Administrator	in	1.0 (account decisions, configuration), 6.0 (audit query)	Yes
14	applications, audit records	System Administrator	out	1.0 (applications), 6.0 (audit records)	Yes
15	approval decisions, work assignments, audit and report requests	Administrator	in	4.0 (approval decisions), 5.0 (approval decisions), 5.0 (work assignments), 6.0 (audit and report requests)	Yes
16	requests for approval, audit records and reports	Administrator	out	4.0 (requests for approval), 5.0 (requests for approval), 6.0 (audit records and reports)	Yes
17	inspection and repair results	Maintenance Staff	in	5.0	Yes
18	work assignments	Maintenance Staff	out	5.0	Yes
19	safety reviews, category defaults, report requests and approvals	Sustainability Officer	in	5.0 (safety reviews), 6.0 (category defaults, report requests and approvals)	Yes

#	Context flow	External entity	In/ out	Level 1 process (part)	Same name and direction
20	end-of-life proposals, action recommendations, indicators, report drafts	Sustainability Officer	out	5.0 (end-of-life proposals, action recommendations), 6.0 (indicators, report drafts)	Yes
21	prices and costs, report requests and approvals	Finance Officer	in	6.0	Yes
22	financial reports, report drafts	Finance Officer	out	6.0	Yes
23	asset text, report figures, assistant prompts	AI Provider	out	2.0 (asset text), 6.0 (report figures), 7.0 (assistant prompts)	Yes
24	classification output, report drafts, assistant replies	AI Provider	in	2.0 (classification output), 6.0 (report drafts), 7.0 (assistant replies)	Yes

In/out is seen from ReSource: "in" means the entity sends the flow to ReSource. For the AI Provider, ReSource sends asset text, report figures and assistant prompts (out) and receives the outputs (in).

2.4 Level 2 of 3.0 Match Requests to Assets

Prepared by: Mohamed Almarzooqi

We open process 3.0 because it holds the two AI calls that matter most and it carries the highest risk in the system. A wrong or unfair ranking can send a department the wrong asset or favor one department over another (LEG-9, QUA-13), and a match chosen on stale data could promise units that are no longer free (FR-8.8). Level 2 shows where each check happens and where the AI output is stored before a person decides.

Table 2-4: Level 2 processes of 3.0

ID	Process	What it does	Requirements
3.1	Validate and Record Request	Checks the required fields, the role and own department; saves a request for a listing as Submitted or a need as Open; returns the request status; writes the audit entry.	FR-3.1 to FR-3.7, FR-3.13
3.2	Embed Request Text	Builds the text from category, purpose and specifications and turns it into a 384-number vector with the local embedding model (run by the worker, pattern B).	FR-8.1
3.3	Filter Eligible Assets	Keeps only published, available assets that the requester may see and that pass the quantity, safety and essential-specification checks.	FR-8.2, FR-5.9, FR-5.27, SAF-1, SAF-5
3.4	Find Similar Assets	Compares the request vector with the vectors of the eligible assets (cosine similarity) and keeps the 20 most similar.	FR-8.1
3.5	Score, Rank and Explain	Scores each candidate on eight documented factors, sets the confidence level, writes the explanation, stores the match records and shows the ranked list.	FR-8.3 to FR-8.6, LEG-9, AI-10, AI-5
3.6	Record Match Decision	Records the Procurement Team choice and reason, rechecks the chosen asset and permissions, attaches the asset to the request and sends it to release.	FR-8.7, FR-8.8, FR-7.A4, QUA-12

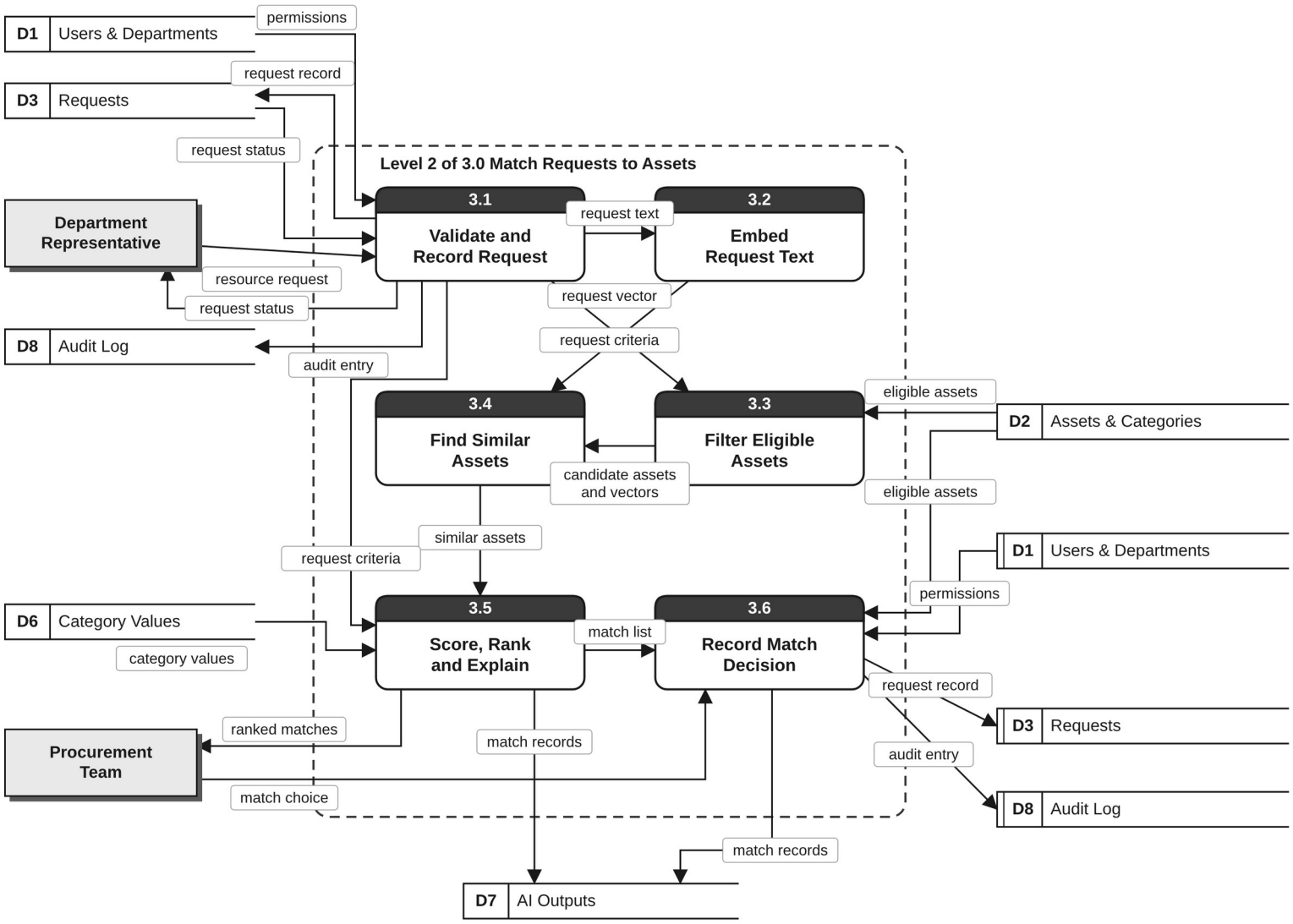


Figure 2-10: Level 2 of process 3.0

Table 2-5 balances Level 2 against the flows of 3.0 on Level 1: all 11 flows appear with the same names and directions, and no new external flow is added. The flows between 3.1 and 3.6 (request text, request criteria, request vector, candidate assets and vectors, similar assets and match list) are internal to 3.0.

Table 2-5: Balancing Level 2 against process 3.0

Level 1 flow of 3.0	From or to	Direction	Level 2 process	OK
resource request	Department Representative	in	3.1	Yes
request status	Department Representative	out	3.1	Yes
match choice	Procurement Team	in	3.6	Yes
ranked matches	Procurement Team	out	3.5	Yes
eligible assets	D2 Assets & Categories	in	3.3, 3.6	Yes
category values	D6 Category Values	in	3.5	Yes
request status	D3 Requests	in	3.1	Yes
request record	D3 Requests	out	3.1, 3.6	Yes
match records	D7 AI Outputs	out	3.5, 3.6	Yes
permissions	D1 Users & Departments	in	3.1, 3.6	Yes
audit entry	D8 Audit Log	out	3.1, 3.6	Yes

2.5 Rule Checks

Prepared by: Mohamed Almarzooqi

Table 2-6 counts the inputs and outputs of each Level 1 process, including the common flows. Every process has at least one input and one output, so there is no black hole (input with no output) and no miracle (output with no input). We also checked each output against the inputs it needs to avoid grey holes: for example, ranked matches from 3.0 need the request (resource request, request status), the assets (eligible assets) and the cost and carbon values (category values), and all three reach 3.0. Every flow has a process at one end or both, so no data moves directly from store to store.

Table 2-6: Inputs and outputs of each Level 1 process

ID	Process	Inputs	Outputs	Has input and output	Outputs derivable
1.0	Manage Accounts and Access	4	5	Yes	Yes
2.0	Register and Publish Assets	5	6	Yes	Yes
3.0	Match Requests to Assets	6	5	Yes	Yes
4.0	Approve and Transfer Assets	7	7	Yes	Yes
5.0	Maintain and Retire Assets	9	6	Yes	Yes
6.0	Report and Measure Impact	14	11	Yes	Yes
7.0	Assist Users in Conversation	6	4	Yes	Yes

Counts include the common flows: one input (permissions from D1) and one output (audit entry to D8).

3. Architecture

Prepared by: Mohamed Almarzooqi

3.1 How We Chose

Prepared by: Mohamed Almarzooqi

Each major choice below is recorded as a decision record. A record gives the driver (the requirements that force a choice), at least two real options, the choice and why ("because"), what we give up, how we checked the choice and the links to the rest of the design. The checks are design walkthroughs and calculations done on this design; they are not test results of running software. Table 3-1 lists the six records.

Table 3-1: Decision records

ID	Decision	Choice
DR-1	How to split the system	Modular monolith with an AI gateway (AIGateway) and a background worker
DR-2	Application stack	Django and MySQL (Phase 2 stack)
DR-3	Where to keep and search text vectors	MySQL table plus in-memory search with NumPy
DR-4	Language model and where data goes	Hosted Claude Haiku 5.5 through the gateway, no personal data
DR-5	Background jobs	Django-Q2 with the database as the queue
DR-6	Hosting	One Linux virtual machine in the UAE with Docker Compose

3.2 Decision Records

Prepared by: Mohamed Almarzooqi

Table 3-2: DR-1 How to split the system

Driver	Many business areas (FR-1 to FR-13) built by four students in three-week phases on one server; records must stay correct when several users act at once (QUA-4); standard functions must keep working when AI is down (PER-6).
Options	A. Modular monolith: one Django project, one app per business module, one database; the AI services sit behind AIGateway inside the same code base, and slow AI jobs run in a separate worker process. B. Service-oriented: the same modules, but the AI part is a separate Python web service called over HTTP. Microservices were also considered and ruled out at once: a team of four cannot run and monitor many services.
We choose	A, modular monolith.
Because	Django and the AI libraries are both Python, so the AI part needs no separate service to use its own language. One database lets an approval, its reservation and its audit entry commit in one transaction (FR-4.5, FR-7.A6). The worker gives the AI work its own process, so a slow embedding job never blocks a web page.
We give up	Scaling the AI part on its own machine; module borders are kept by our rules (Section 3.4) rather than by the network; a fault in the web process stops all modules until it restarts.
Verify	Change walk: moving the AI services to a separate service later changes only AIGateway, because no module calls an AI service directly (Rule 3). Result: 1 component changes. Failure walk: with the worker and the AI Provider both stopped, each of UC-1 to UC-13 still has a working path (form, standard search or standard report) through the fallbacks in Table 4-1. Result: 13 of 13.
Links	Figure 3-1, Figure 13-1, Section 4; QUA-4, QUA-5, PER-6

Table 3-3: DR-2 Application stack

Driver	Sign-in, password, session and permission rules (SEC-1 to SEC-6); safe input handling (SEC-13); reservations that never double-book (FR-4.5); the Phase 2 plan and skills [5].
Options	A. Django (Python) with MySQL and server-rendered pages, the Phase 2 stack. B. Node.js with a React single-page front end and Firebase as the database.
We choose	A, Django and MySQL.
Because	Django provides hashed passwords, sessions, permission checks, forms with validation, protection against cross-site request forgery and safe database queries [6], which cover most of SEC-2 to SEC-6 and SEC-13 without custom security code. MySQL transactions and row locks let the final approval recheck and reserve units safely (FR-4.5). Python lets us use the embedding library directly. Phase 2 costed and planned this stack.
We give up	The smoother page updates of a single-page React app; Firebase live updates for notifications (a page refresh or a short poll shows new notifications within the five minutes FR-7.N3 allows).
Verify	Mapping check: each of SEC-1 to SEC-6 and SEC-13 maps to a named Django feature or a stated design rule in Table 6-6. Result: 7 of 7 mapped.
Links	Section 6; Figure 13-2; SEC-1 to SEC-6, SEC-13, FR-4.5

Table 3-4: DR-3 Where to keep and search text vectors

Driver	Matching must find assets by meaning (FR-8.1) and return within the AI time limit (PER-4) for a prototype of at least 100 assets (PER-1); we plan for up to 5,000.
Options	A. Store each 384-number vector in a MySQL table and compare vectors in memory with NumPy. B. Move to PostgreSQL with the pgvector extension, or add a separate vector database.
We choose	A, MySQL table and NumPy.
Because	At 5,000 assets all vectors fit in 7.68 MB of memory and one comparison is about 1.9 million multiplications, a few milliseconds on a normal CPU. We keep one database (DR-2) and nothing new to run, back up or secure.
We give up	Indexed search for very large catalogues. Above about 100,000 assets we would move to option B.
Verify	Arithmetic: 5,000 assets x 384 numbers x 4 bytes = 7,680,000 bytes; 5,000 x 384 = 1,920,000 multiplications per search. Both are far inside the 2-second budget we set for ranking (Table 4-1).
Links	Section 2.4 (3.4 Find Similar Assets), Section 14, ASSET_EMBEDDING in Section 15; FR-8.1, PER-4

Table 3-5: DR-4 Language model and where data goes

Driver	Classification must read titles, descriptions and photos (FR-9.1); the assistant and AI reports need fluent text (FR-11, FR-12); no names, university IDs or e-mail addresses may be sent (SEC-11), and no personal data may leave the UAE until university IT confirms (LEG-3, TBD-9).
Options	A. A hosted model, Claude Haiku 5.5 [7], called only through AIGateway with a field allow-list. B. An open model run on our own server. C. No language model: rule-based templates only.
We choose	A, with C kept as the fallback for every feature.

Because	Haiku 5.5 accepts text and images, costs about AED 0.37 per million input tokens and AED 1.84 per million output tokens for prompts up to 100,000 tokens [7] (converted at the fixed rate of AED 3.6725 per US dollar), and needs no graphics card, which our budget does not include. At an estimated 2,000 AI calls a month (3,000 tokens in and 500 out each) the cost is about AED 4 a month. The gateway sends only the asset and request fields listed in Section 16.2 and removes e-mail addresses, university ID patterns and the names of registered users from free text, so no personal data is sent. Matching uses the local model (DR-3), so most AI work never leaves the server.
We give up	Independence from an outside service (covered by the fallbacks); a small running cost; model retirements by the provider. Phase 2 costed Claude Haiku 4.5, which the provider now lists as a legacy model, so the design uses Haiku 5.5 and records the model version with every output (AI-5).
Verify	Allow-list check: none of the six request schemas in Section 16.2 has a name, university ID or e-mail field. Result: 0 of 6 schemas carry personal fields. Change walk: switching to another provider changes only the provider adapter inside the gateway. Result: 1 component changes.
Links	Section 4, Section 16.2, DPIA in Section 6.7; SEC-11, LEG-3, AI-5, TBD-9

Table 3-6: DR-5 Background jobs

Driver	Embedding on save, retries, notification retries (FR-7.N5), expiry checks and reminders (FR-3.9, FR-7.N6), nightly backup (SAF-8) and weekly monitoring (Section 19) must run without a user waiting.
Options	A. Celery with a Redis server as the queue. B. Django-Q2 using the existing MySQL database as the queue [8].
We choose	B, Django-Q2.
Because	It needs no extra server to run, secure and back up; queued jobs survive a restart because they are stored in the database; it has retries and scheduled jobs built in.
We give up	The higher job throughput of Redis. Our load is a few hundred small jobs a day for five users at a time (PER-1), which the database handles easily.
Verify	Job list check: all seven job types (embed asset, embed request, AI retry, notification retry, expiry and reminder checks, nightly backup, weekly monitoring) are low volume and none needs an answer within one second.
Links	Figure 13-2; SAF-8, FR-3.9, FR-7.N5

Table 3-7: DR-6 Hosting

Driver	Personal data stays in the UAE (LEG-3); daily backups and a 60-minute restore (SAF-8, QUA-6); a university IT review before real data (ORG-1); 99% availability during testing (QUA-5).
Options	A. One Linux virtual machine in a UAE data center or on university servers, running the containers with Docker Compose. B. A hosted platform service in a data center outside the UAE.
We choose	A, one UAE virtual machine.
Because	We control where the data and the backups are kept, the whole system starts with one command, and the same files can be given to university IT for review.
We give up	Automatic scaling and managed updates; we apply operating-system updates and renew the TLS certificate ourselves.
Verify	Restore walk: the restore steps in Section 6.5 rebuild the system from the last backup in an estimated 35 minutes, inside the 60-minute target; the weekly restore test measures the real time.
Links	Section 6.5, Figure 13-2; LEG-3, SAF-8, QUA-5, QUA-6, ORG-1

3.3 Architecture View

Prepared by: Mohamed Almarzooqi

Figure 3-1 shows the architecture. It has four layers (presentation, application, AI and analytics, and data), a security column that applies to every layer and the outside systems ReSource integrates with. The eight components named in the project description are numbered (1) to (8) on the figure.

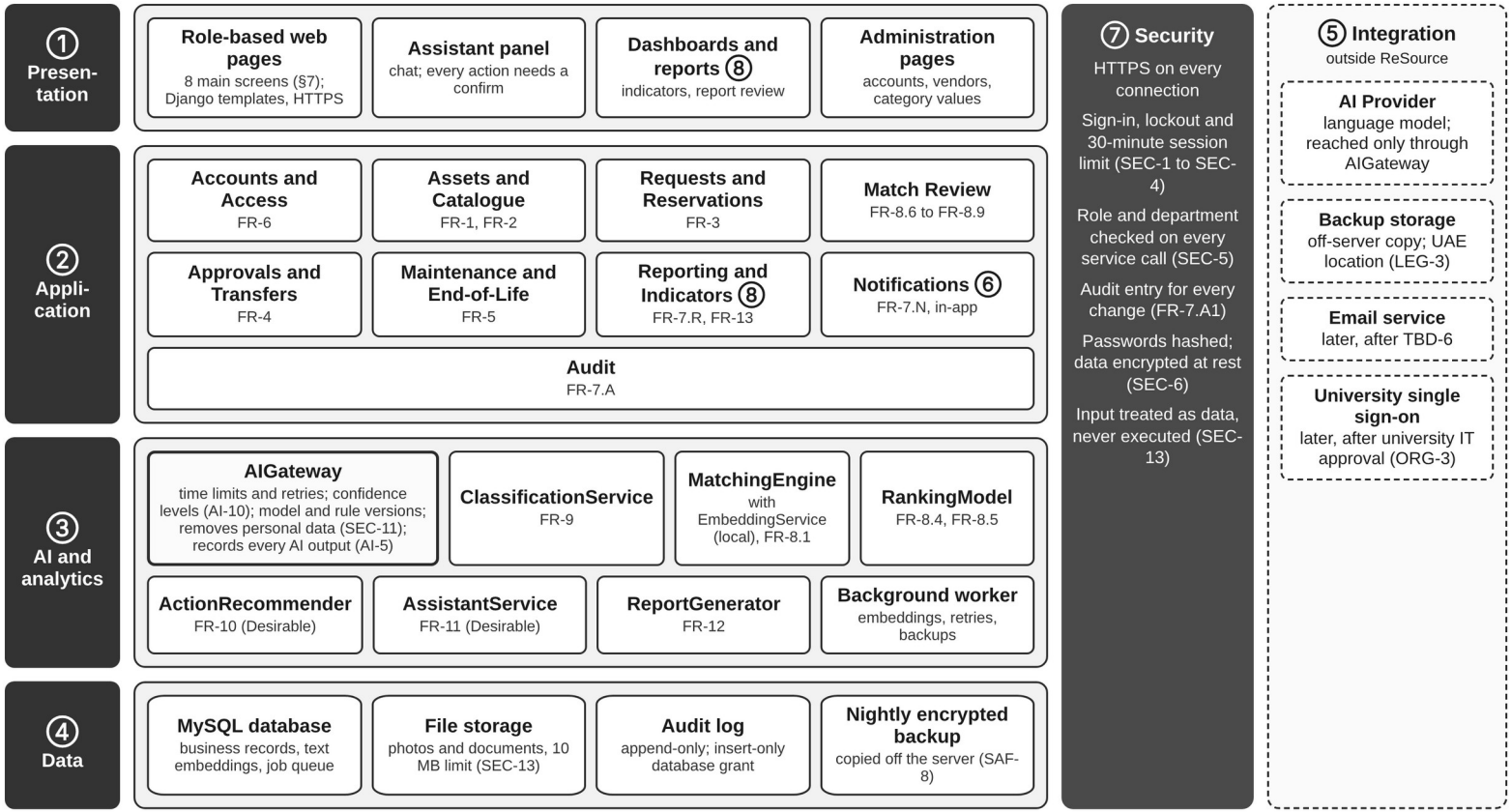


Figure 3-1: Architecture view

Table 3-8: Components

Component	Contents	Requirements	When AI is down
(1) Presentation	Role-based web pages (Django templates), the assistant panel, dashboards and administration pages; keyboard use and 200% zoom [9]	QUA-1 to QUA-3	Pages show the standard form or list and an "AI unavailable" note
(2) Application	Nine business modules: Accounts and Access, Assets and Catalogue, Requests and Reservations, Match Review, Approvals and Transfers, Maintenance and End-of-Life, Reporting and Indicators, Notifications, Audit	FR-1 to FR-8, FR-13, BR-1 to BR-12	Unchanged; every module works without AI (PER-6)
(3) AI and analytics	AIGateway; classification, semantic matching, ranking, sustainable-action recommender, assistant and report generator; background worker	FR-8 to FR-12, AI-1 to AI-10	Gateway returns "unavailable" within 20 s or at once when the service is marked down
(4) Data	MySQL database (business records, text vectors, job queue), file storage, append-only audit log, nightly encrypted backup	FR-7.A, SAF-8, SEC-6	Unchanged
(5) Integration	AI Provider; off-server backup storage; later the e-mail service (TBD-6) and university single sign-on (ORG-3)	LEG-3, SAF-8	Provider down: fallbacks in Table 4-1
(6) Notification	In-application notifications created after each committed event, with retry	FR-7.N1 to FR-7.N6	Unchanged
(7) Security	HTTPS, sign-in and lockout, role and department checks on every call, audit entries, encryption, input handling	SEC-1 to SEC-13	Unchanged
(8) Reporting	Standard reports and the nine indicators calculated by fixed formulas; AI report review	FR-7.R, FR-12, FR-13	Standard reports and indicators replace the AI report (FR-12.11)

3.4 How the Parts Communicate

Prepared by: Mohamed Almarzooqi

Five rules keep the module borders and make the AI part replaceable. Rule 1: calls only go down: a layer may call any layer below it, never one above it. Rule 2: a module uses another module through its service interface and never reads or writes the other module's tables directly. Rule 3: only AIGateway talks to the AI services and the AI Provider. Rule 4: the role and department are checked on every service call, not only on the page. Rule 5: no AI output changes a business record without a decision by an authorized person (BR-9, SAF-2).

Table 3-9: Communication paths

From	To	How	Example
Web pages	Module services	Direct Python call in the same request; HTTPS from the browser	Request form calls submit (Requests and Reservations)
Assistant panel	AIGateway	Synchronous call, 20 s limit (pattern A); the only page that calls AIGateway	A question in the Conversational Interaction screen
Module service	Another module	Service interface call (Section 16.1)	approve (Approvals and Transfers) calls reserve (Assets and Catalogue)
Module service	AIGateway	Synchronous call, 20 s limit (pattern A)	Match Review asks AIGateway to rank a request
Module service	Worker	Job placed in the queue (patterns B and C)	Assets and Catalogue queues "embed asset" after a save
AIGateway	AI Provider	HTTPS API call with the field allow-list	Classification text, report draft

From	To	How	Example
Any module	Notifications, Audit	Audit entry in the same transaction; notification after commit	Approval writes its audit entry, then notifies

4. AI Services

Prepared by: Ahmed AlAbdouli

ReSource has six AI services: the five named in the project description (classification, semantic matching, ranking, recommendation engine and LLM assistant) and the AI report generator required by FR-12. Table 4-1 gives, for each service, the call pattern, why we chose it, the time limit, what happens when the service is down and what happens when it is unsure. The patterns are those of the Phase 4 guide [2]: A, ask and wait; B, compute ahead; C, queue and notify.

Table 4-1: AI services, call patterns and fallbacks

Service	Pattern	Why	Time limit	If down	If unsure (where Low starts)
ClassificationService (FR-9, UC-10): category, tags, description, missing fields	A	The Asset Coordinator is on the form and can always choose the category from the list without it.	Progress shown within 1 s; result within 15 s for 90% (PER-4); cut off at 20 s (PER-5)	Fields stay empty, entries are kept; the coordinator picks from the approved list (FR-9.8)	Calibrated confidence below 0.50, or the local vote and the language model disagree: shown as Low, nothing pre-filled; never pre-accepted at any level (FR-9.6)
EmbeddingService and MatchingEngine (FR-8.1): text vectors for assets and requests	B	Each asset is embedded once when saved and read by every later request; the request is embedded on save.	Search over vectors under 1 s; embedding job retried 3 times (after 1, 5 and 15 min)	Keyword and rule search, results labeled "not ranked"; the asset waits for the retry	Cosine similarity below 0.30 gives a function score of 0, so the asset cannot reach the top of the list
RankingModel (FR-8.3 to FR-8.6, UC-9)	A	The Procurement Team member (or the requester on the request form) is waiting for the list.	Ranking itself under 2 s; whole match page within 15 s for 90%, cut off at 20 s	Keyword and rule matches only, labeled "not ranked", no scores (AI-6)	No candidate scores 60 or more: "weak matches" warning and no top pick; each match also has its own confidence level (Low below 0.50)
ActionRecommender (FR-10, Desirable, UC-7)	C	Nobody waits: it runs when an inspection starts, a repair is closed as not repairable, the cost flag is raised (FR-5.11) or an end-of-life proposal is opened; the suggestion is ready when a person reviews it.	Suggestion ready within 5 minutes of the trigger; 3 retries	No suggestion; the proposal shows the asset summary for manual review (FR-10.8)	Low (below 0.50) is marked; for recycling, retirement or disposal the approver must record their own reason (SAF-6); never applied by itself
AssistantService (FR-11, Desirable, UC-11)	A	The Department Representative waits in the chat.	Progress within 1 s; first answer within 15 s for 90%; cut off at 20 s	"The assistant is unavailable"; the draft is kept and opens in the request form; search still works (FR-11.9)	Answers only from permitted records returned by its read tools; if none support an answer it says so (QUA-15); if the request is unclear (below 0.50) it asks a question instead of drafting; every action needs a confirm (FR-11.3)
ReportGenerator (FR-12, UC-12)	A	The officer waits for the draft; PER-4 includes AI reports in the 15-second target.	Progress within 1 s; draft within 15 s for 90%; cut off at 20 s	Message, then the standard report and indicators for the same period open (FR-12.11)	Any figure that differs from the standard figures rejects the draft (FR-12.4); each recommended action shows its confidence and Low ones are marked

4.1 AIGateway

Prepared by: Ahmed AlAbdouli

Every AI call goes through AIGateway (Rule 3). The gateway does the same six things for every service, so no module has to repeat them:

1. **Checks access first.** It receives the user with each call and passes on only records that user may see (AI-7).
2. **Removes personal data.** It builds the payload from the field allow-list in Section 16.2 and removes e-mail addresses, university ID patterns and names of registered users from free text (SEC-11).
3. **Applies the time limits.** It shows progress within 1 second and stops waiting at 20 seconds (PER-5). After three failures of the same service within five minutes it marks the service as down for two minutes and answers "unavailable" at once, so users do not wait 20 seconds each time.
4. **Sets the confidence level.** It turns the calibrated confidence into High, Medium or Low with the versioned thresholds in Table 4-2 (AI-10).
5. **Treats text as data.** Instructions found inside asset descriptions, messages or files cannot change permissions or approval rules (AI-9); the services can only read through permission-filtered tools and can never write a business record.
6. **Records every output.** It saves the feature, input record references, model and rule versions, result, confidence, explanation and time taken, and later the human decision (AI-5, FR-7.A4).

Table 4-2: Confidence levels (rule set v3)

Level	Calibrated confidence	Shown to the user as	Effect
High	0.80 or more	Text "High" and the number	Normal suggestion; still needs a person to accept
Medium	0.50 to below 0.80	Text "Medium" and the number	Normal suggestion with its main limitation
Low	Below 0.50, or required information missing or conflicting	Text "Low", the number and the reason	Not pre-filled; disposal-type advice needs the approver's own reason (SAF-6)

Confidence is always shown as text as well as color (QUA-3). The thresholds are those of AI-10 and are stored with the model version (Section 19); they are confirmed against evaluation data before release (TBD-12).

4.2 How Each Service Works

Prepared by: Ahmed AlAbdouli

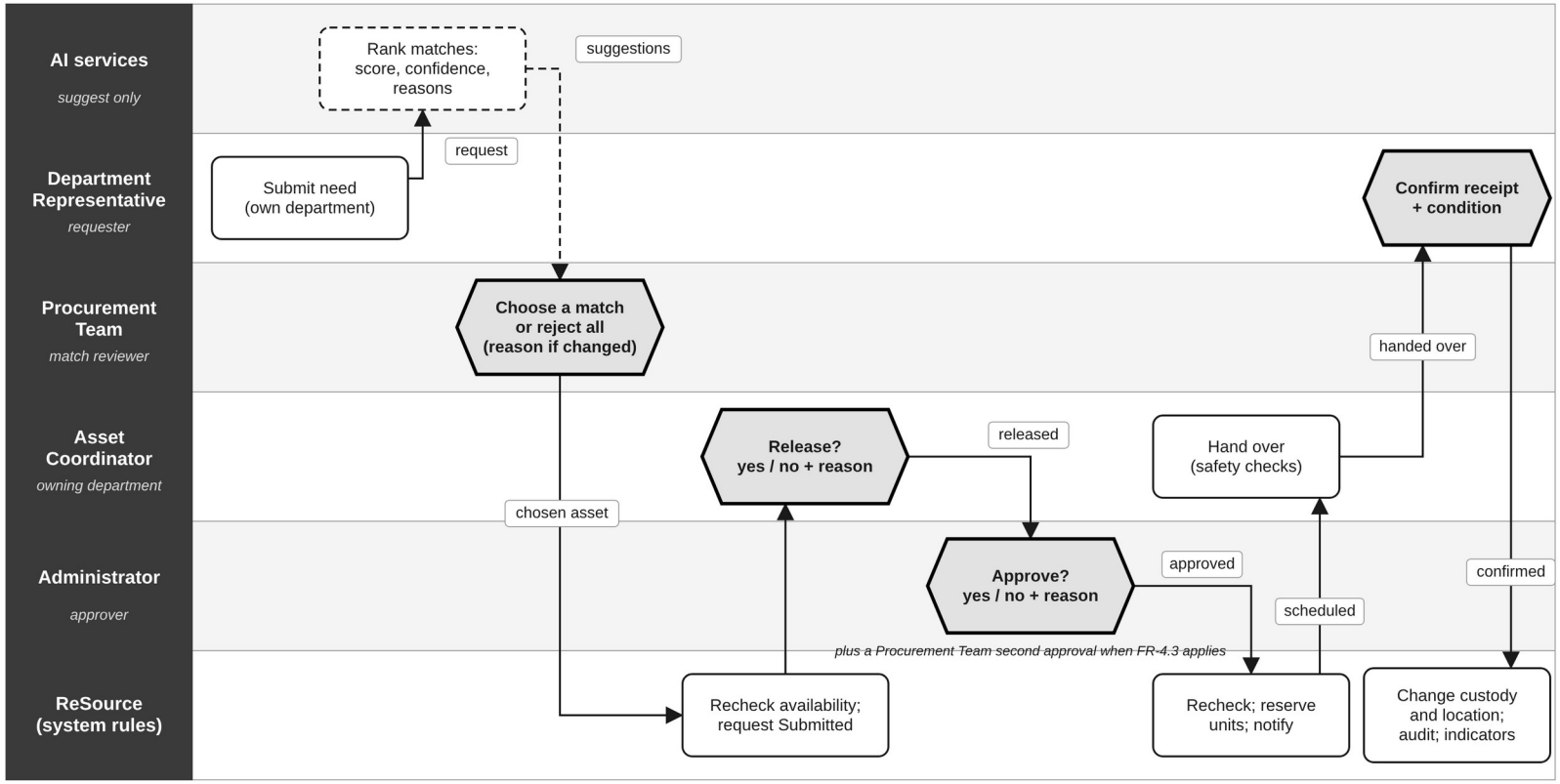
- **ClassificationService.** The local embedding model finds the five most similar already-classified assets and votes for a category; the language model suggests a category from the approved list, tags, a factual description and the missing fields. If both agree, the confidence is the calibrated vote; if they disagree, the level is Low and both options are shown with the reason (FR-9.5). With fewer than five labeled neighbours the level is at most Medium.

- **EmbeddingService and MatchingEngine.** The worker embeds name, description, specifications, category and tags with all-MiniLM-L6-v2 [10], [11], a 384-dimension sentence model released under the Apache 2.0 license, and stores the vector with a hash of the text, so an unchanged asset is not embedded again.
- **RankingModel.** A documented, deterministic score of eight factors (Section 17) ranks the 20 most similar eligible assets. The weights are visible to Administrators and do not include the department name or size (LEG-9).
- **ActionRecommender.** It scores the eight actions of FR-10.1 with rules on condition, safety, maintenance cost, demand (open requests in the same category) and remaining life, prefers keeping assets in use (FR-10.3) and shows cost, weight diverted and carbon estimates from Section 18.
- **AssistantService.** The language model gets the question and three read tools (search permitted assets, read request status, read indicators). It can propose a request draft but cannot submit it; submission goes through the normal request service after the user confirms.
- **ReportGenerator.** The report service first calculates every figure with the fixed formulas; the language model only writes text around those figures. The gateway checks that every number in the draft equals a standard figure or a change calculated from them (FR-12.4).

5. Human Approval Points

Prepared by: Ahmed AlAbdouli

We put a person wherever a wrong AI answer would be costly or hard to undo: where an asset could be reserved or moved, where a saved record would change, where something could be disposed of, and where a report could be shared. Figure 5-1 shows the gates on the main reuse path, and Table 5-1 lists all of them. At each gate the system records who decided, when, the decision, the reason and what the person saw: the AI result, its confidence and its explanation (FR-7.A4, AI-5).



Grey hexagons with a thick border are human gates: each records who decided, when, the decision, the reason and what the person saw (the AI score, confidence and explanation, FR-7.A4). Check: no path from the AI lane reaches a ReSource record change without passing a gate. No person approves their own request or fills two approval stages (FR-4.4, BR-4).

Figure 5-1: Human approval gates

Table 5-1: Human approval points

ID	Decision	Who decides	What they see	Log entry	Requirements
H-1	Accept, edit or reject each AI classification field	Asset Coordinator of the owning department	Each suggestion, its confidence and reason, conflicting sources	AI output, decision per field, final values	FR-9.6, FR-9.9, FR-1.12
H-2	Choose a match, choose another or reject all	Procurement Team member	Ranked list, score, factor values, confidence, explanation, rule checks	Recommendation decision and reason when not the top match	FR-8.7, AI-3, QUA-12
H-3	Accept a partial or split quantity	Requesting Department Representative	Units available and units requested	Confirmation on the request	FR-8.6
H-4	Release or decline a request	Owning Asset Coordinator (not the requester)	Request, quantity, requester department, the match score if any	Approval record (Release)	FR-4.1, FR-4.4, BR-3
H-5	Approve or reject the released request	Administrator (did not submit or release it)	Request, asset, recheck result, AI score and reasons if from a match	Approval record (Approval)	FR-4.2, FR-4.5, BR-4
H-6	Second approval for high or unknown value	Procurement Team member (third distinct person)	Estimated value and the limit	Approval record (Second approval)	FR-4.3, TBD-2
H-7	Confirm receipt or report a problem; decide the dispute	Receiving DR or AC; Administrator for disputes	Transfer, handling notes, expected condition	Transfer state, condition, decision	FR-4.10, FR-4.12
H-8	Safety review of an end-of-life proposal	Sustainability Officer other than the proposer	Hazard flag, inspection results, AI action advice if any	Safety review, handling party	FR-5.23, FR-5.24, SAF-3
H-9	Approve the end-of-life action	Administrator other than the proposer	Proposal, safety review, AI advice and its confidence	Approval with reason (own reason if advice is Low)	BR-8, SAF-4, SAF-6, FR-10.7
H-10	Approve an AI report before sharing	Finance Officer or Administrator; Sustainability Officer for reports without financial figures	Draft, figure check, assumptions, model version	Original draft, edits and approved version	FR-12.9, FR-12.10, BR-11
H-11	Confirm an assistant action	The Department Representative who asked	Proposed action and final values	Assistant action, confirmed or cancelled	FR-11.3, FR-11.10
H-12	Human review of an AI match or recommendation	Administrator	The AI output, its inputs and the decision taken	Human review record	SEC-10
H-13	Put a new model version or threshold set into use	System Administrator after the evaluation run passes	Evaluation results against the targets in Table 19-1	Model version, approved by	AI-10, QUA-7, QUA-9, QUA-10

The AI cannot do any of the following on its own: grant access, approve a request, reserve units, change custody or location, certify safety, complete an end-of-life action, share a report or change its own thresholds (BR-9, SAF-2). In Figure 5-1, no path from the AI lane reaches a change to a ReSource record without passing through a gate. The system also stops one person from filling two stages of the same approval (FR-4.4, BR-4, SAF-3).

6. Security

Prepared by: Rashid Alzaabi

6.1 Security Boundaries

Prepared by: Rashid Alzaabi

Table 6-1 lists the five places where data crosses from one level of trust to another, and the control at each. The numbers match the security column (7) and the integration column (5) in Figure 3-1.

Table 6-1: Security boundaries

Boundary	Crossing	Controls	Requirements
B1 Internet to server	Browser to the reverse proxy	HTTPS only (TLS 1.2 or later), HTTP redirected; secure, HTTP-only session cookie; 30-minute idle limit; lockout after 5 failed sign-ins	SEC-1, SEC-3, SEC-4, SEC-6
B2 User to function	Every service call inside the application	Default deny; the role, department and account status are checked on each call (Table 6-2)	FR-6.7, SEC-5
B3 Application to database	Web and worker containers to MySQL	Private network only; separate database accounts; the audit table can only be inserted into	FR-7.A2, SEC-12
B4 ReSource to AI Provider	AIGateway to the AI Provider	HTTPS; field allow-list; personal data removed; API key kept in a server secret, never in code	SEC-11, LEG-3, AI-9
B5 Server to backup storage	Nightly backup job	Encrypted before it leaves the server; storage in the UAE; separate credentials with write-only access for the job	SAF-8, LEG-3

6.2 Access Matrix

Prepared by: Rashid Alzaabi

Table 6-2 shows who may perform each controlled action, for all nine roles, the Applicant and the AI services. The rule is default deny: anything not marked is refused, and the refusal is audited (FR-7.A1). "Own" means the user's own department, and the other conditions in the cells are checked by the system. DR, AC, AD, PT, MS, FO, SO, SA and UM are the nine roles of Table I-1, and "-" means refused. Sources: Phase 3 Table 5.1-1, SEC-5, FR-7.R1, FR-7.R3, FR-7.A3, FR-12.1, FR-12.9 and BR-2 to BR-11.

Table 6-2: Access matrix (default deny)

Action	DR	AC	AD	PT	MS	FO	SO	SA	UM	Applicant	AI services
Apply for an account, give or refuse consent	-	-	-	-	-	-	-	-	-	Yes	never
Activate accounts; assign roles and departments	-	-	-	-	-	-	-	Yes	-	-	never
View own personal data, request correction, withdraw consent	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	-	never
Search the catalogue	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	-	rank only
See financial values (SEC-5)	-	own assets	Yes	Yes	own work	Yes	totals only	-	Yes	-	never
Register, edit, publish or withdraw an asset	-	own	-	-	-	-	-	-	-	-	suggest only
Accept AI classification suggestions	-	own	-	-	-	-	-	-	-	-	never
Submit, edit or cancel a request	own	-	-	-	-	-	-	-	-	-	draft only
Choose a match for an open need	-	-	-	Yes	-	-	-	-	-	-	rank only
Release or decline a request	-	own assets, not own request	-	-	-	-	-	-	-	-	never
Approve or reject a released request	-	-	not own or released	second approval	-	-	-	-	-	-	never
Arrange a transfer and record handover	-	own assets	-	-	-	-	-	-	-	-	never
Confirm receipt or report a problem	own	own	-	-	-	-	-	-	-	-	never
Decide a transfer dispute	-	-	Yes	-	-	-	-	-	-	-	never
Report a defect	Yes	Yes	-	-	Yes	-	-	-	-	-	never
Request an inspection	-	Yes	-	-	Yes	-	-	-	-	-	never
Assign maintenance work	-	-	Yes	-	Yes	-	-	-	-	-	never
Record inspection and repair results	-	-	Yes	-	assigned	-	-	-	-	-	never
Propose an end-of-life action	-	own	-	-	-	-	Yes	-	-	-	recommend only
Safety review of an end-of-life proposal	-	-	-	-	-	-	not proposer	-	-	-	never
Approve an end-of-life action	-	-	not proposer	-	-	-	-	-	-	-	never
Maintain the approved vendor list	-	-	Yes	-	-	-	-	-	-	-	never
Maintain category values	-	-	-	-	-	prices, costs	weights, factors	-	-	-	never
Generate standard reports	own	own	Yes	Yes	-	Yes	Yes	-	Yes	-	never
Request an AI report	-	-	Yes	Yes	-	Yes	Yes	-	-	-	draft only
Approve an AI report	-	-	Yes	-	-	Yes	no financial figures	-	-	-	never
Use the assistant	Yes	-	-	-	-	-	-	-	-	-	answer, draft
Request a human review of an AI result	affected	affected	-	affected	affected	-	affected	-	-	-	never
Decide a human review	-	-	Yes	-	-	-	-	-	-	-	never
Read the audit log	-	-	Yes	-	-	-	-	Yes	-	-	never
Manage AI model versions and thresholds; set approval limits	-	-	-	-	-	-	-	Yes	-	-	never

Holding several roles does not let a person skip a separation rule: the checks compare user IDs across the stages of the same record, not role names (BR-4). An account that is suspended or has withdrawn consent is refused on its next call, even in an open session (FR-6.7, FR-6.10).

6.3 Audit Log

Prepared by: Rashid Alzaabi

The audit log records every action listed in FR-7.A1. Each entry is written in the same database transaction as the change it describes; if the entry cannot be written, the change is rolled back and the user sees an error (FR-7.A6). Table 6-3 lists the fields.

Table 6-3: Audit entry fields

Field	Content	Example
Who	Internal user ID and the role used (never a name or e-mail, SEC-8)	user 1047, Administrator
What	Action and record type and ID	APPROVE, request R-2026-0391
When	Date and time to the second, server clock (UAE time)	27/10/2026 14:02:31
Before	Old values of the changed fields	state: Released
After	New values of the changed fields	state: Approved; reserved: 2
Reason	Reason entered by the user, where required	Not needed for approval
AI link	AI output ID when the action used an AI result	AI-2026-08812
Chain	Hash of the previous entry and of this entry	9f3a...c21

- **Cannot be changed.** The application connects to the audit table with a database account that can only insert and read; no screen edits or deletes entries, even for System Administrators (FR-7.A2). Each entry stores a hash of the previous entry, so any change outside the application breaks the chain and is found by the nightly check.
- **No secrets or personal values.** Passwords, recovery secrets and personal-data field values are never stored; for those fields the entry records only that the field changed (FR-6.12, FR-7.A1).
- **Who can read it.** Only System Administrators and Administrators, searching by user, record, action and date (FR-7.A3). Every search of the log is itself audited.
- **How long.** No entry is deleted until university IT sets the retention period (FR-7.A5, TBD-4).

6.4 Data Storage and Encryption

Prepared by: Rashid Alzaabi

Table 6-4: Where data is stored and how it is protected

Data	Where	Protection
Business records, text vectors, job queue	MySQL 8 on an encrypted disk volume	Encryption at rest; private network; least-privilege database accounts (SEC-6)
Passwords	USER table	Stored only as salted PBKDF2 hashes, Django default [6] (SEC-6)
Photos and documents	File volume outside the web root	Image or PDF, 10 MB or less, type checked by content; served only after a permission check (SEC-13)
Personal data	USER and CONSENT_RECORD only	Five fields only (SEC-7); removed on consent withdrawal (SEC-8); never sent to AI services (SEC-11)

Data	Where	Protection
Secrets (database password, provider API key)	Server environment file, readable by the service account only	Never in the code repository; changed if exposed
Backups	Off-server storage in the UAE	Encrypted before upload (AES-256); separate key held by the System Administrator and one other named person

6.5 Backup and Recovery

Prepared by: Rashid Alzaabi

Table 6-5: Backup and restore plan

Item	Design
How often	Every night at 02:00 a full database dump and a copy of new or changed files; at most 24 hours of data can be lost (SAF-8)
Where	Encrypted copy on off-server storage in the UAE (LEG-3); the last 14 daily and 8 weekly copies are kept
Check	After each backup the job checks the file size and checksum and notifies the System Administrator if it failed (FR-7.N5)
Restore steps	1. Start a clean virtual machine from the deployment files. 2. Download and decrypt the last backup. 3. Load the database dump and the files. 4. Start the containers and run the health check. 5. Rebuild missing text vectors in the background (matching shows "not ranked" until done).
Restore test	Every week the System Administrator restores the latest backup into a separate test machine, times it against the 60-minute target (QUA-6), checks the record counts and the audit hash chain, and records the result
Estimated time	About 35 minutes: 10 to start the machine, 10 to download and decrypt, 10 to load, 5 to check

6.6 Integration Points

Prepared by: Rashid Alzaabi

Table 6-6: Integration points and security features

Integration or feature	Status	How it is secured or provided
AI Provider (language model)	Prototype	Through AIGateway only: HTTPS, field allow-list, secret API key, time limits, recorded outputs (DR-4)
Backup storage	Prototype	Encrypted upload, write-only job credentials, UAE location (Section 6.5)
E-mail copies of notifications	Later (TBD-6)	Desirable copy of in-application notifications; no change to the notification module interface
University single sign-on	Later (ORG-3, TBD-3)	Replaces the local sign-in after university IT approval; roles stay in ReSource
Sign-in, passwords, sessions (SEC-1 to SEC-4)	Prototype	Django authentication and session framework, lockout rule in the Accounts module, 30-minute idle expiry
Permissions and encryption (SEC-5, SEC-6)	Prototype	Access matrix checks in every service; HTTPS; encrypted volume and backups
Input handling (SEC-13)	Prototype	Django forms and parameterized queries; template escaping; upload type and size checks [12]

6.7 Draft Data Protection Impact Assessment

Prepared by: Rashid Alzaabi

LEG-2 requires a data protection impact assessment to be drafted during design, because AI matching evaluates requests automatically (UAE PDPL [13]). Table 6-7 is the draft. The university must approve it before any real personal data is used (LEG-2, ORG-2, TBD-11).

Table 6-7: Draft data protection impact assessment

Item	Assessment
Personal data processed	Name, university ID, university e-mail, department and role of each user (SEC-7); consent records. No sensitive personal data.
Purpose	Running accounts, permissions, approvals and the audit trail for campus asset exchange.
Automated evaluation	Matching ranks assets for a request; it does not evaluate people. Department name and size are not inputs (LEG-9); every match is decided by a person (H-2), and any affected user can ask for a human review (SEC-10).
Transfers outside the UAE	None of personal data: the AI Provider receives only allow-listed asset and request fields (SEC-11, LEG-3). Whether non-personal asset text may go to the provider is confirmed under TBD-9.
Main risks	1. Personal data typed into free text reaches the provider. 2. A user sees records of another department. 3. Audit history conflicts with an erasure request. 4. A data breach of the server or backups.
Measures	1. Gateway redaction and a warning on free-text fields. 2. Default-deny checks on every call and tests per role. 3. Audit entries hold only internal user IDs, so erasing the user record leaves no name (SEC-8). 4. Encryption at rest and in transit, encrypted backups, lockout and session limits.
Remaining risk and decision	Low after the measures, subject to university IT and legal review (TBD-4, TBD-11).

7. Mockups

Prepared by: Mohammad Alsuwaidi

The eight screens below are the ones listed in the project description. Each screen shows the role, real fields with sample data, the AI result with its confidence and reason, the decision the person takes and the state shown when the AI is unavailable. Numbered markers on each screen point to these five parts: (1) AI result, (2) confidence, (3) reason, (4) human decision and (5) AI-down state. Table 7-1 summarizes the screens.

Table 7-1: Mockup screens

Screen	Role	Use case	AI result shown	Human decision	AI-down state	Requirements
7.1 Asset Registration	Asset Coordinator, Electrical Engineering	UC-1, UC-10	Category, tags and description suggestions with confidence and reason	Accept, edit or reject each field	Fields stay empty; choose from the list	FR-1.2, FR-9.1 to FR-9.9
7.2 Request Submission	Department Representative, Physics	UC-3, UC-2	Existing listings that may meet the need, with fit score and reason	Request a listing or submit an open need	Standard search, results not ranked	FR-3.2, FR-3.3, FR-3.13, FR-8.1
7.3 Recommendation Review	Procurement Team	UC-9	Ranked matches, factor values, confidence, explanation	Select, choose another or reject with a reason	Keyword and rule matches, not ranked	FR-8.3 to FR-8.9, LEG-9
7.4 Transfer Approval	Administrator	UC-4	Match score and reasons that led to the request	Approve or reject with a reason	Approval works; AI details hidden	FR-4.2 to FR-4.6
7.5 Maintenance Tracking	Maintenance Staff	UC-6, UC-7	Sustainable-action suggestion with confidence and estimates	Record results; propose or reject the action	No suggestion; manual review	FR-5.4 to FR-5.12, FR-10
7.6 Sustainability Dashboard	Sustainability Officer	UC-12	AI highlights of large changes	Review indicators and request a report	Indicators shown, no highlights	FR-13.1 to FR-13.12, FR-12.8
7.7 AI Report Review	Finance Officer	UC-12	AI report draft with a figure check and assumptions	Approve, edit or discard	Standard report opens	FR-12.1 to FR-12.11, BR-11
7.8 Conversational Interaction	Department Representative	UC-11	Answer with linked listing and a request draft	Confirm, edit or cancel	Assistant unavailable; draft kept	FR-11.1 to FR-11.10

7.1 Asset Registration

Prepared by: Mohammad Alsuwaidi

The Asset Coordinator of Electrical Engineering registers the surplus oscilloscopes. The AI suggestions appear in a separate panel, none is pre-accepted, and each field is accepted, edited or rejected on its own (FR-9.6). The hazard flag is entered by the person, never suggested (FR-1.15).

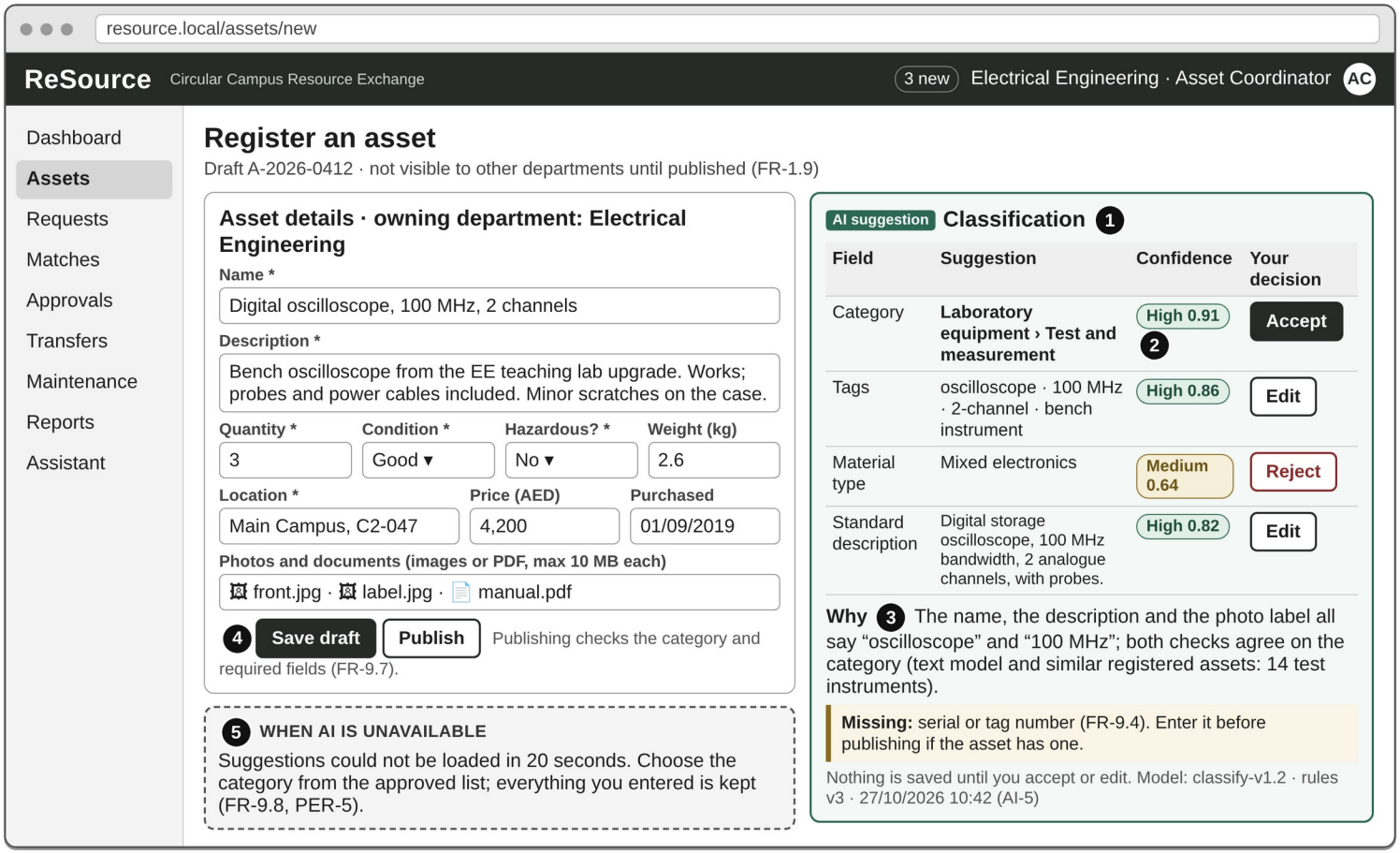


Figure 7-1: Asset Registration screen

7.2 Request Submission

Prepared by: Mohammad Alsuwaidi

A Physics Department Representative describes the need. While typing, the page shows listings that may already meet it, with a fit score, a confidence level and the reason (FR-8.1). The representative can request a listing directly (UC-3) or submit an open need for the Procurement Team (UC-9).

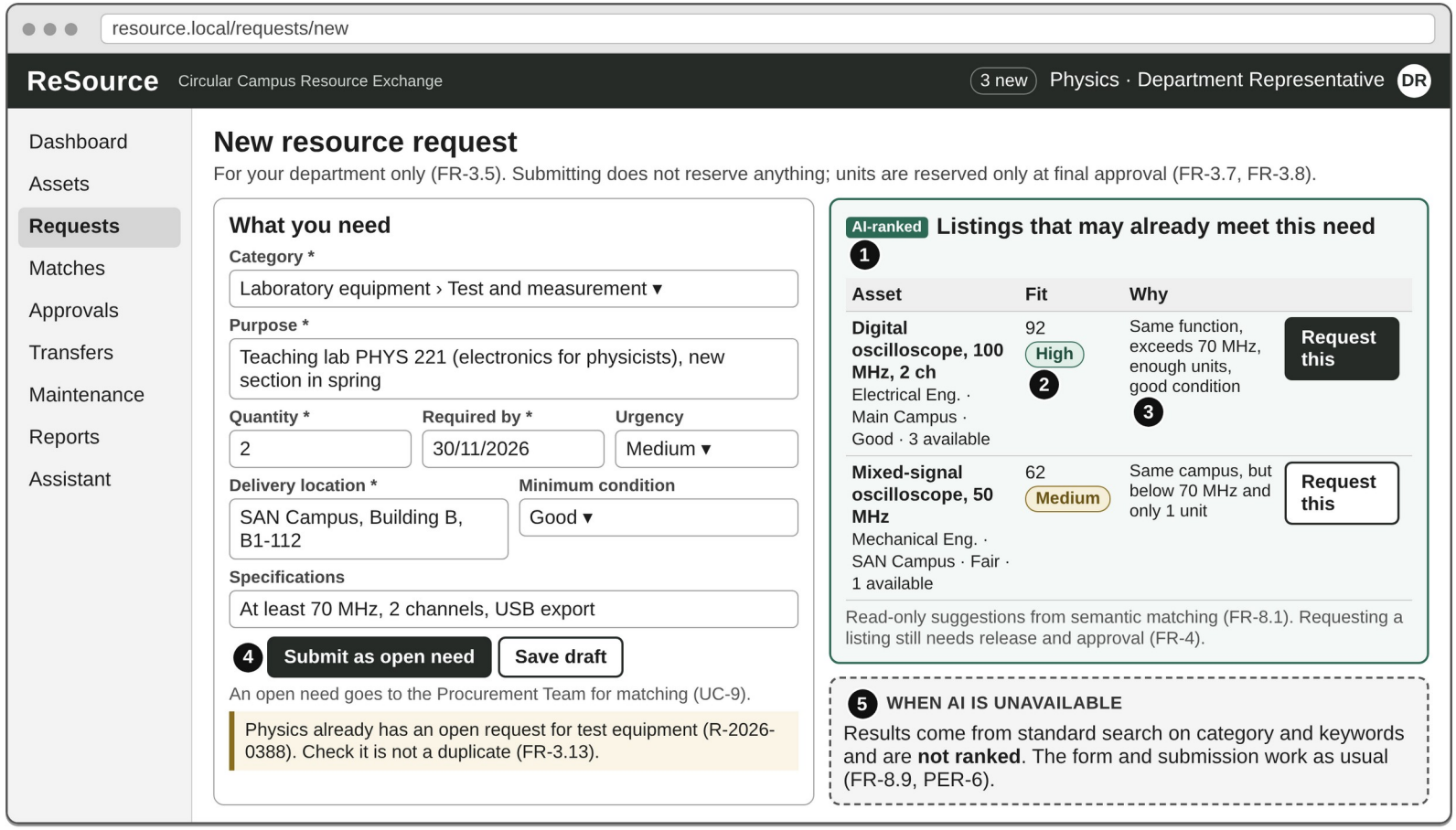


Figure 7-2: Request Submission screen

7.3 Recommendation Review

Prepared by: Ahmed AlAbdouli

The Procurement Team member sees the ranked matches for R-2026-0391. The top match shows its eight factor values, the confidence and a plain-language reason; the second match shows its main limitation. Choosing any other option asks for a reason (FR-8.7).

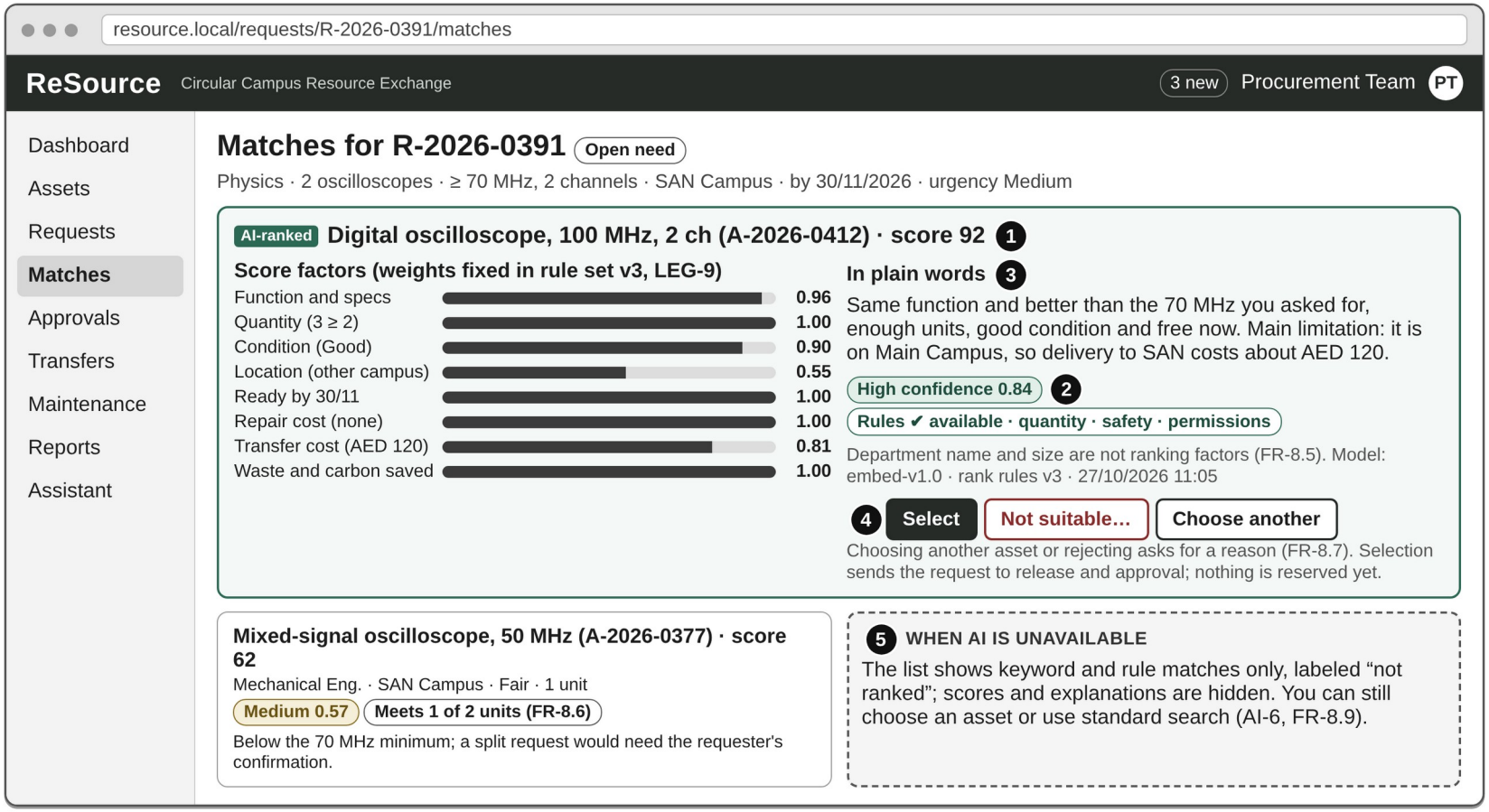


Figure 7-3: Recommendation Review screen

7.4 Transfer Approval

Prepared by: Mohammad Alsuwaidi

The Administrator approves the request released by the owning Asset Coordinator. The system shows that the quantity has just been rechecked and that the approver did not submit or release the request (FR-4.4, FR-4.5). The AI score is shown as background only; the decision is the approver's.

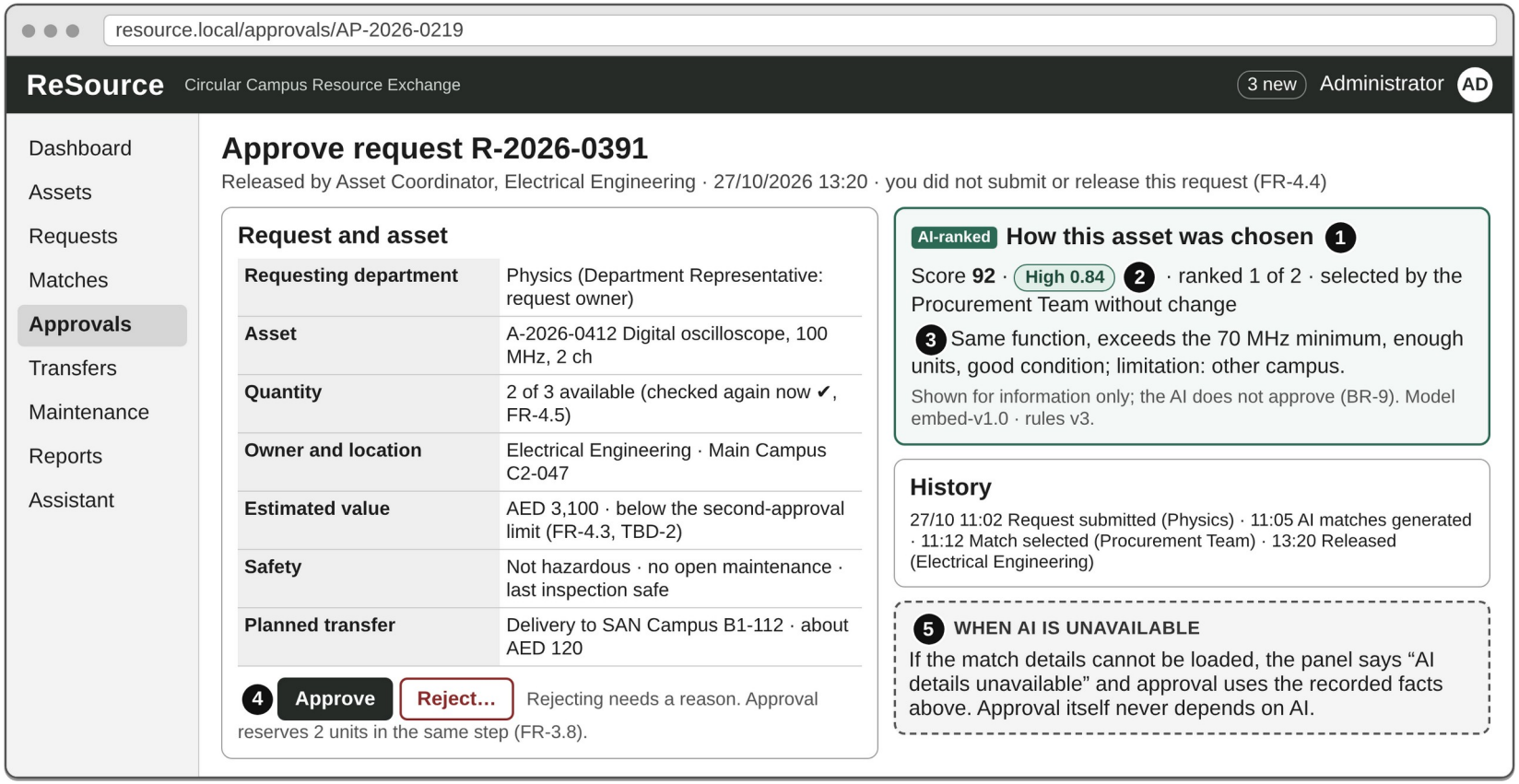


Figure 7-4: Transfer Approval screen

7.5 Maintenance Tracking

Prepared by: Rashid Alzaabi

Maintenance Staff record the inspection of a Chemistry fume hood whose airflow sensor is faulty. The asset history shows the cumulative cost against the 50% review threshold (FR-5.10, FR-5.11). The ActionRecommender suggests repair with Medium confidence, gives its reason and main limitation, and never changes the asset by itself; the staff member chooses the next step.

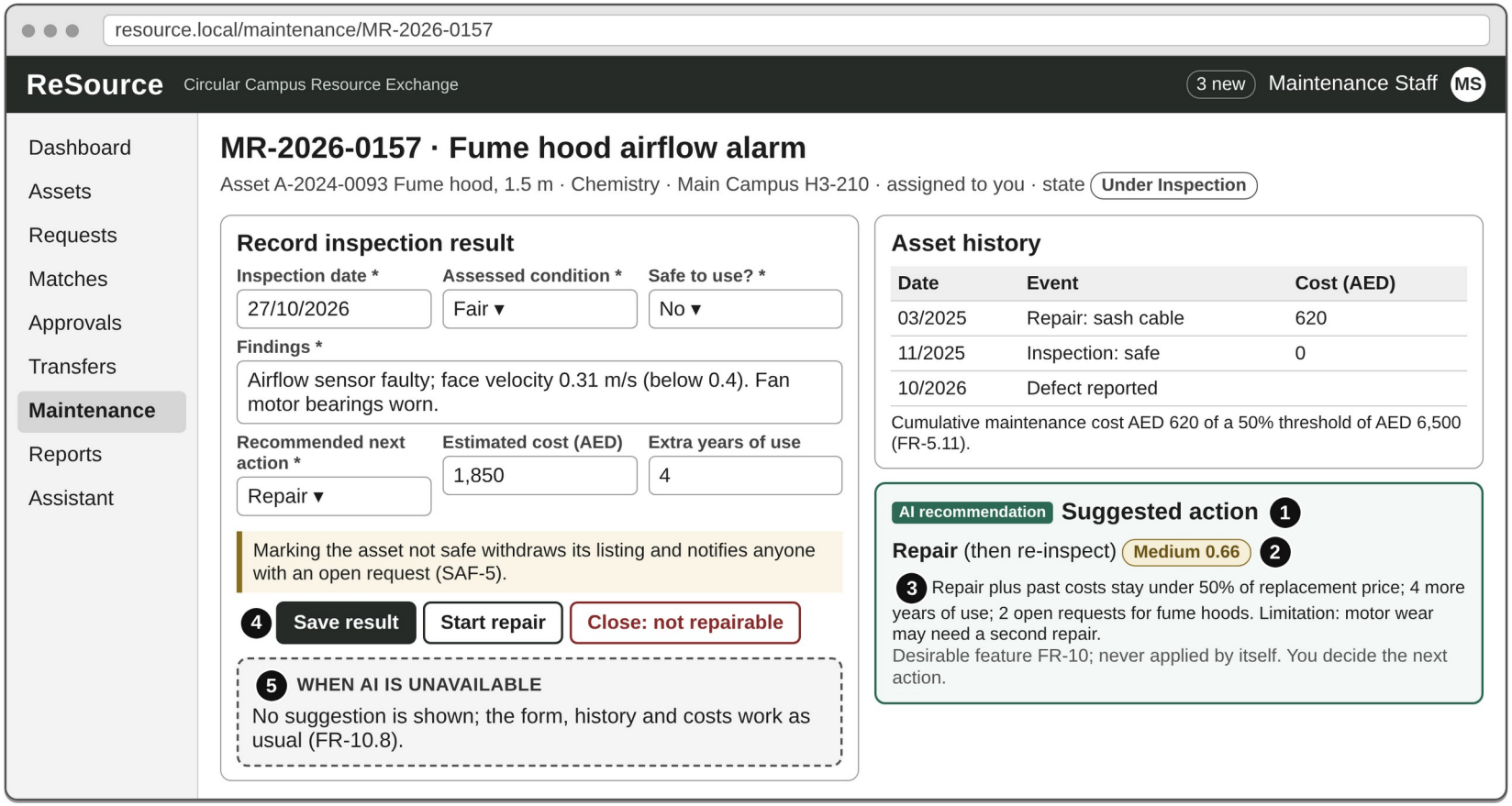


Figure 7-5: Maintenance Tracking screen

7.6 Sustainability Dashboard

Prepared by: Rashid Alzaabi

The Sustainability Officer sees the September indicators for the whole campus (the screen shows five of the nine), their trend, the share of default values and the "estimate" labels (FR-13.7, FR-13.12). AI highlights point out large changes and are calculated from the same indicator values.

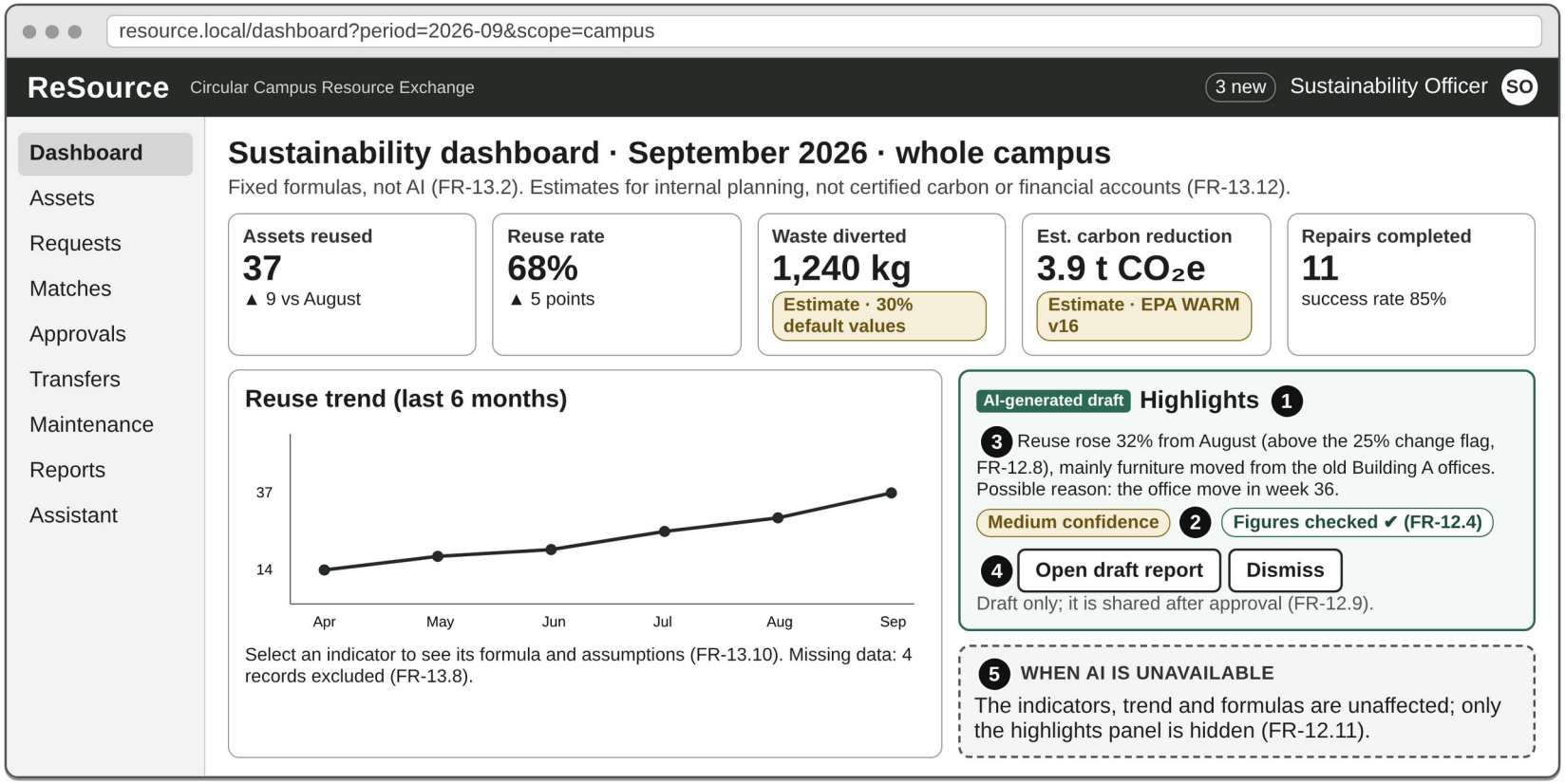


Figure 7-6: Sustainability Dashboard screen

7.7 AI Report Review

Prepared by: Rashid Alzaabi

The Finance Officer reviews the AI report draft for September. Every figure has been checked against the standard reports, the assumptions are listed, and the report stays a draft until it is approved (FR-12.9).

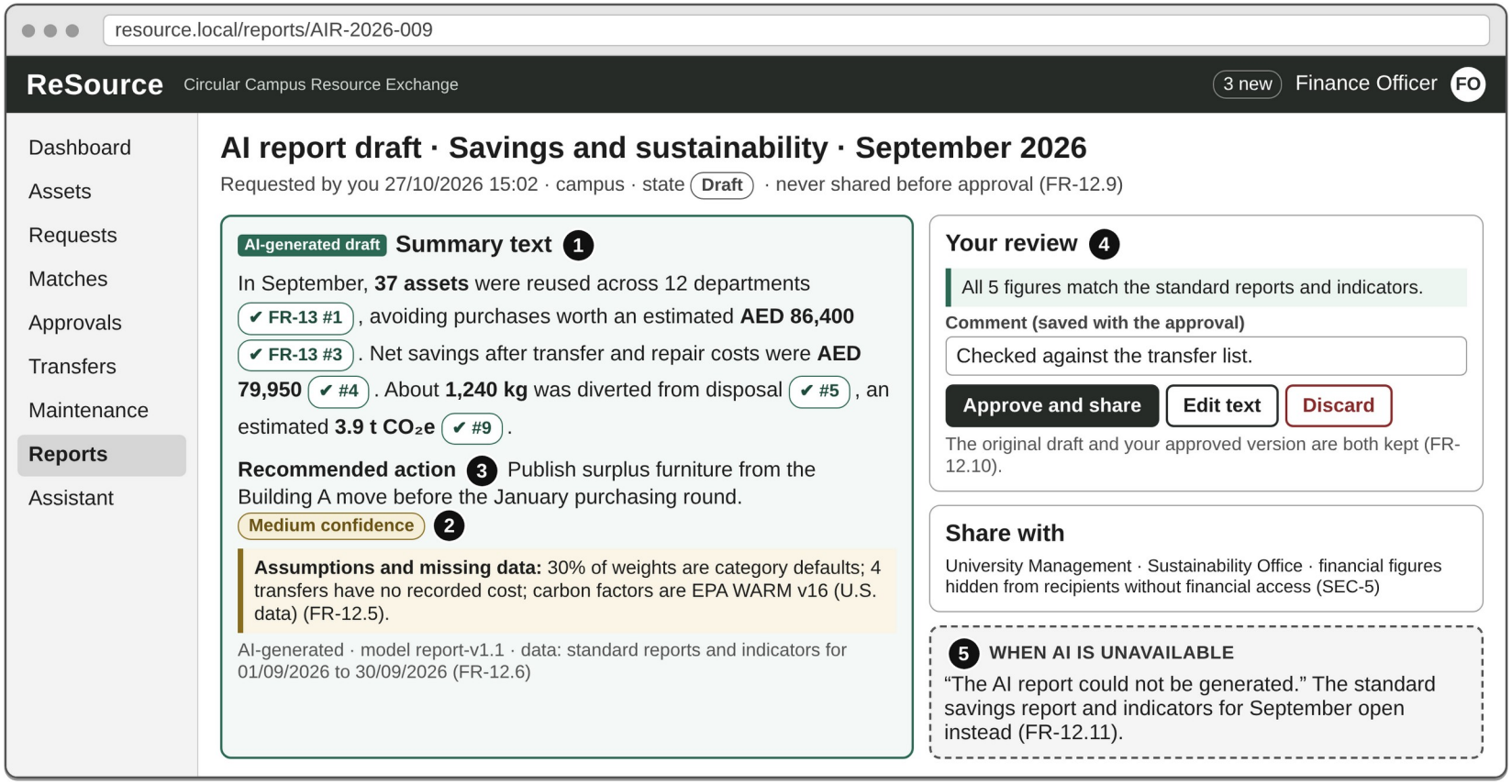


Figure 7-7: AI Report Review screen

7.8 Conversational Interaction

Prepared by: Ahmed AlAbdouli

The Department Representative asks the assistant in everyday language. The assistant answers from records the user may see, links the listing and prepares a request draft that is submitted only after the user confirms (FR-11.3).

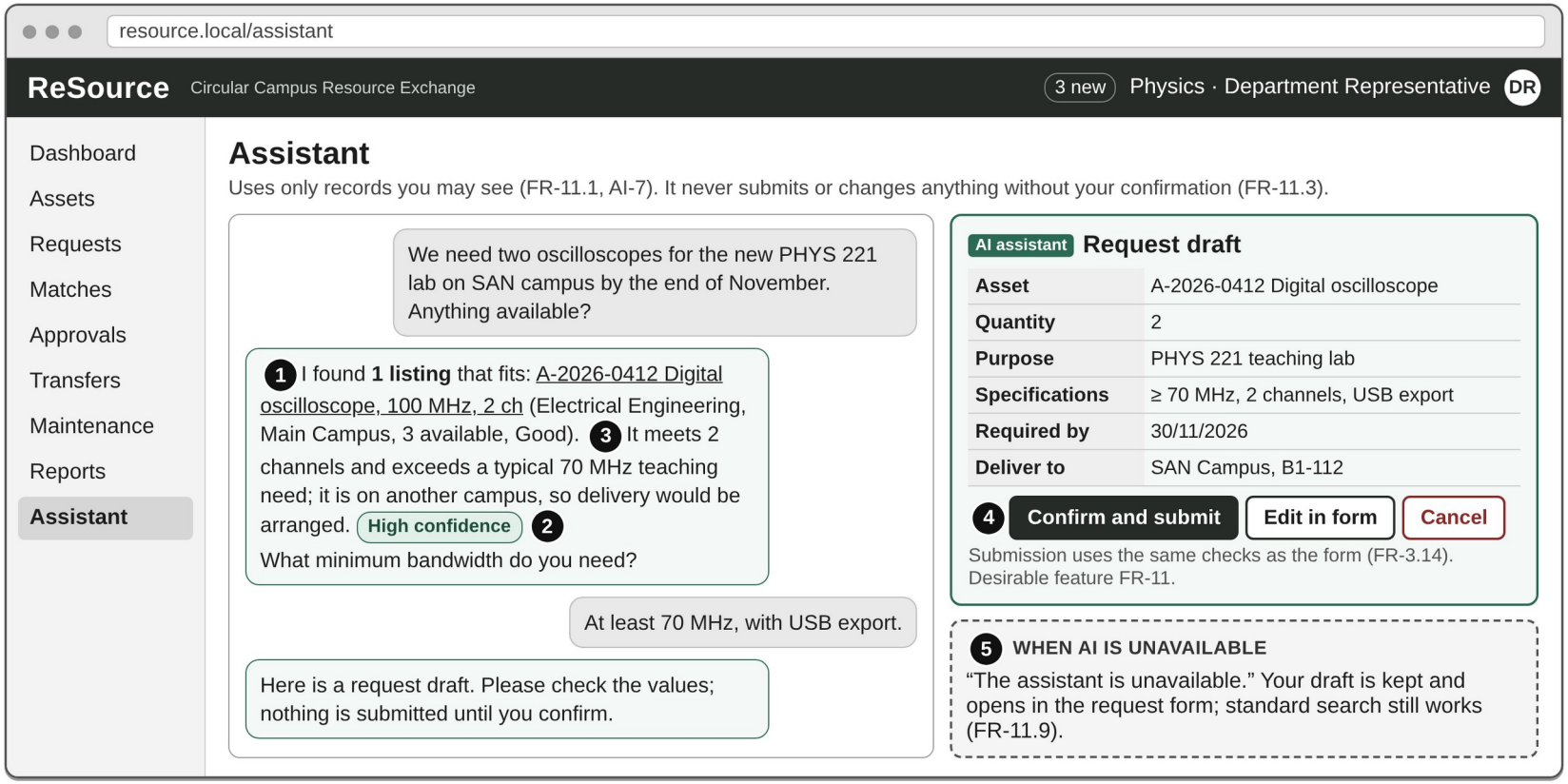


Figure 7-8: Conversational Interaction screen

8. Use Case Diagram

Prepared by: Mohamed Almarzooqi

Figure 8-1 is the use case diagram from the Phase 3 document [4]; the 13 use cases and the actors have not changed. Table 8-1 shows where each use case is designed in this document, so a reader can follow one use case from the data flow to the screen.

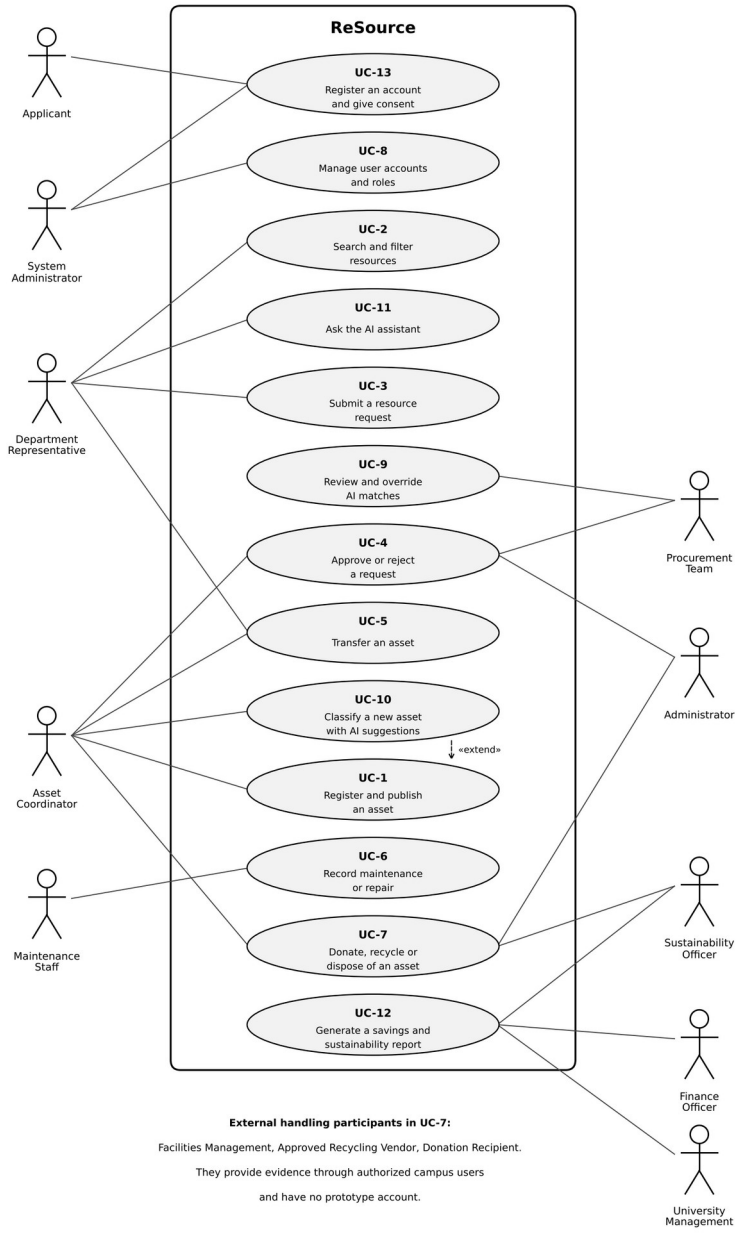


Figure 8-1: Use case diagram

Table 8-1: Where each use case is designed

Use case	DFD process	Class	Sequence	Activity	State	Mockup
UC-1 Register and Publish an Asset	2.0	9.1, 9.3	10.3	11.1	12.1	7.1
UC-2 Search and Filter Resources	2.0	9.1	10.7	11.1	12.1	7.2
UC-3 Submit a Resource Request	3.0 (Level 2)	9.1	10.6, 10.7	11.1	12.3	7.2
UC-4 Approve or Reject a Request	4.0	9.2	10.1	11.1	12.3, 12.4	7.4
UC-5 Transfer an Asset	4.0	9.2	10.2	11.2	12.1, 12.4	-
UC-6 Record Maintenance or Repair	5.0	9.2	10.9	11.3	12.2, 12.5	7.5
UC-7 Donate, Recycle or Dispose of an Asset	5.0	9.2, 9.4	10.10	11.3	12.2, 12.6	7.5
UC-8 Manage User Accounts and Roles	1.0	9.1, 9.4	10.8	-	-	-
UC-9 Review and Override AI Matches	3.0 (Level 2)	9.3	10.4	11.1, 14	12.3	7.3
UC-10 Classify a New Asset with AI Suggestions	2.0	9.3	10.3	11.1	-	7.1
UC-11 Use the AI Assistant	7.0	9.3	10.6	-	-	7.8
UC-12 Generate a Savings and Sustainability Report	6.0	9.3, 9.4	10.5	-	-	7.6, 7.7
UC-13 Register an Account and Give Consent	1.0	9.1	10.8	-	-	-

Every use case has classes in Section 9 and at least one sequence diagram in Section 10. UC-8 and UC-13 have no activity or state diagram because their steps are short; Figure 10-9 shows them in full.

9. Class Diagrams

Prepared by: Mohammad Alsuwaidi

The class model is drawn in six diagrams: the core domain (Figure 9-1); approvals and transfers (Figure 9-2); maintenance and end of life (Figure 9-3); the AI services (Figure 9-4) and the AI records they keep (Figure 9-5); and notifications, audit and sustainability (Figure 9-6). Every association has a name and a multiplicity at both ends; the name is the relationship name used in the ERD (Section 15) and is read in the direction of the small arrow. Every class that a person creates, decides on or receives is linked to User by a named association. Attributes are private (-) and operations public (+); the operations change state and match the transitions in Section 12. A class drawn in more than one diagram is the same class, shown with only the members needed there, and each class has the name of its table in Section 15 (for example Request and REQUEST).

9.1 Core Domain

Prepared by: Mohammad Alsuwaidi

Figure 9-1 shows accounts, the catalogue and requests.

- **Aggregation.** A Department employs its Users, but a user record survives a department change and a department cannot be deactivated while it has active users (FR-6.9).
- **Association class.** UserRole records who assigned a role and why (FR-6.8, BR-4). A User gives one or more ConsentRecords, and a withdrawal is a new state of the record, not a deletion (FR-6.2, SEC-8).
- **Composition.** An Asset owns its Location value and its files (up to 5 photos and 3 documents, so 0..8, FR-1.6); a Request owns its RequestSpecification. The parts are deleted only with their owner.
- **Generalization.** ITAsset and FurnitureAsset specialize Asset with their own fields. They are stored in the one ASSET table with an asset_kind column (single-table inheritance), so search and matching treat all assets alike.
- **Navigability.** A Request points to 0..1 Asset: an open need (FR-3.2) has no asset until the Procurement Team selects a match (UC-9).

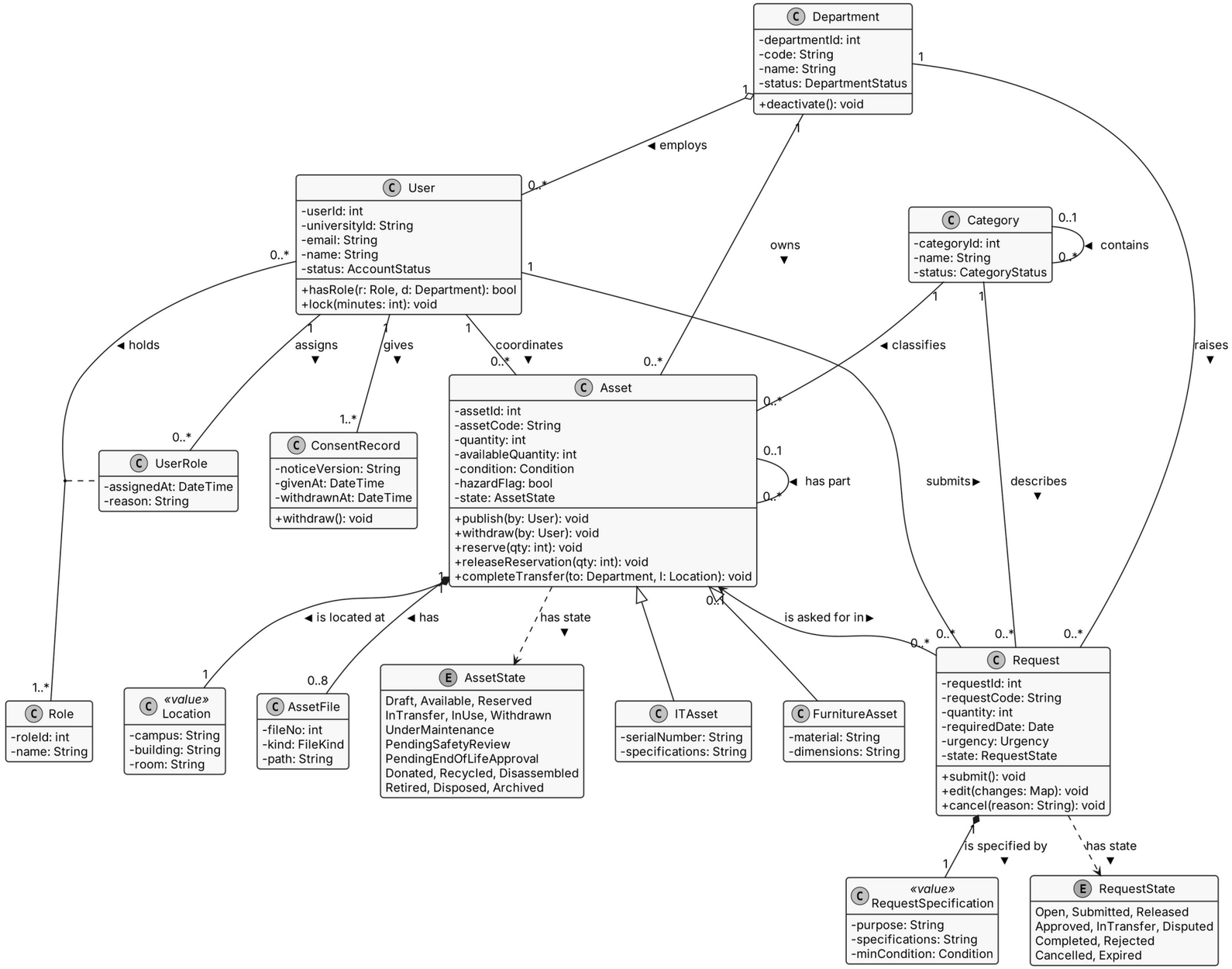


Figure 9-1: Class diagram, core domain

9.2 Approvals, Transfers and Maintenance

Prepared by: **Mohammad Alsuwaidi**

A Request has 0 to 3 Approvals: release, approval and, when FR-4.3 applies, the second approval. Each Approval is linked to the User who decides it, and `Approval.isBy(user)` lets the service refuse a person who already filled another stage (FR-4.4). A Reservation exists only after the final approval (BR-5); a Transfer holds up to 5 documents (FR-4.8) and is linked to the User who receives it.

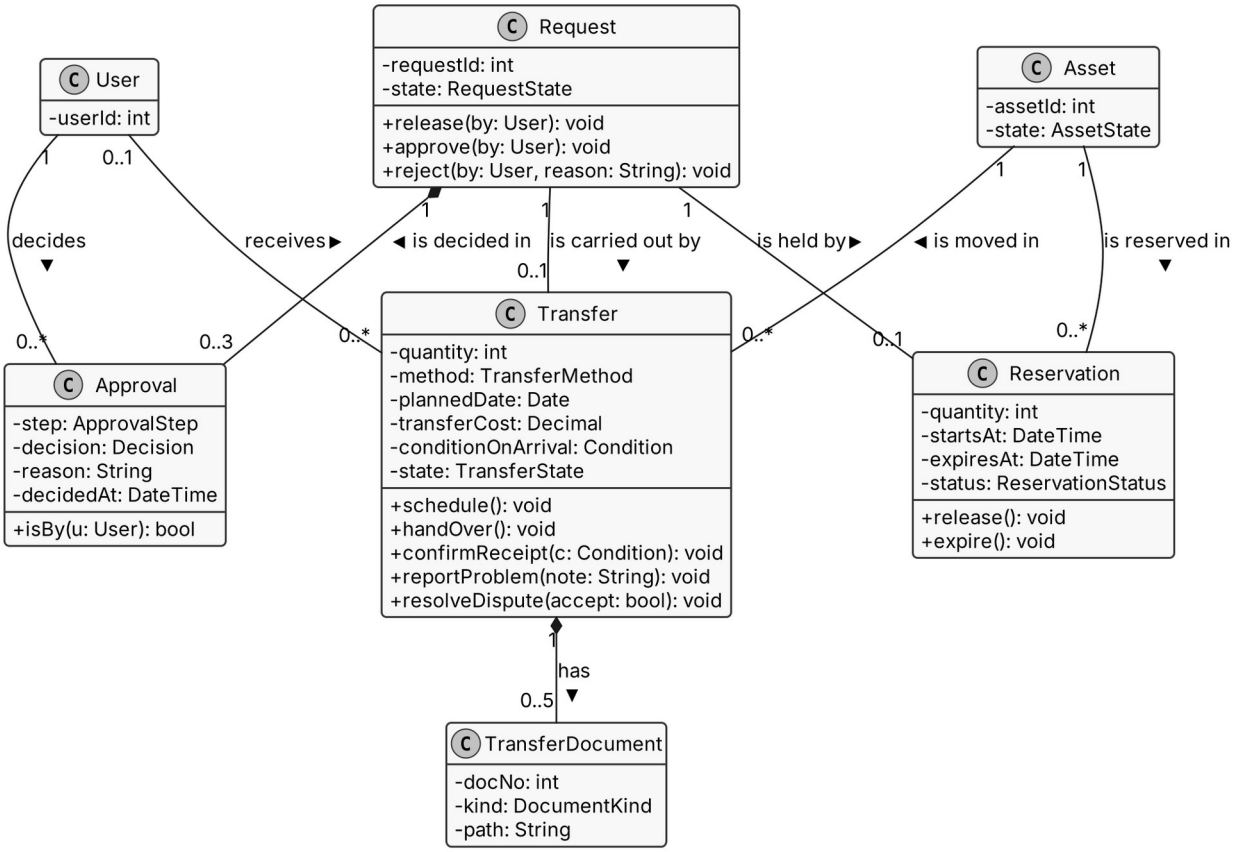


Figure 9-2: Class diagram, approvals and transfers

Inspections, MaintenanceRecords and LifecycleEvents are composed into their owners and have no update operation, because recorded results and life-cycle events cannot be edited (FR-5.18, SEC-12); a mistake is fixed by a new event that corrects the old one. An EndOfLifeProposal is linked to three different Users (proposes, reviews safety of, approves), which is how BR-8 is checked, and to a Vendor only when a vendor handles it; `Vendor.isApproved(date)` blocks expired vendors (FR-5.25).

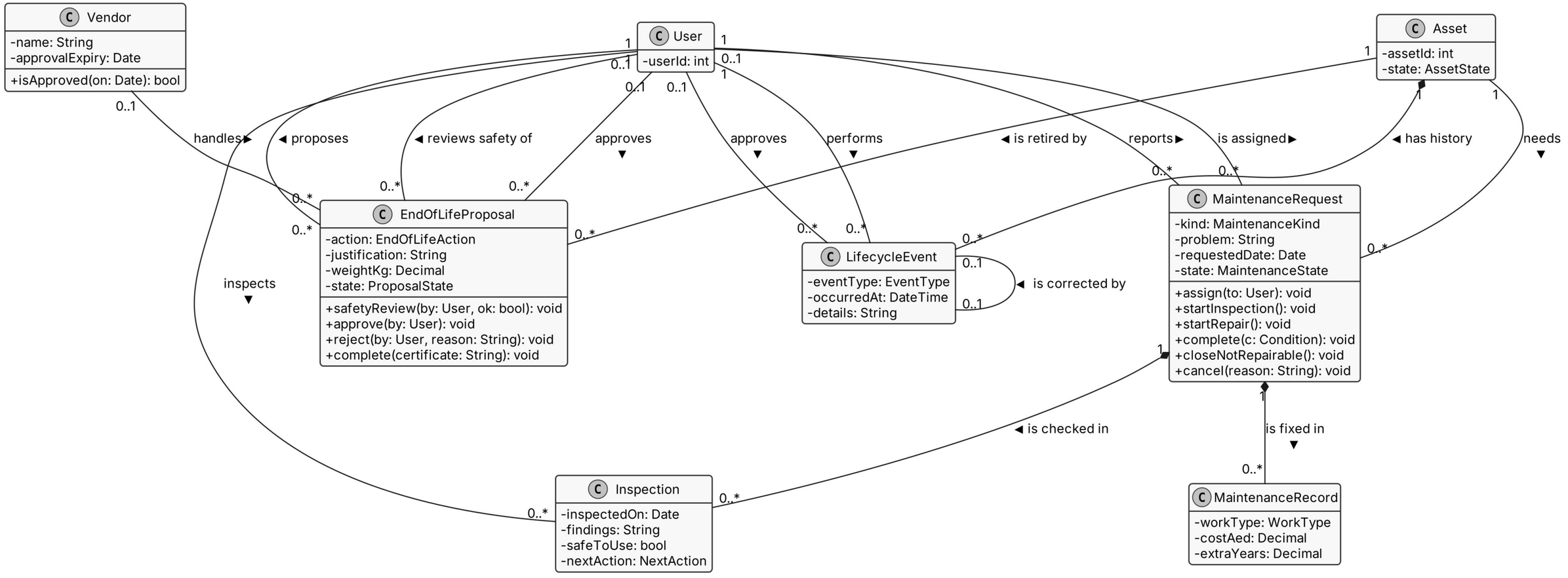


Figure 9-3: Class diagram, maintenance and end of life

9.3 AI Services

Prepared by: Ahmed AlAbdouli

AIGateway is the only class that calls the AI service interfaces and the AI Provider. Every call creates an AIOutput, which always carries a confidence, a level from ConfidencePolicy, an Explanation and the ModelVersion that produced it. `recordDecision` stores the human decision and reason (FR-7.A4). The interfaces are the contracts in Section 16.2.

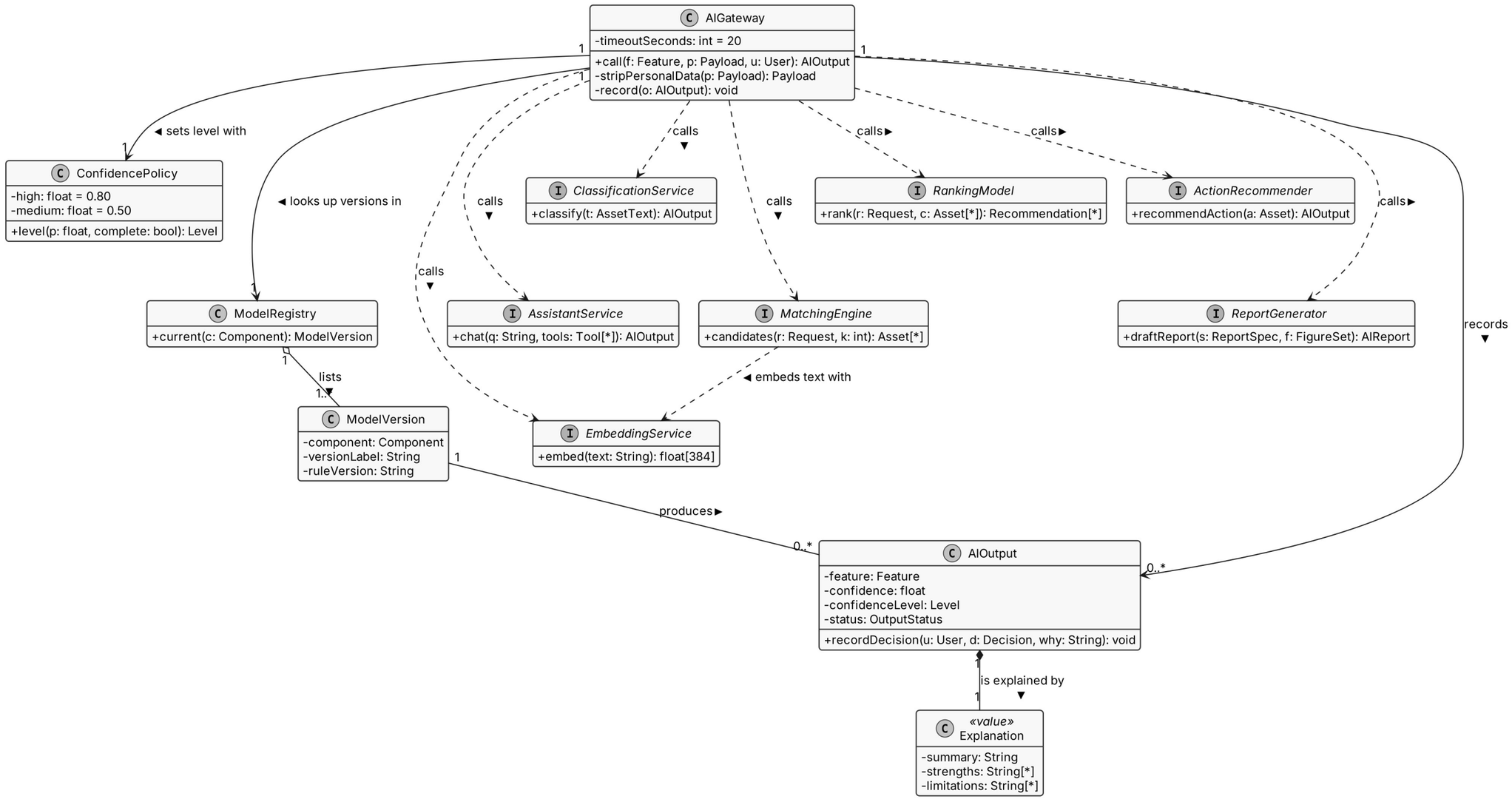


Figure 9-4: Class diagram, AI services

Figure 9-5 shows what the AI services leave behind. One ranking AIOutput ranks up to 20 Recommendations, each linked to the Request, the Asset and the User who reviewed it; an AIReport is drafted by one AIOutput and approved by a User; a HumanReview links the person who asked for it and the Administrator who decides (SEC-10). EvaluationRuns and MonitoringRecords belong to one ModelVersion (QUA-9, QUA-12).

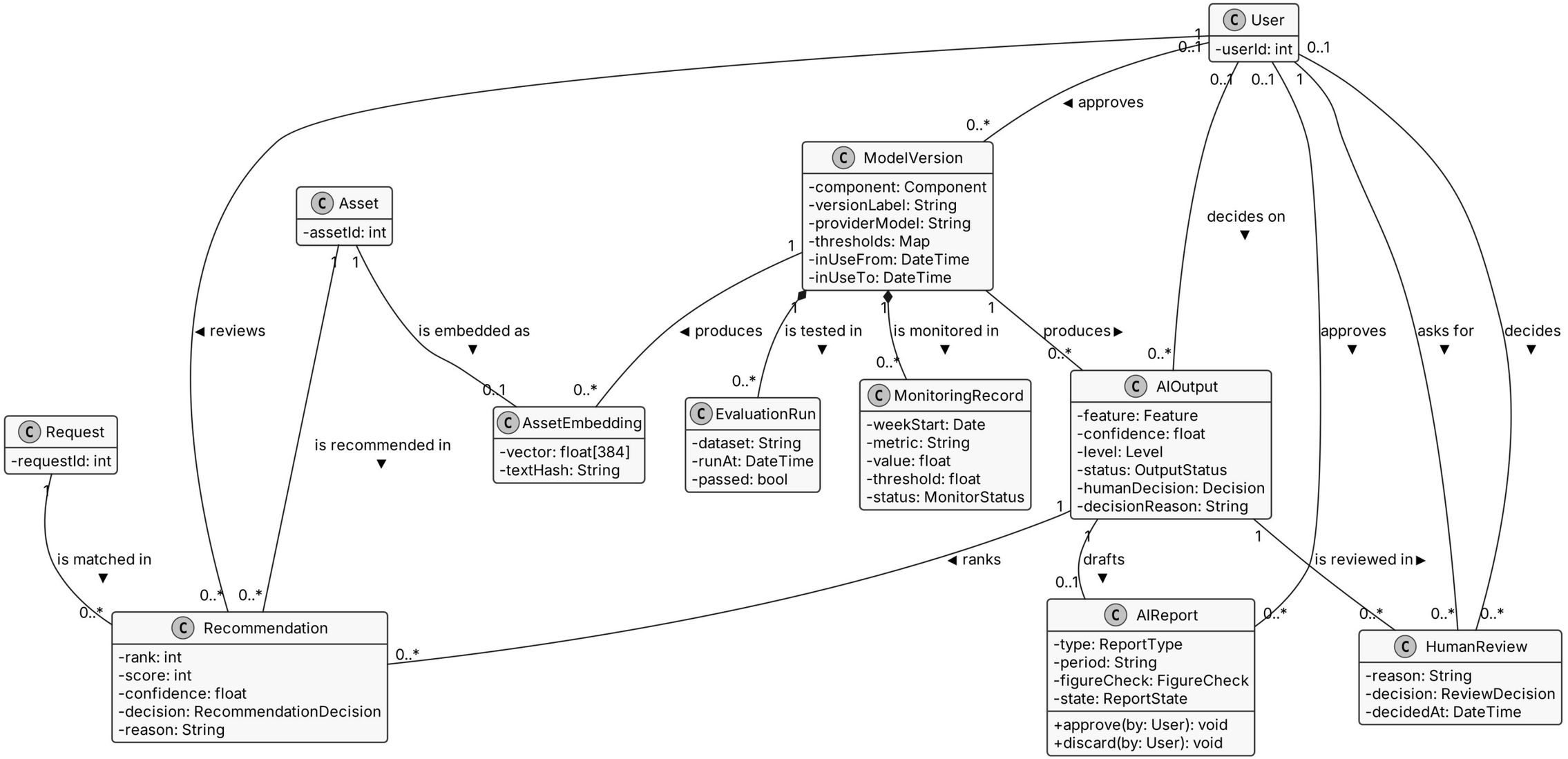


Figure 9-5: Class diagram, AI records

9.4 Notifications, Audit and Sustainability

Prepared by: Rashid Alzaabi

A Notification is received by one User and points to the record it is about (FR-7.N1). An AuditEntry is performed by a User, or by the system when no user is linked, and records the AI decision when one was involved; `verifyChain` checks the hash chain of Section 6.3 (FR-7.A6). A Category has versioned CategoryValues, and each SustainabilityImpact uses the value versions it was calculated with (FR-13.5), for the whole campus or for one Department.

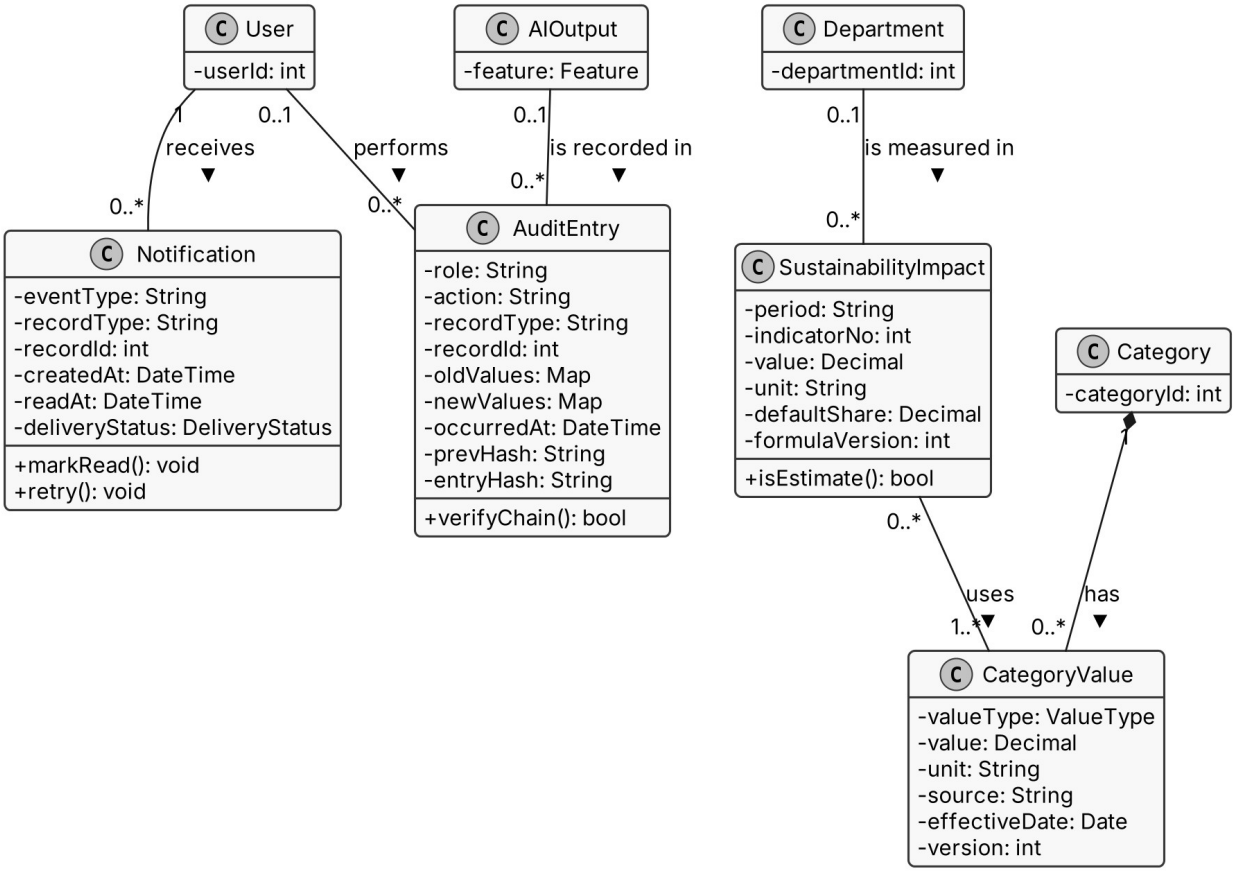


Figure 9-6: Class diagram, notifications, audit and sustainability

10. Sequence Diagrams

Prepared by: **Ahmed AlAbdouli**

The sequence diagrams cover all 13 use cases (Table 8-1). Participants are the actors (Phase 3 role names), the screen of Section 7, the modules and AI classes of the component diagram (Figure 13-1), the AI Provider and the MySQL database. Messages to modules are the operations of Table 16-1, and messages to AI services are the operations of Table 16-2. Solid arrows with filled heads are calls, dashed arrows are replies and open-headed arrows are messages to Notifications. "alt" and "opt" frames show the alternative paths, including every fallback. A reject, decline or refusal is always the last branch of its path: after it the diagram shows only the record, the notification and the reply.

Table 10-1: Sequence diagrams

Section	Interaction	Use cases	Figure
10.1	Release and approve a request; second approval	UC-4	10-1, 10-2
10.2	Transfer an asset, including a dispute	UC-5	10-3
10.3	Register, classify and publish an asset	UC-1, UC-10	10-4
10.4	Review AI matches	UC-9	10-5
10.5	Generate an AI report	UC-12	10-6
10.6	Ask the assistant and submit a request	UC-3, UC-11	10-7
10.7	Search and request a listing	UC-2, UC-3	10-8
10.8	Register and activate an account	UC-13, UC-8	10-9
10.9	Record maintenance or repair	UC-6	10-10
10.10	Propose and complete an end-of-life action	UC-7	10-11

10.1 Release and Approve a Request (UC-4)

Prepared by: **Mohammad Alsuwaidi**

The release and the approval are separate decisions by different people (FR-4.4, BR-4), and each may end the request. A decline or rejection records the reason, closes the request and notifies the requester; nothing else happens after it. When the value is above the limit or unknown, the Procurement Team gives a second approval (Figure 10-2) before any units are reserved (FR-4.3). The final approval locks the asset row, rechecks the units and reserves them in one transaction, so the last unit can never go to two requests (FR-4.5, QUA-4).

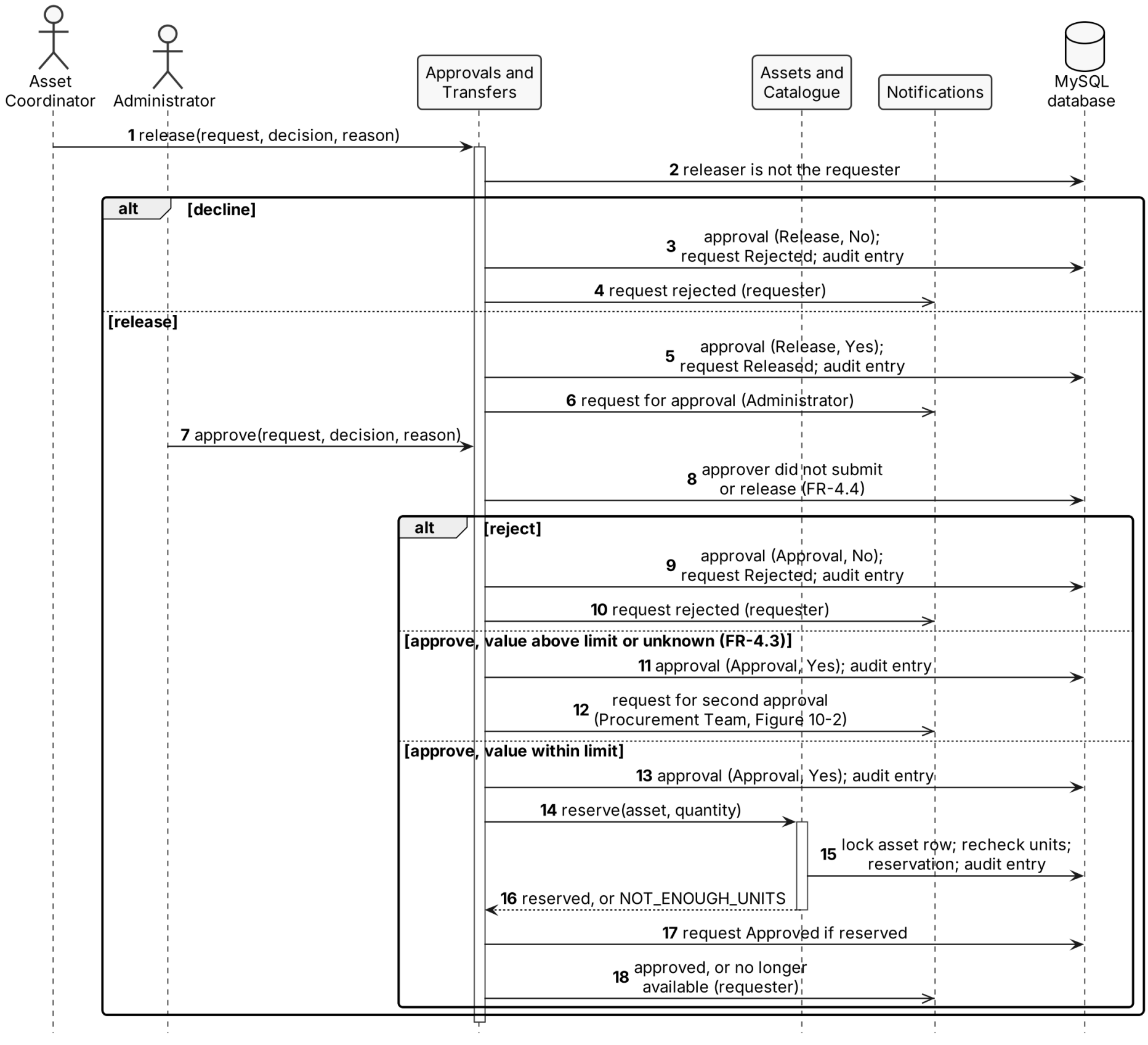


Figure 10-1: Release and approve a request

The second approval is given by a third person, who did not submit, release or approve the request (BR-4). A rejection closes the request; an approval reserves the units exactly as in Figure 10-1.

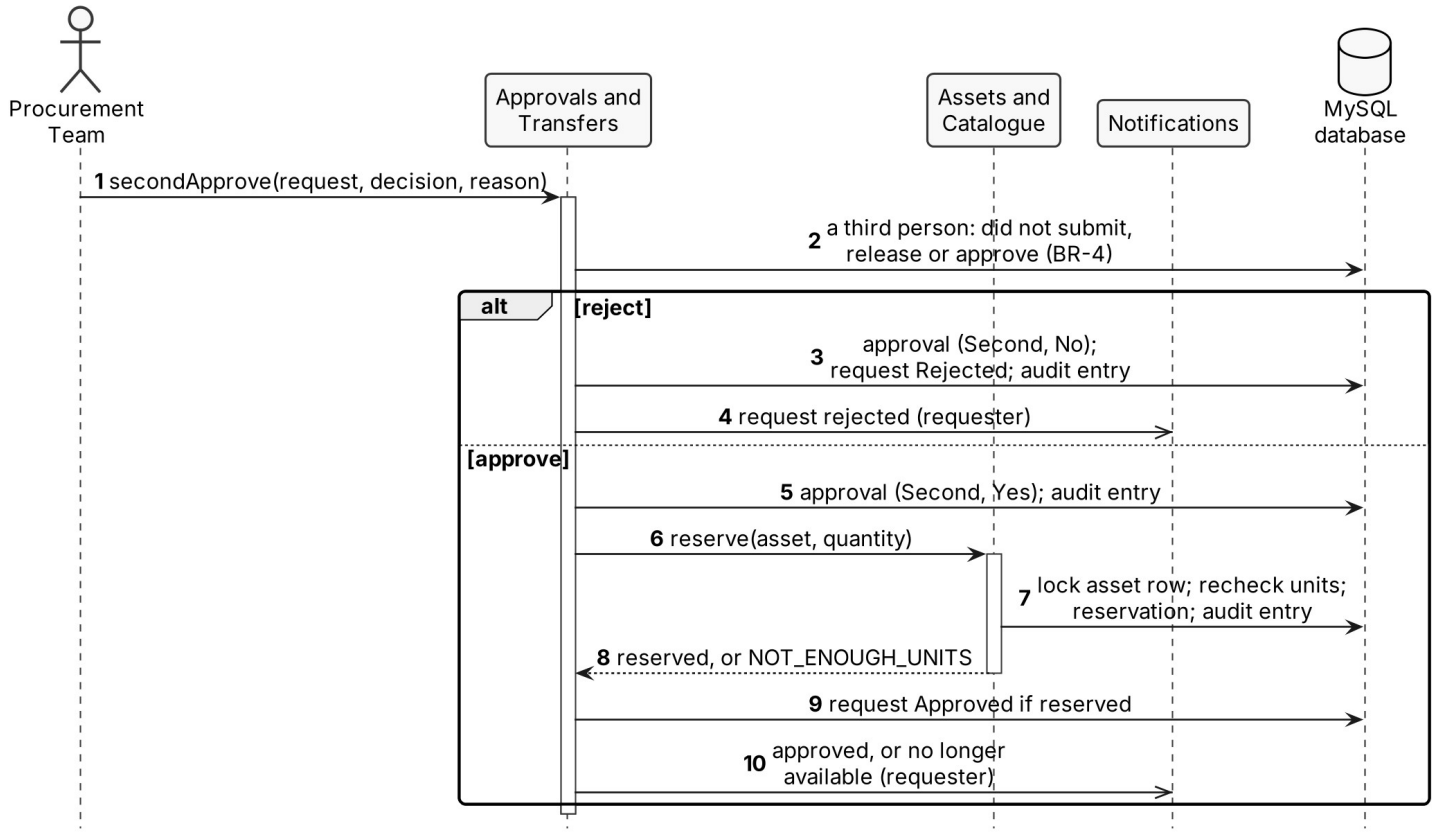


Figure 10-2: Second approval

10.2 Transfer an Asset (UC-5)

Prepared by: Mohammad Alsuwaidi

Handover is blocked for unsafe or uninspected risky assets (FR-4.16, SAF-9), and handling notes are shown to the people involved (SAF-7). Custody, owning department and location change only when receipt is confirmed or the Administrator accepts a disputed transfer (FR-4.11, FR-4.12); a returned asset becomes available again and the request is cancelled.

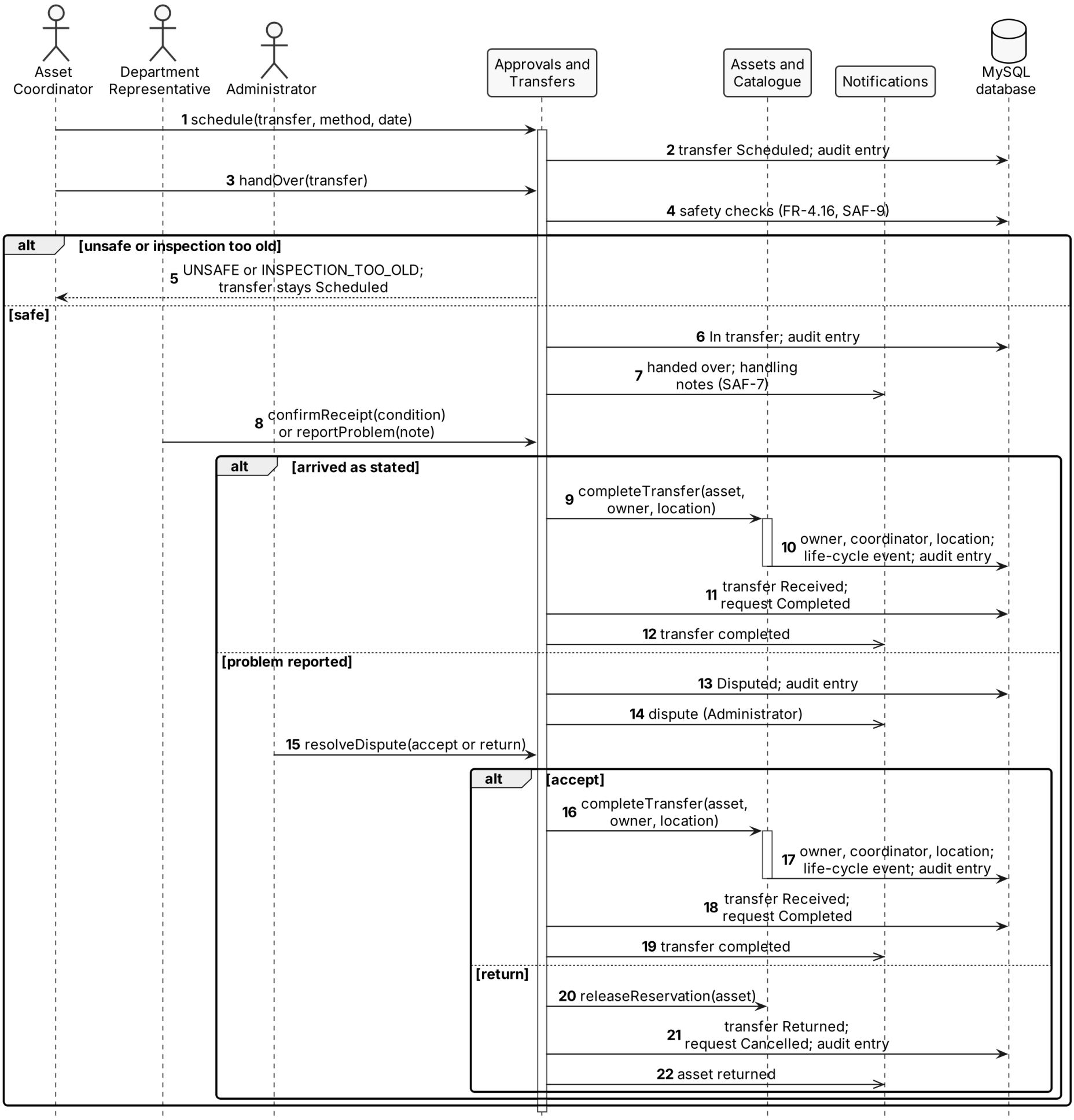


Figure 10-3: Transfer an asset

10.3 Register, Classify and Publish an Asset (UC-1, UC-10)

Prepared by: Ahmed AlAbdouli

The draft is saved before the AI is called, so a failed or slow AI call never loses the entries (FR-9.8). AIGateway removes personal data and is the only part that sends the prompt to the AI Provider. The suggestion is shown without being accepted; the asset changes only when the Asset Coordinator confirms each field (FR-9.6) and publishes it, and publishing is refused while a required field or a safety check is missing.

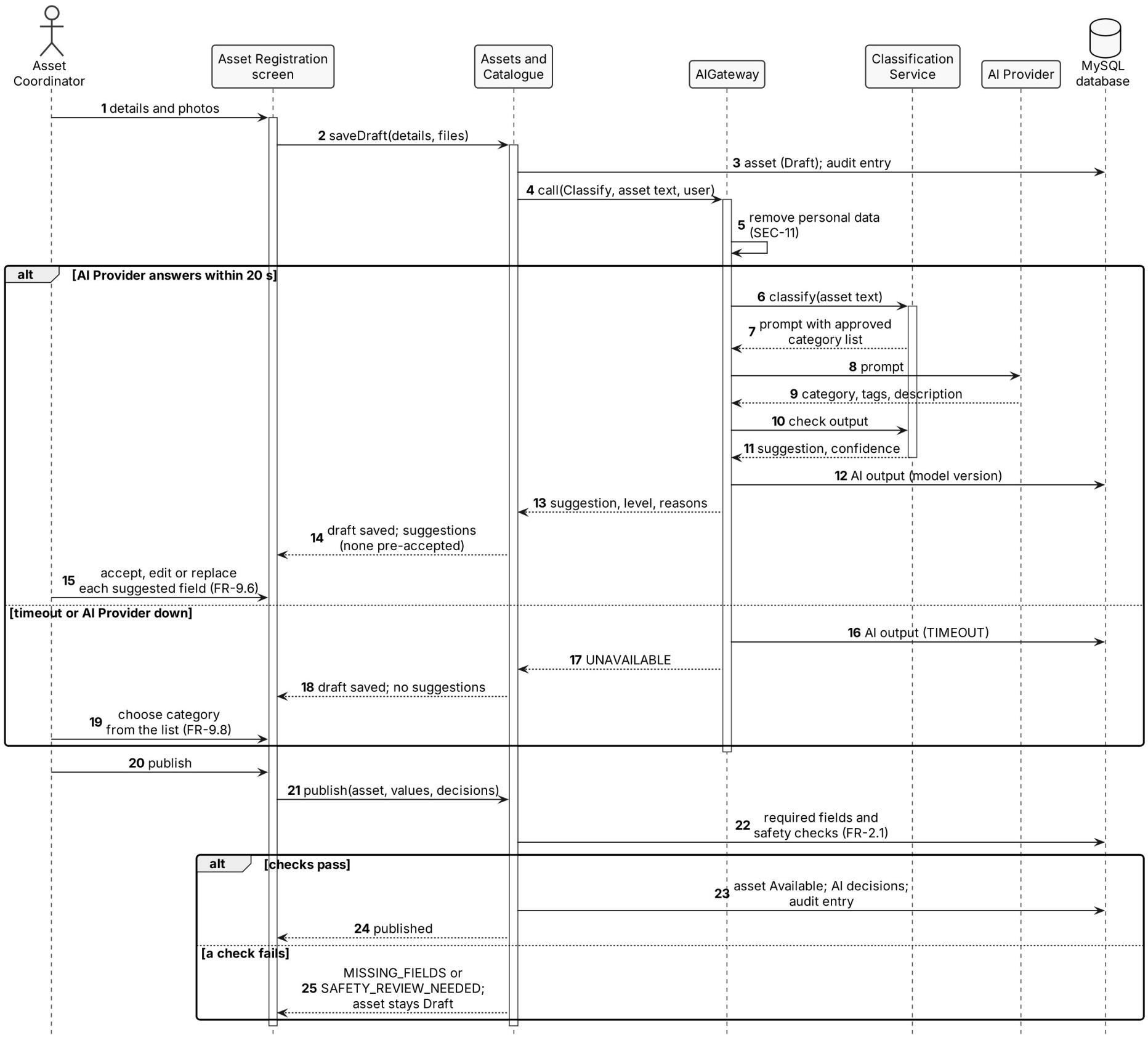


Figure 10-4: Register, classify and publish an asset

10.4 Review AI Matches (UC-9)

Prepared by: Ahmed AlAbdouli

AIGateway gets the 20 most similar eligible assets from MatchingEngine, RankingModel scores them, and each recommendation is stored before the list is shown. If the AI is down or too slow, keyword and rule matches are shown labeled "not ranked" (AI-6). Selecting a match rechecks the asset before the request goes to release (FR-8.8); rejecting all matches closes the request with the reason, and nothing runs after it.

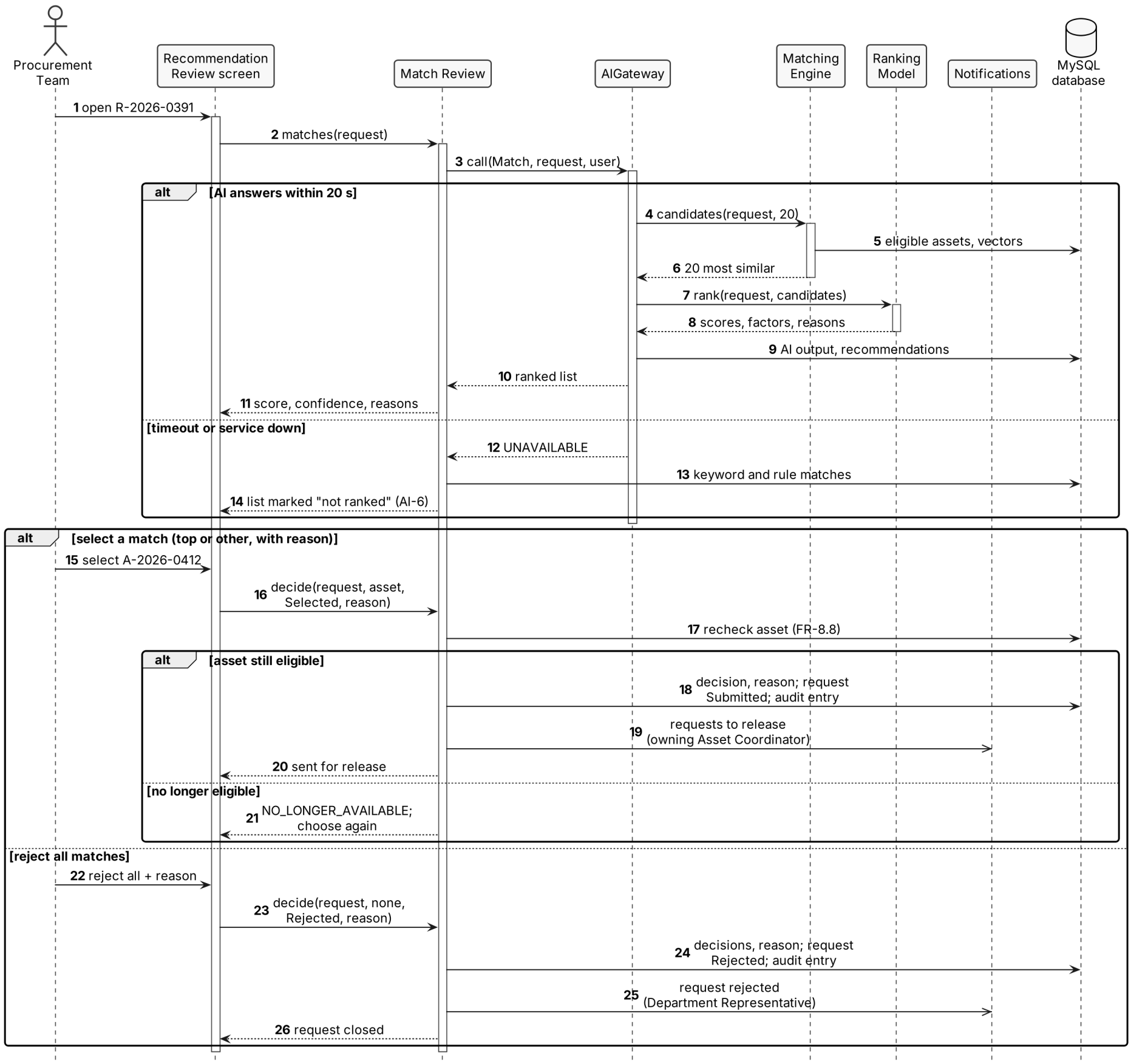


Figure 10-5: Review AI matches

10.5 Generate an AI Report (UC-12)

Prepared by: Rashid Alzaabi

Figures are calculated with fixed formulas before the AI is called; the language model only writes the text, and ReportGenerator checks every number against the standard figures (FR-12.4). The report stays a draft until an authorized reviewer approves or discards it (FR-12.9).

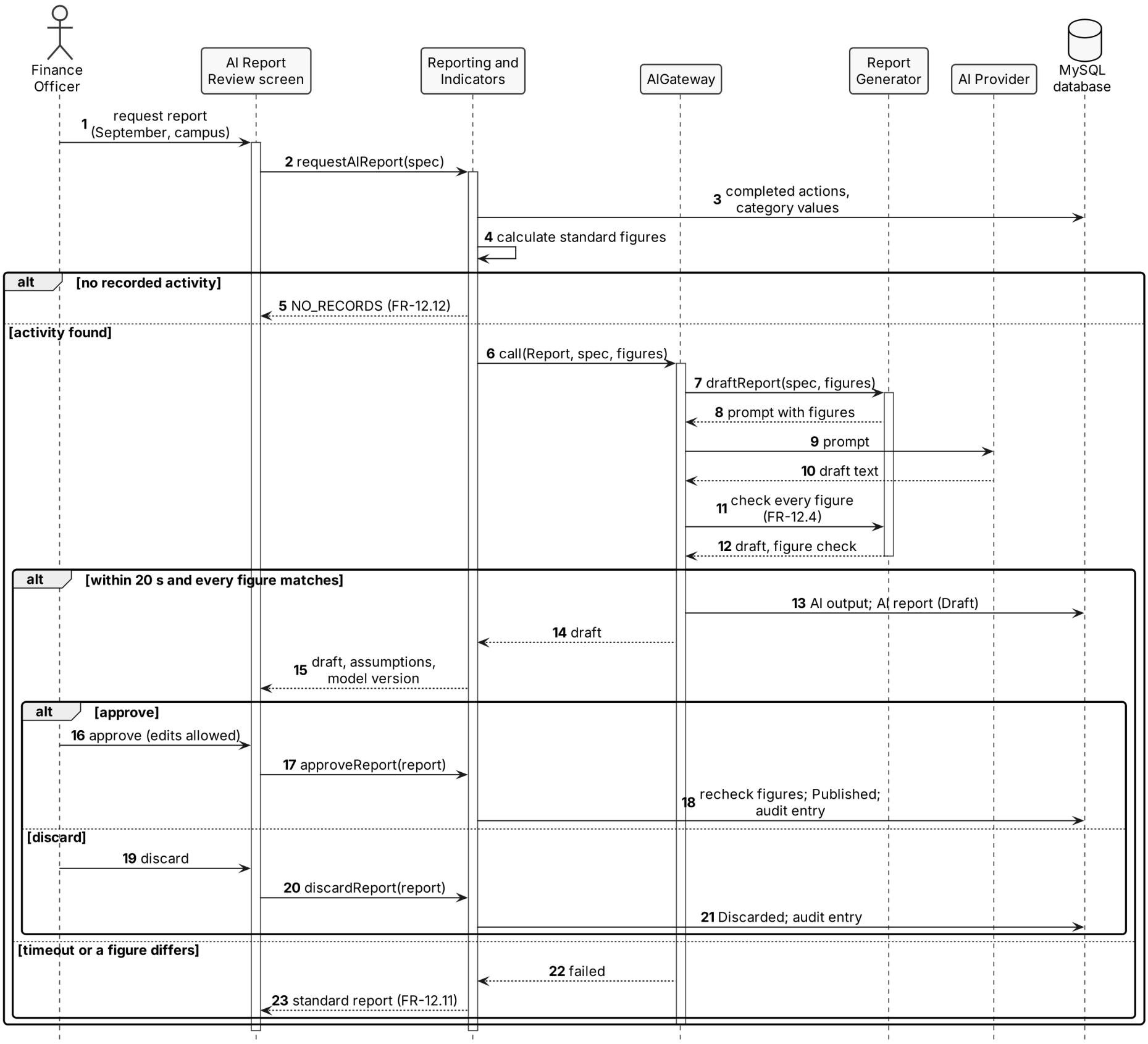


Figure 10-6: Generate an AI report

10.6 Ask the Assistant and Submit a Request (UC-3, UC-11)

Prepared by: **Ahmed AlAbdouli**

The question goes from the screen to AIGateway, the only part that talks to the AI Provider. The assistant reads only records the user may see, through read-only queries filtered by the user's permissions, and the results are passed to the model as data (AI-9). The draft is submitted through the normal request operation, with the same checks as the form, only after the user confirms (FR-3.14, FR-11.3). If the assistant is unavailable, the draft is kept and opens in the request form (FR-11.9).

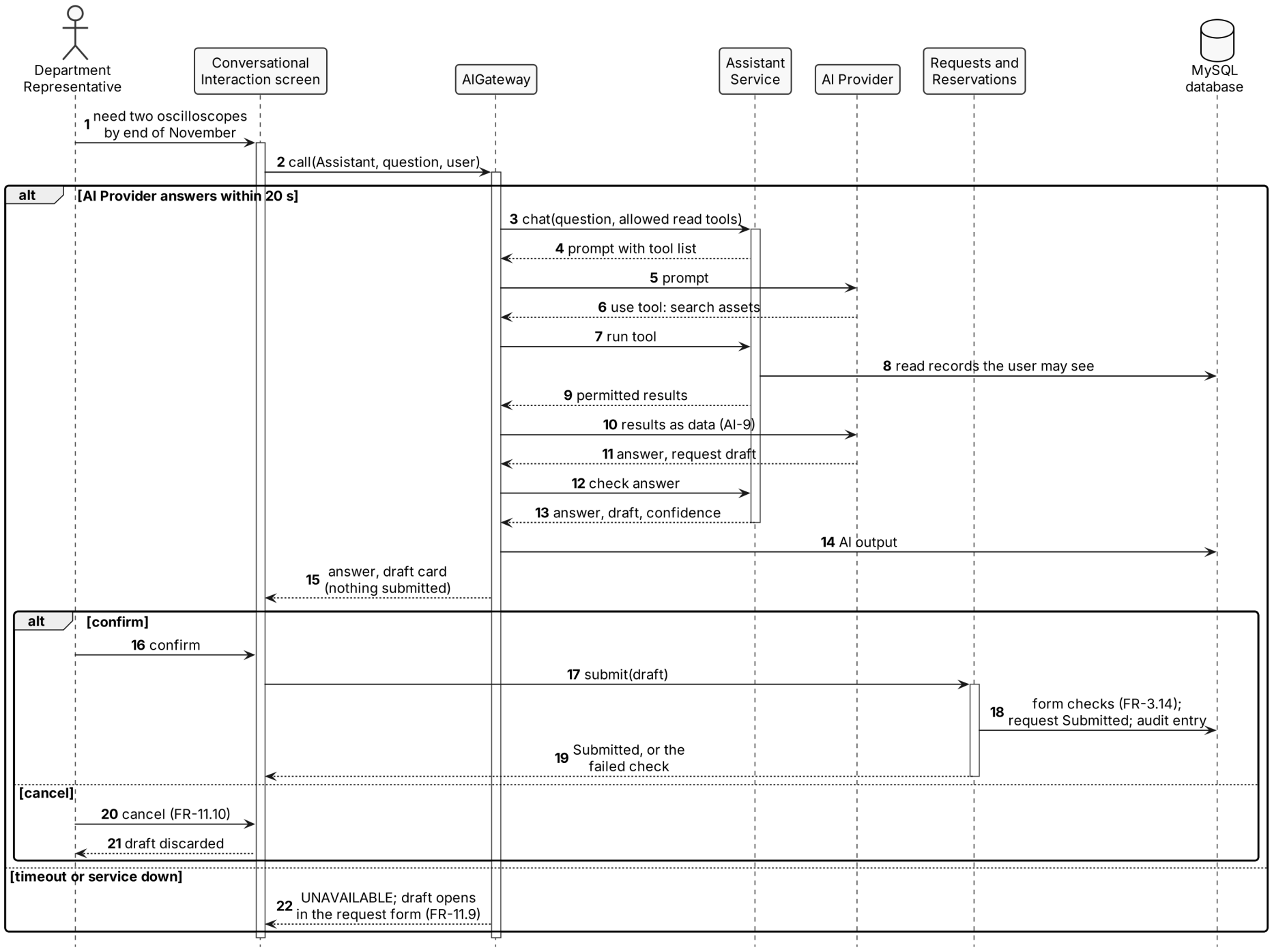


Figure 10-7: Assistant and request draft

10.7 Search and Request a Listing (UC-2, UC-3)

Prepared by: **Mohammad Alsuwaidi**

Search returns only published assets that the user may see (FR-2.2, FR-2.5). A listing that fits is requested directly; every check runs before anything is saved, and a failed check returns its error code. When nothing fits, the user describes the need and it is saved as an open need for matching (FR-3.2).

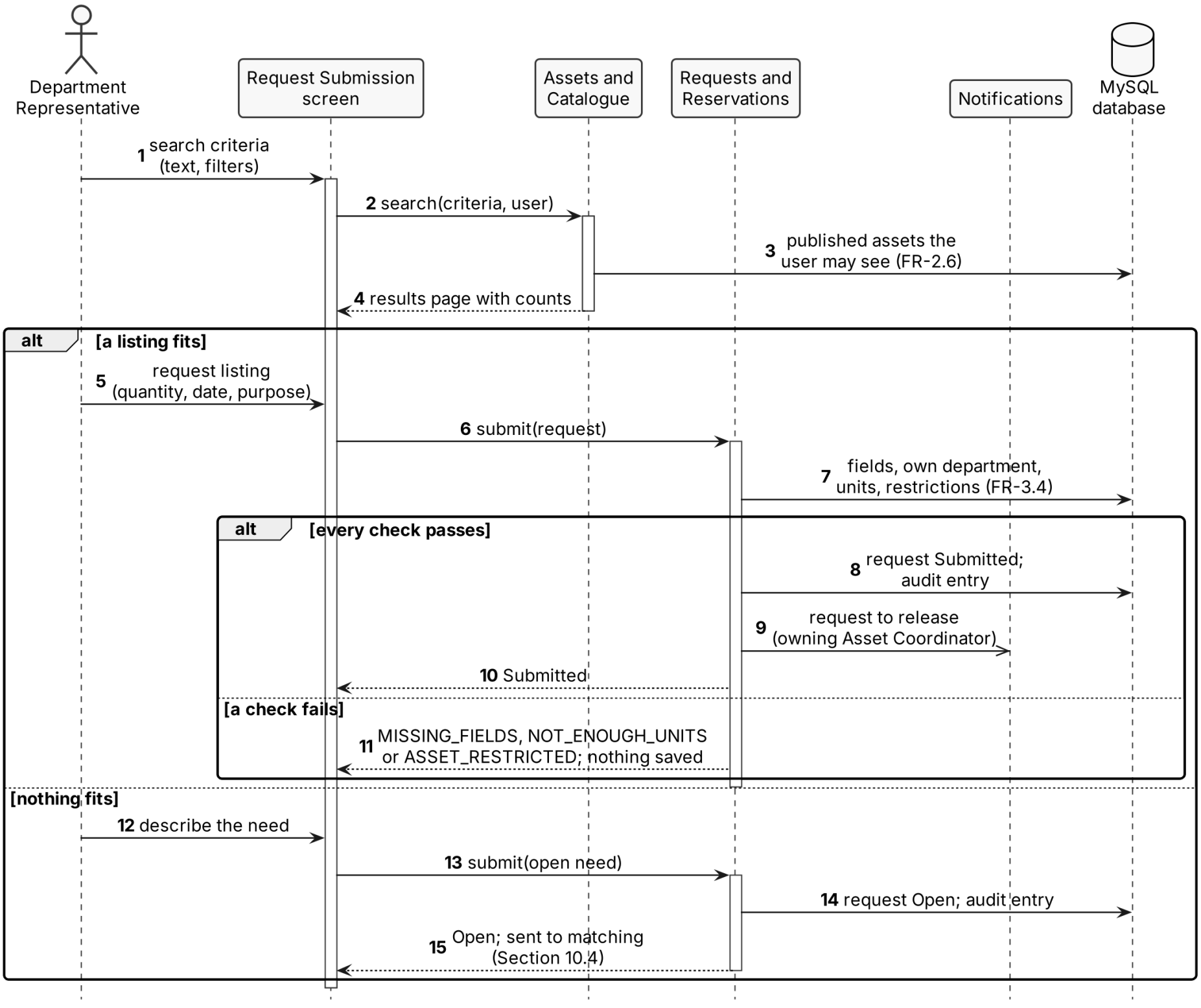


Figure 10-8: Search and request a listing

10.8 Register and Activate an Account (UC-13, UC-8)

Prepared by: Mohamed Almarzooqi

An application is saved only with consent and with a university ID and email that are not already used (FR-6.1, FR-6.2) and waits in Pending (FR-6.3). The System Administrator either refuses it with a reason, which closes the account, or activates it and assigns the role and department (FR-6.8); each outcome is notified to the applicant.

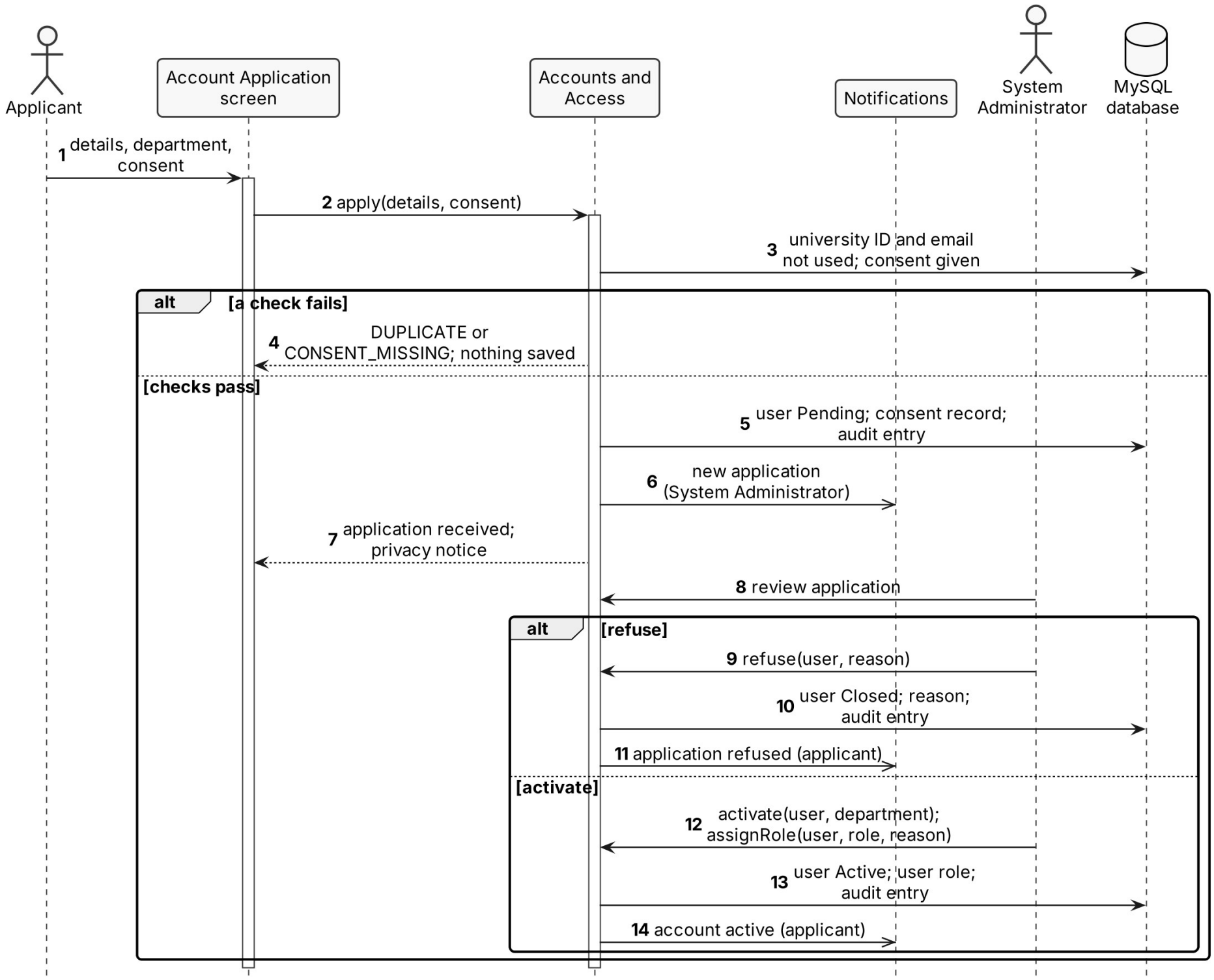


Figure 10-9: Register and activate an account

10.9 Record Maintenance or Repair (UC-6)

Prepared by: Rashid Alzaabi

A defect report goes to the Administrator, who assigns a Maintenance Staff member. While the work is open the asset is Under maintenance and hidden from search (FR-5.9). A repair restores the previous state with the new condition; a repair that brings the total cost to 50% of the replacement price, or a request closed as not repairable, flags the asset for end-of-life review and notifies the Asset Coordinator and the Sustainability Officer (FR-5.11, FR-5.12).

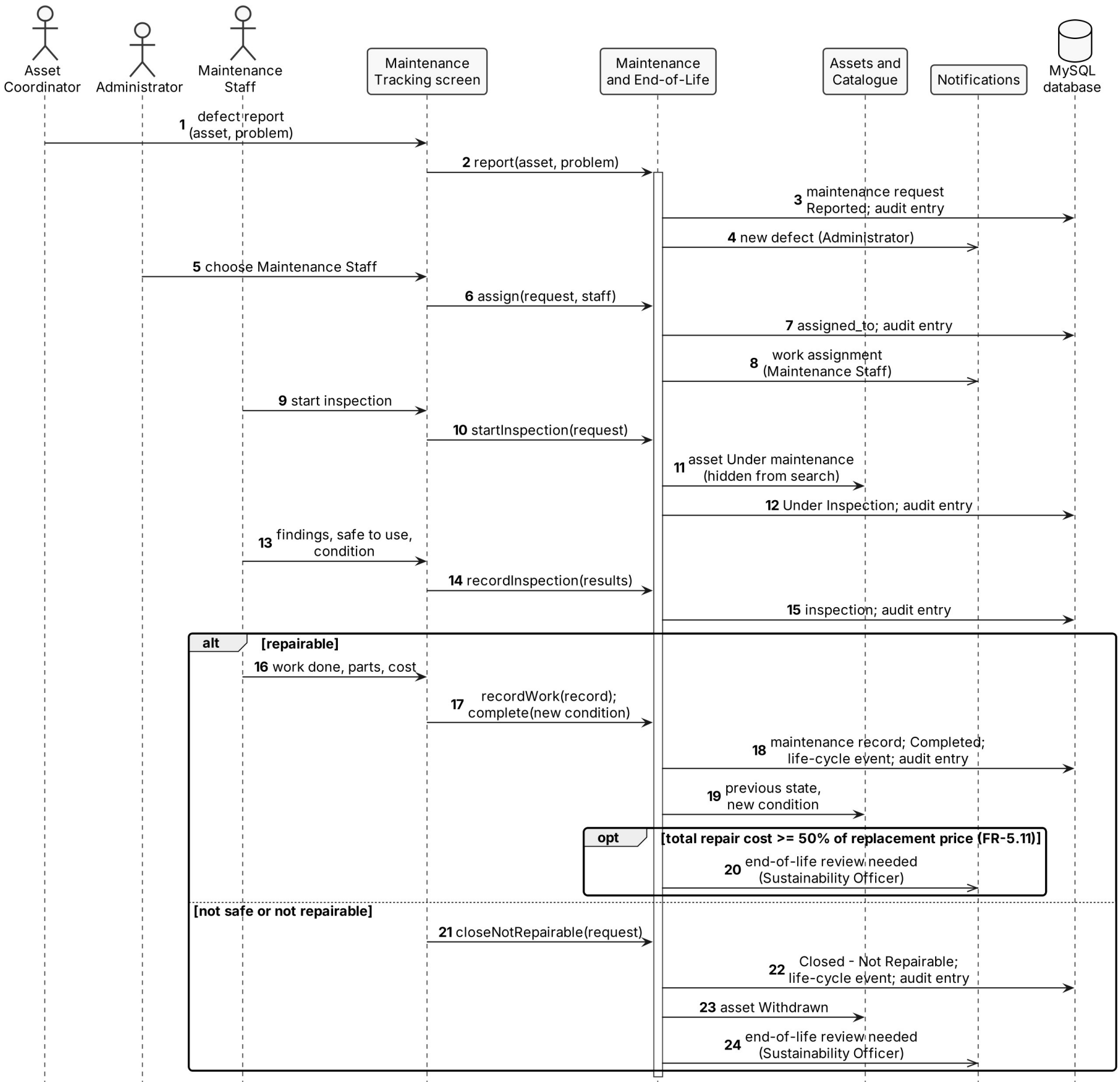


Figure 10-10: Record maintenance

10.10 Propose and Complete an End-of-Life Action (UC-7)

Prepared by: Rashid Alzaabi

The proposal is refused while the asset has a reservation, a transfer or open maintenance (FR-5.21). The AI advice is shown to the proposer but never decides. A second Sustainability Officer and then an Administrator must approve (BR-8); a rejection at either stage returns the asset to its previous state and nothing runs after it. The completion record needs an approved vendor and, for hazardous items, a certificate (FR-5.25, FR-5.30).

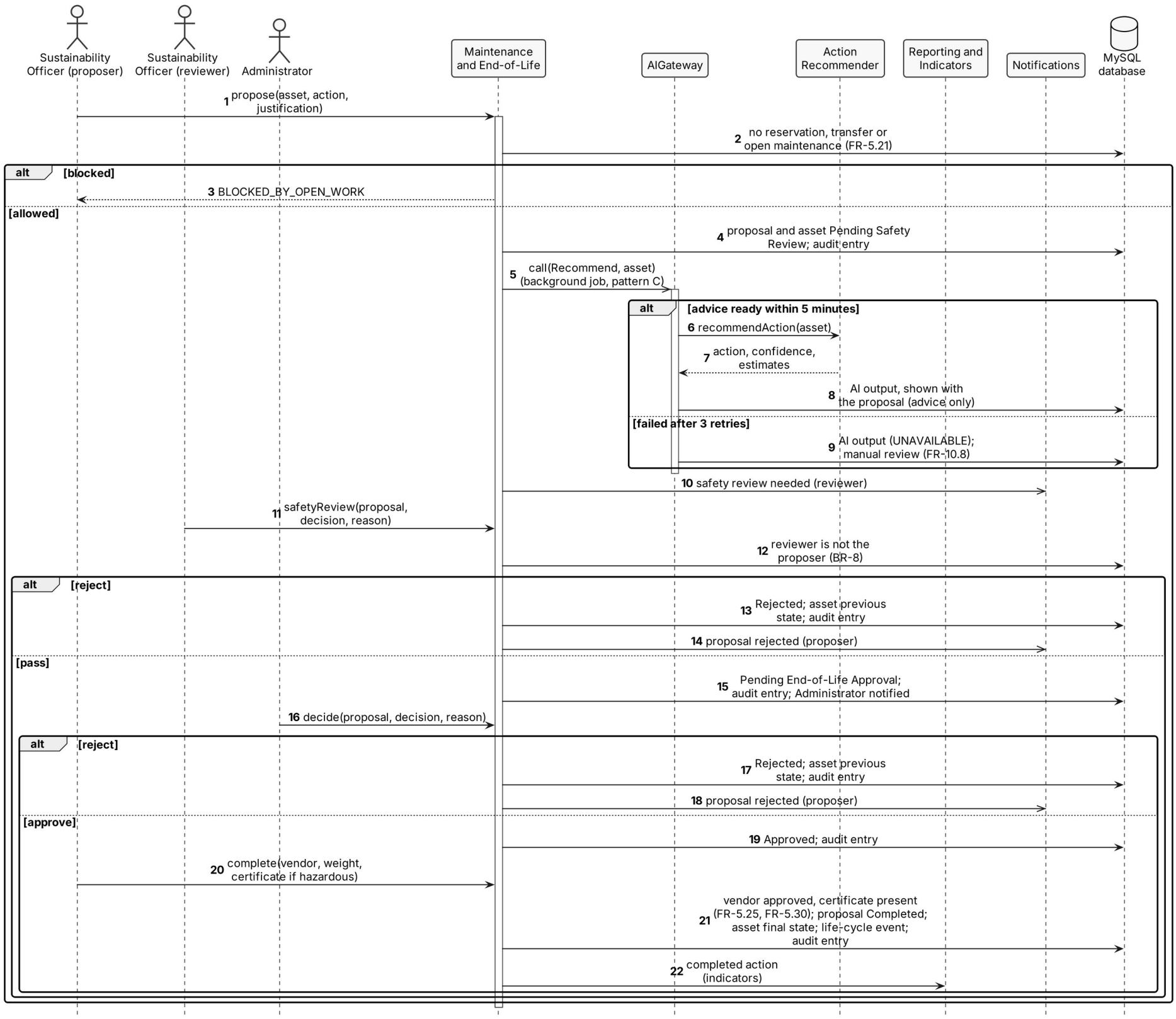


Figure 10-11: End-of-life action

11. Activity Diagrams

Prepared by: Mohammad Alsuwaidi

Each swimlane is a role or ReSource itself; a diamond is a decision by the person in that lane, and every "no" branch that ends the flow records a reason.

11.1 From Listing to Approved Request (UC-1 to UC-4, UC-9)

Prepared by: Mohammad Alsuwaidi

A listing can be requested directly; an open need goes to the Procurement Team, who choose an AI-ranked match or reject all. Both paths then need release and approval, and units are reserved only after the final recheck (FR-4.5).

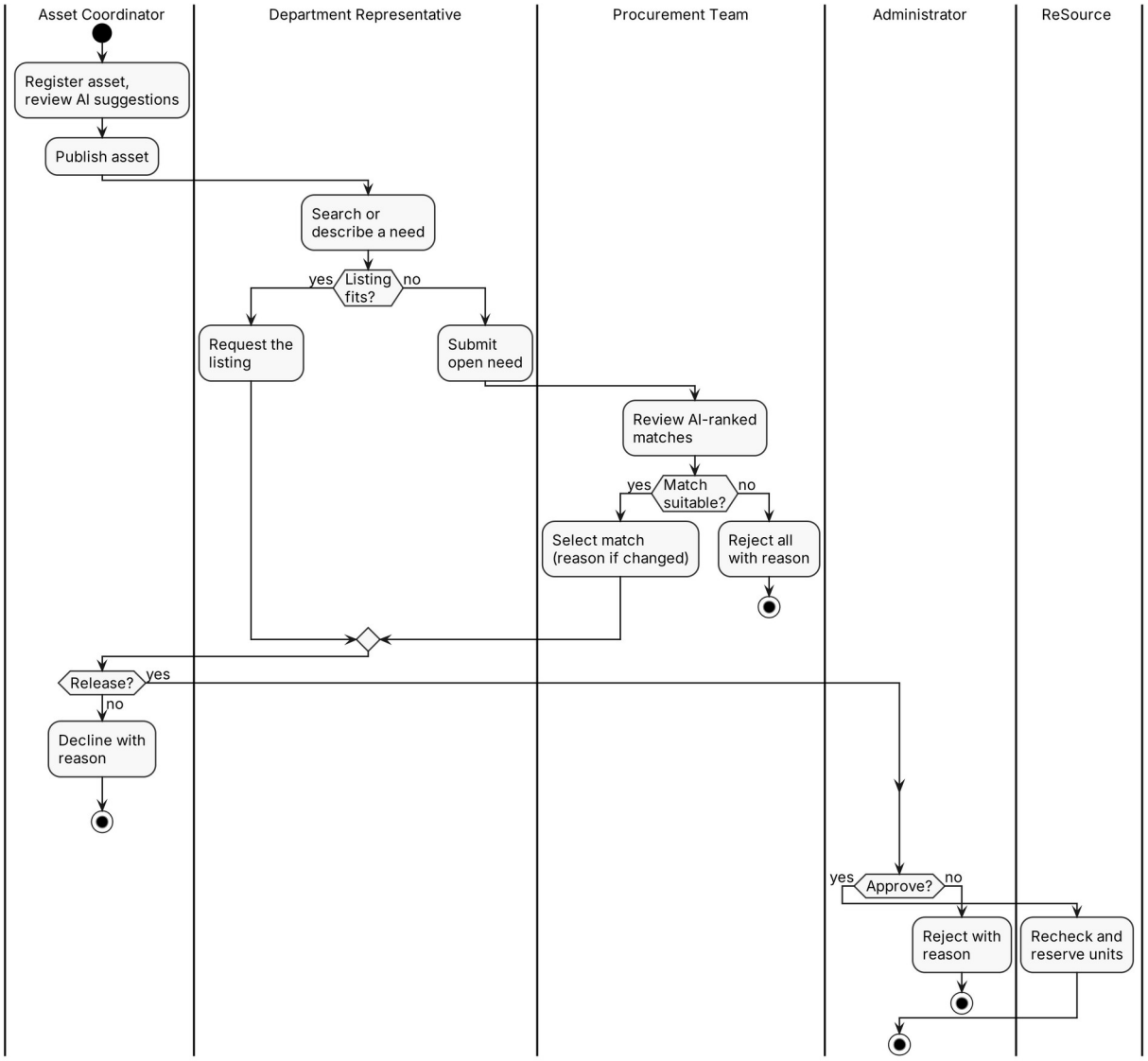


Figure 11-1: Listing to approved request

11.2 Transfer and Receipt (UC-5)

Prepared by: **Mohammad Alsuwaidi**

After approval, the owning Asset Coordinator schedules and hands over the asset. A problem on arrival goes to the Administrator, who accepts the transfer or returns the asset (FR-4.12). Custody, owner and location change only on a confirmed receipt or an accepted dispute, and only then do the indicators count the reuse (FR-4.11, FR-13.3).

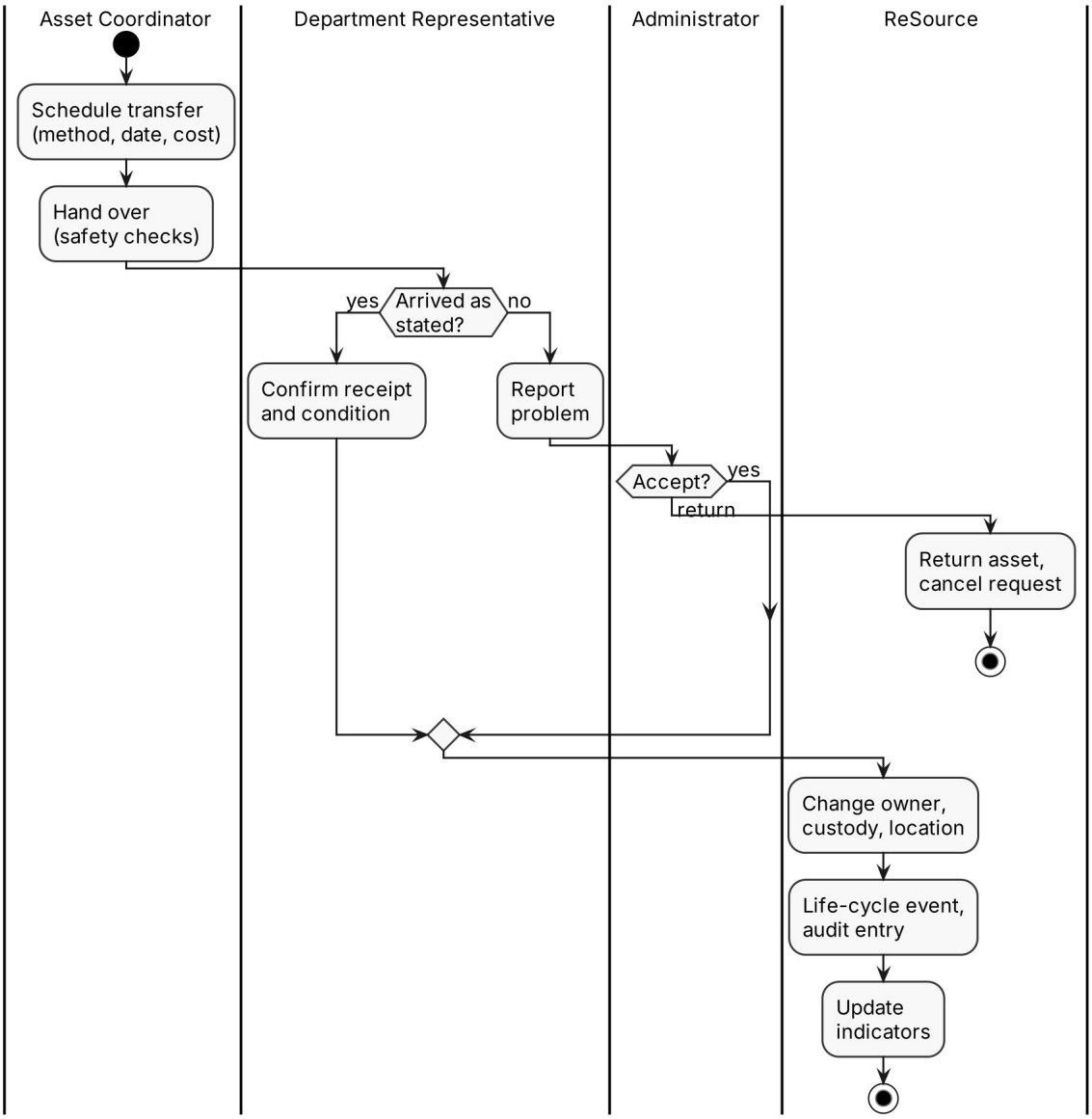


Figure 11-2: Transfer and receipt

11.3 Maintenance and End of Life (UC-6, UC-7)

Prepared by: **Rashid Alzaabi**

A repair whose cumulative cost reaches 50% of the replacement price, or a request closed as not repairable, flags the asset for end-of-life review (FR-5.11, FR-5.12). The proposal passes a safety review by a second Sustainability Officer and an approval by an Administrator before the completion record is entered (BR-8). The AI advice is shown to the proposer but never decides.

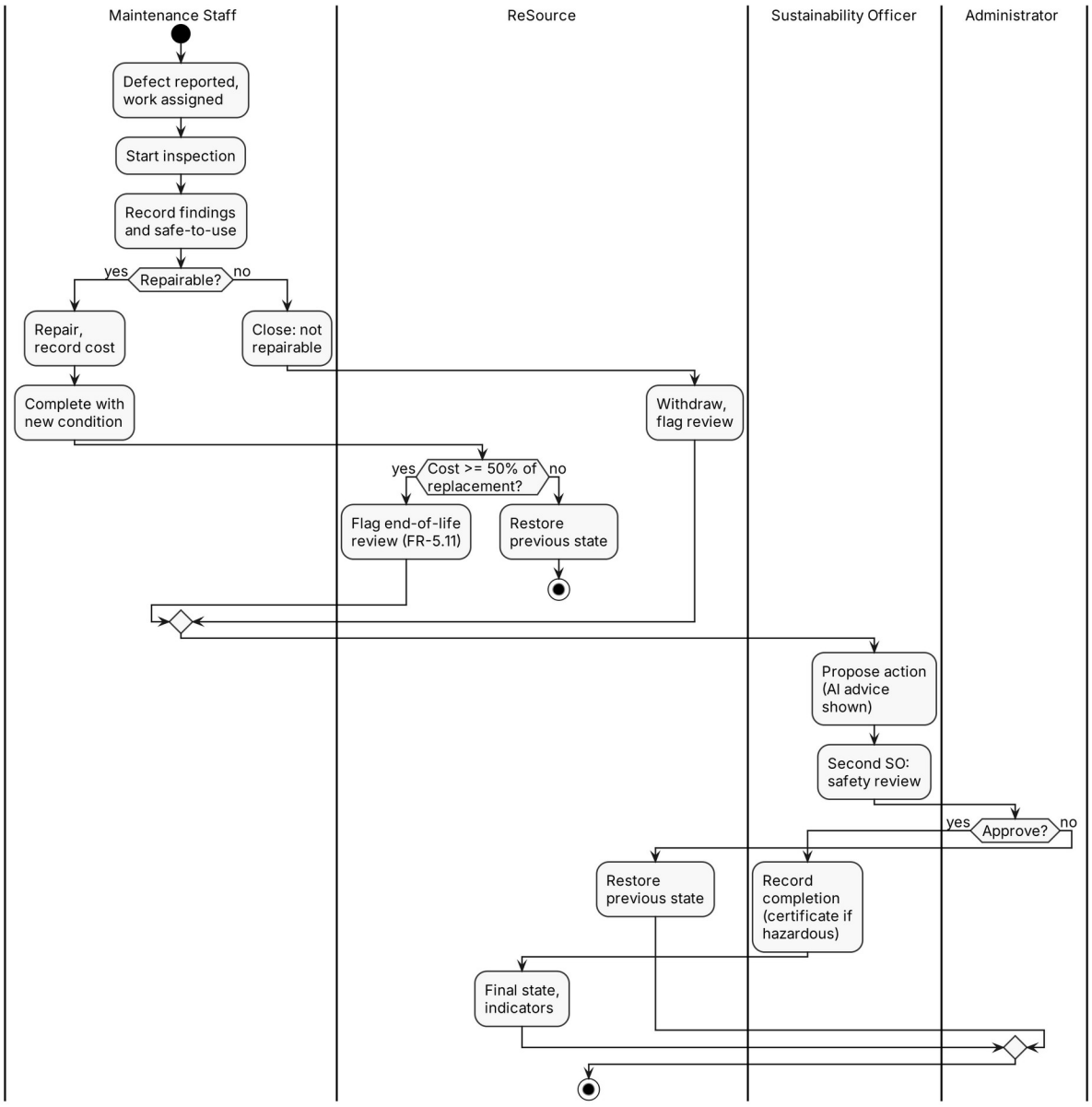


Figure 11-3: Maintenance and end of life

12. State Diagrams

Prepared by: Mohammad Alsuwaidi

The state diagrams use the state lists of Phase 3, Appendix B. Each transition is labeled "event [guard] / action (who)": the guard in brackets must be true, the action after the slash is done by the system, and the role in brackets causes the event, using the abbreviations of Table I-1 ("receiver" is the receiving Department Representative or Asset Coordinator). Approval states are drawn as the first part of the transfer diagram (Figure 12-4), because an approval exists only within one request.

12.1 Asset

Prepared by: Mohammad Alsuwaidi

Availability is never set by hand; it follows from these states (FR-1.4). Category, quantity and condition cannot change while the asset is Reserved or In transfer (FR-1.10).

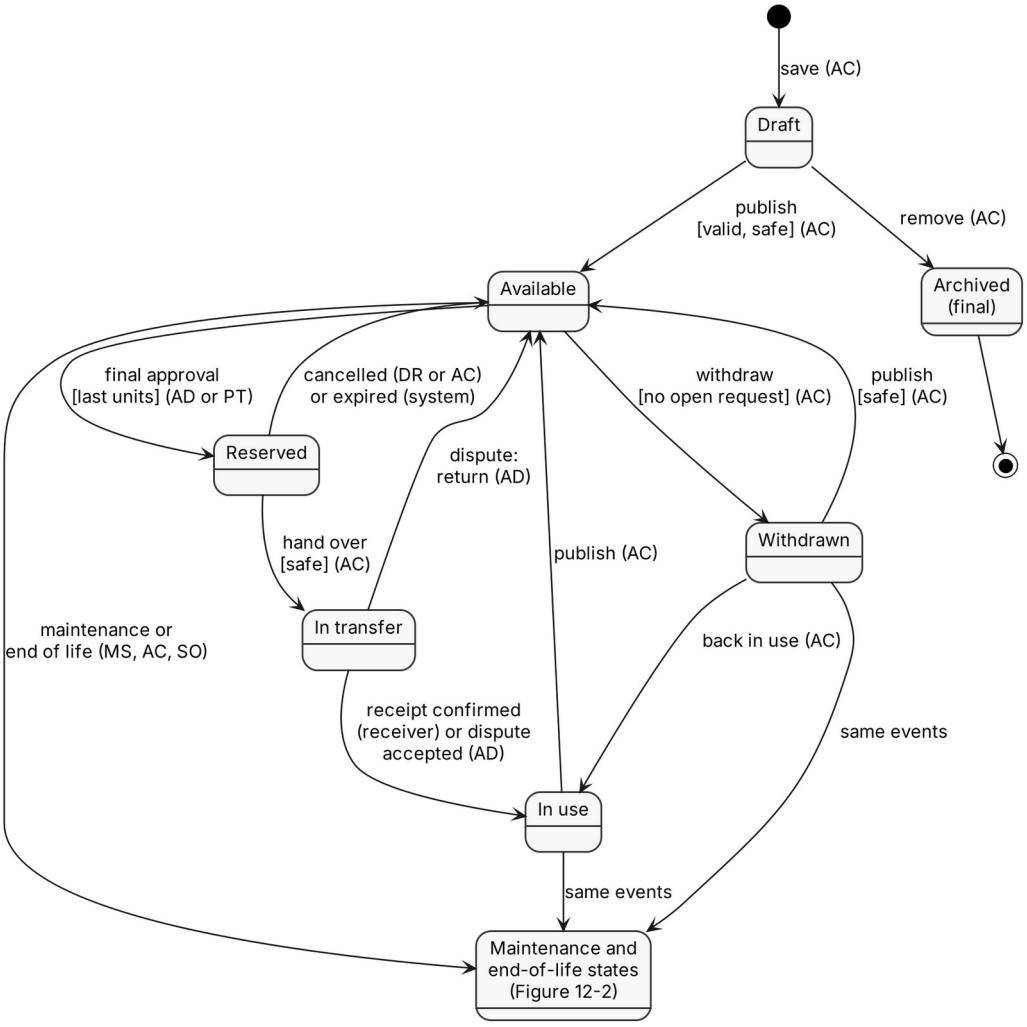


Figure 12-1: Asset states

12.2 Asset Maintenance and End of Life

Prepared by: **Rashid Alzaabi**

While an asset is Under maintenance, Pending Safety Review or Pending End-of-Life Approval it is hidden from search and matching and cannot be requested (FR-5.9, FR-5.27). A rejection returns it to the state it had before (FR-5.28).

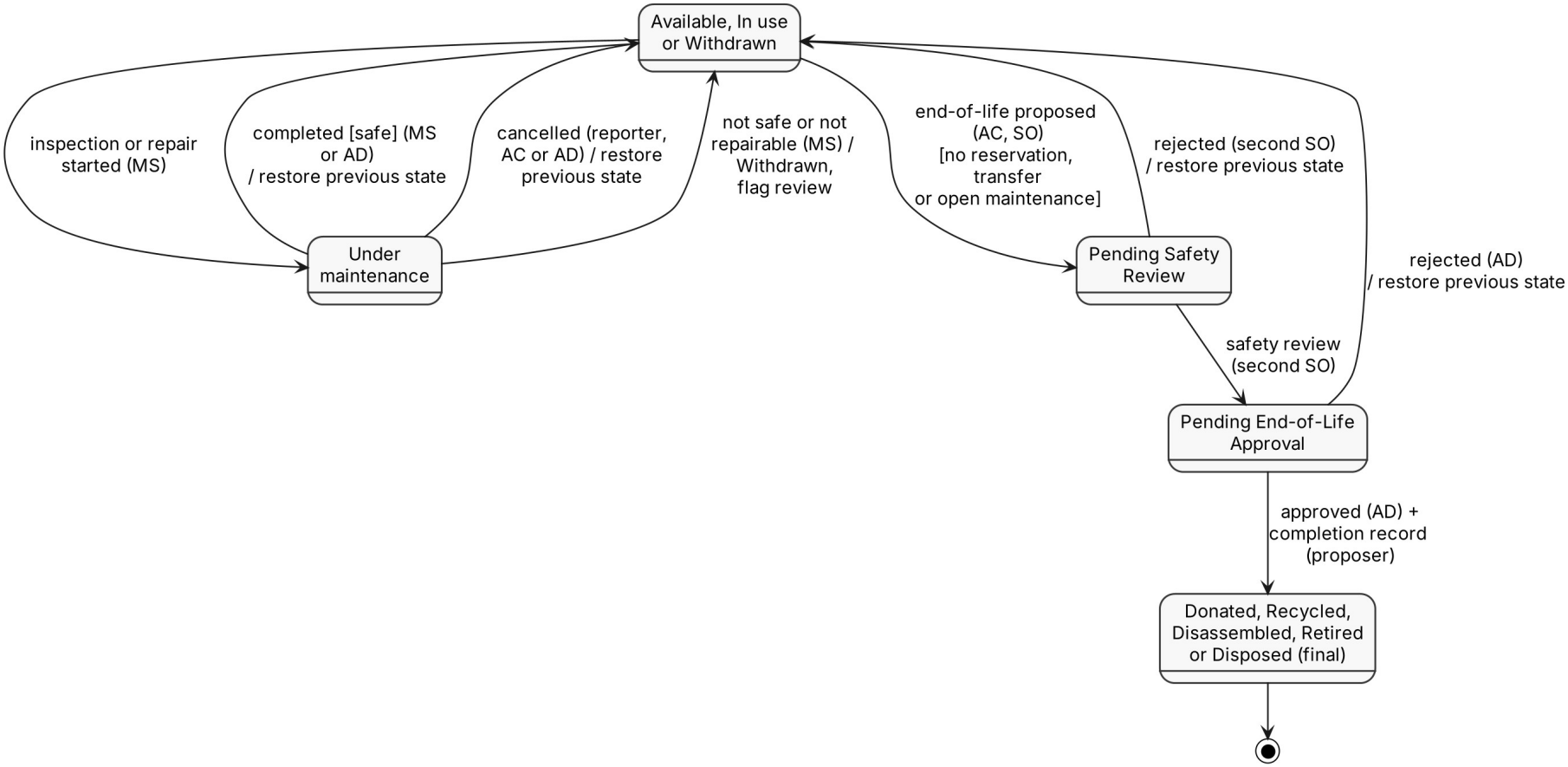


Figure 12-2: Asset maintenance and end-of-life states

12.3 Request

Prepared by: Mohammad Alsuwaidi

A request for a listing starts as Submitted; an open need starts as Open and becomes Submitted when the Procurement Team selects a match. Units are reserved on approval and released on cancellation or expiry (BR-5, FR-3.9).

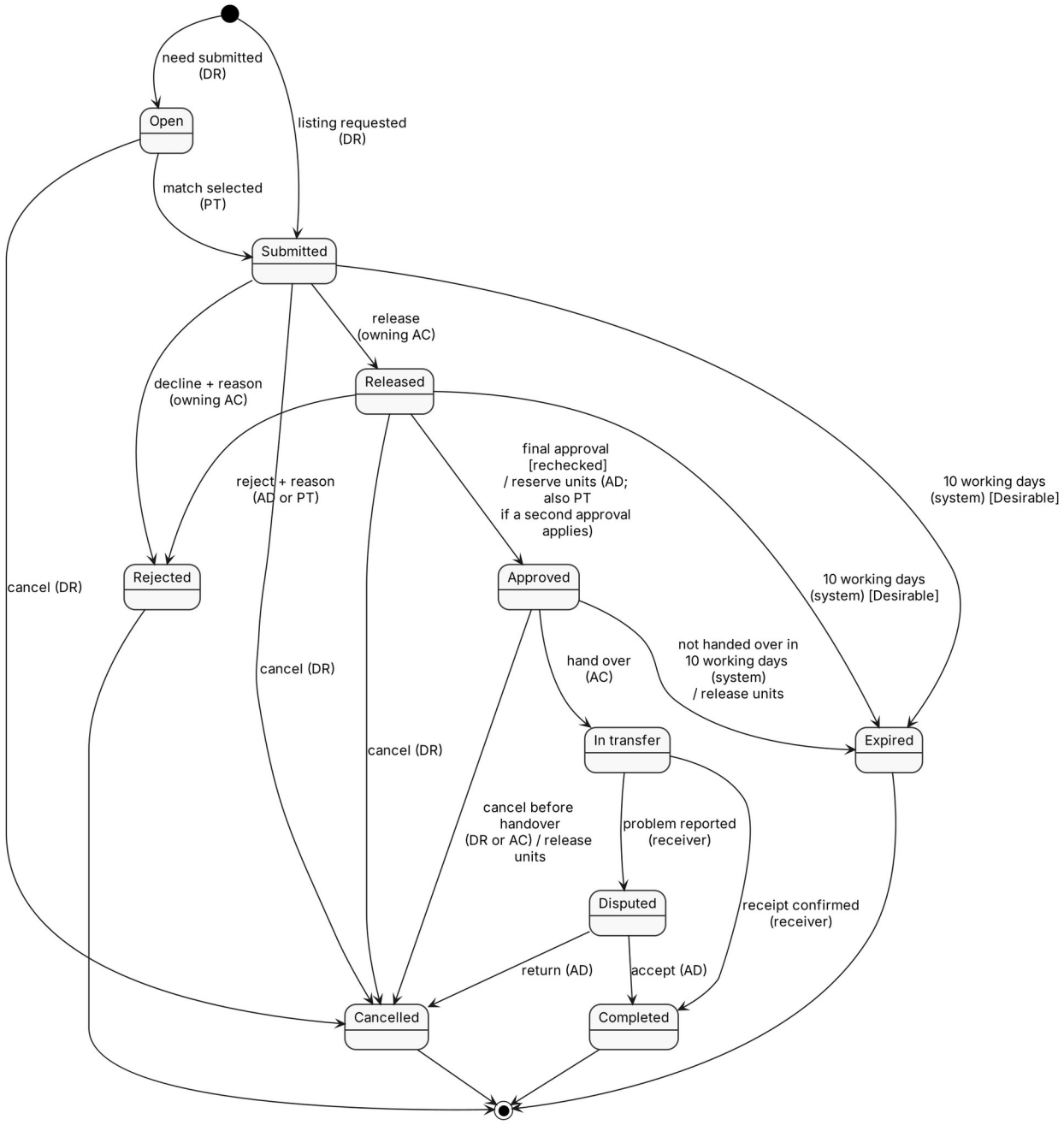


Figure 12-3: Request states

12.4 Approval and Transfer

Prepared by: Mohammad Alsuwaidi

The approval stages are Awaiting release, Awaiting approval and, for a value above the limit or unknown, Awaiting second approval (FR-4.1 to FR-4.3). Units are reserved when the last required stage approves (BR-5).

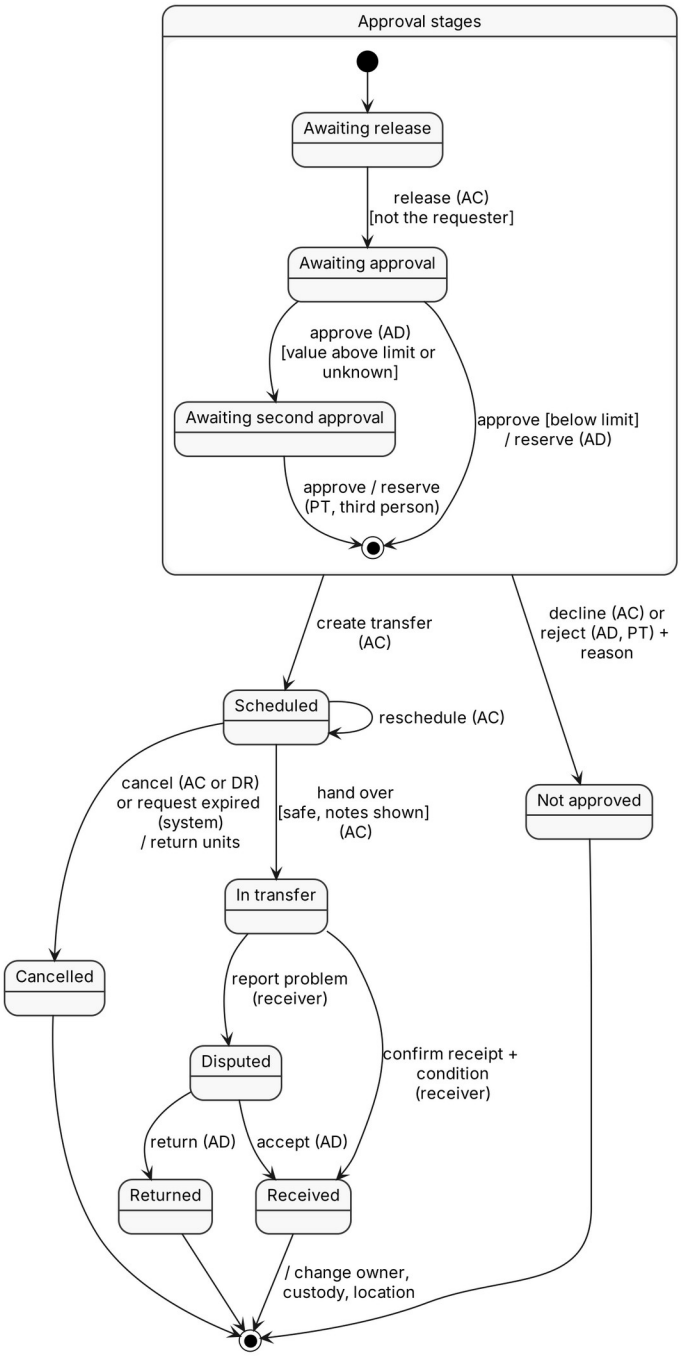


Figure 12-4: Approval and transfer states

12.5 Maintenance Request

Prepared by: **Rashid Alzaabi**

Only the assigned Maintenance Staff member or an Administrator can record inspection and repair results (FR-5.15). Completing requires the new condition (FR-5.8). Every cancel transition is made by the reporting user, the Asset Coordinator or an Administrator, with a reason (FR-5.13).

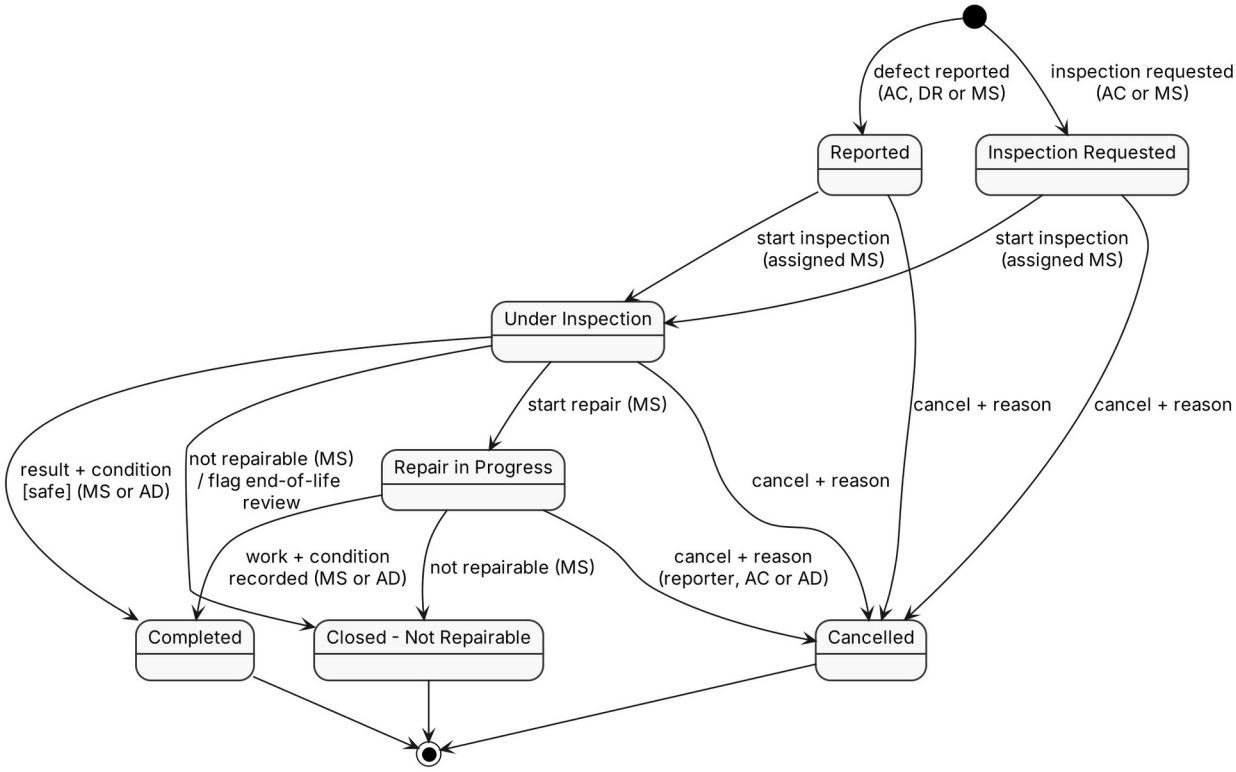


Figure 12-5: Maintenance request states

12.6 End-of-Life Proposal

Prepared by: Rashid Alzaabi

An end-of-life action passes two approval stages by two different people before the asset reaches its final state (BR-8, SAF-3). A rejection at either stage returns the asset to the state it had before (FR-5.28).

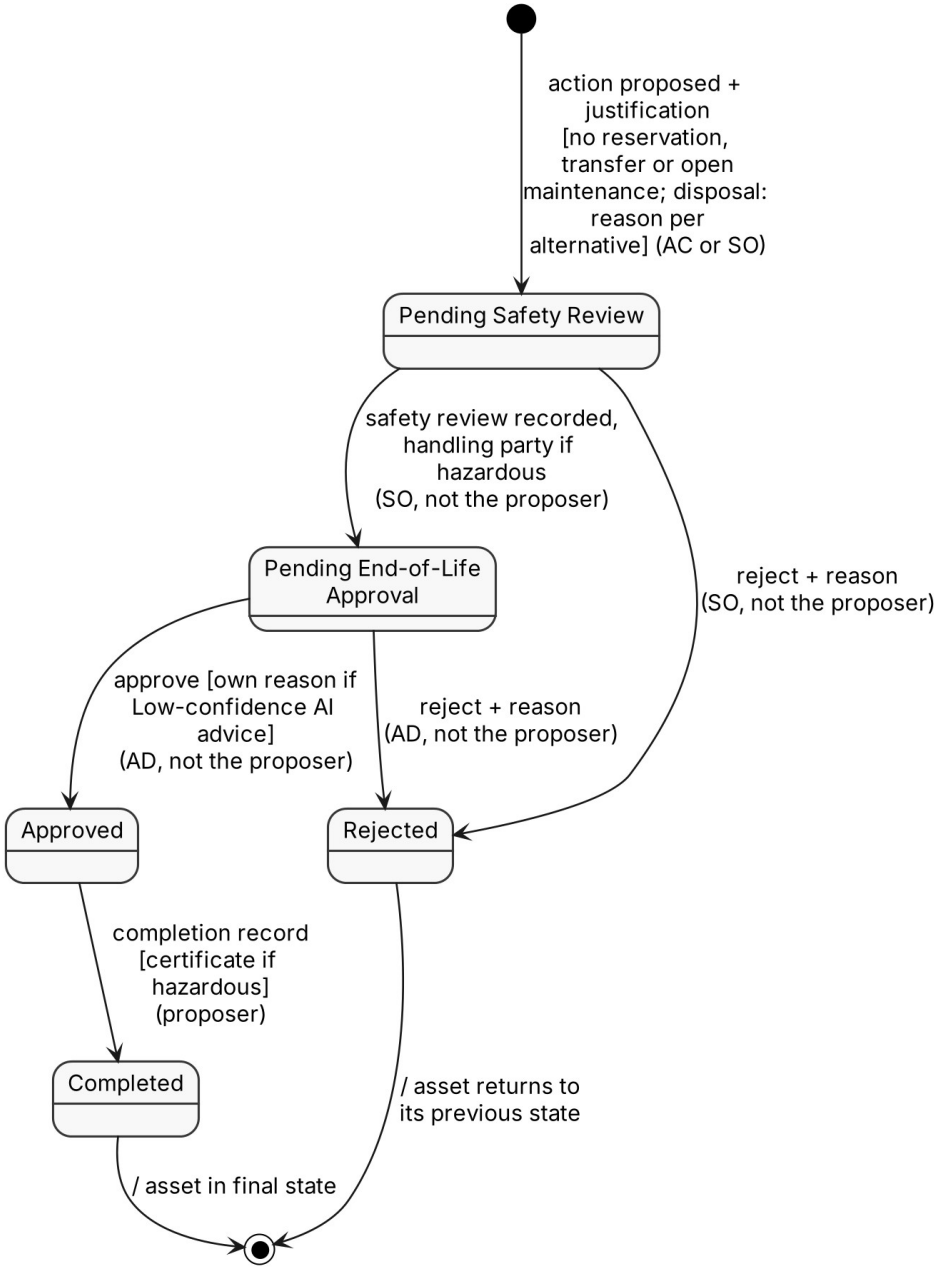


Figure 12-6: End-of-life proposal states

13. Component and Deployment Diagrams

Prepared by: Mohamed Almarzooqi

13.1 Component Diagram

Prepared by: Mohamed Almarzooqi

Business modules offer their operations through module service interfaces (Section 16.1) and require four shared interfaces: AccessCheck, AuditLog, Notify and AIService. Only AIGateway requires ProviderAPI. Replacing an AI service or the AI Provider therefore touches only the AI and analytics package (DR-1, DR-4).

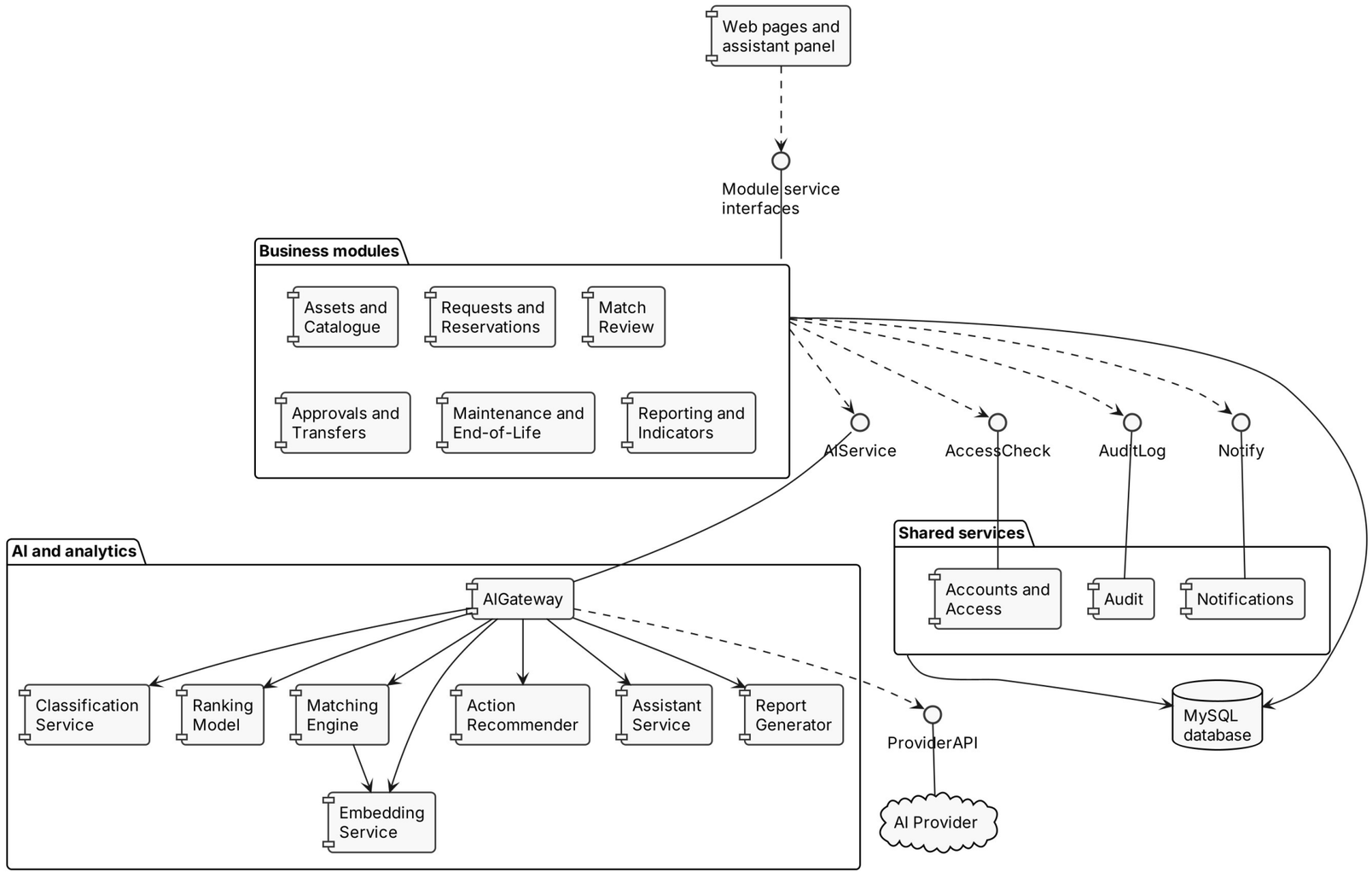


Figure 13-1: Component diagram

13.2 Deployment Diagram

Prepared by: Rashid Alzaabi

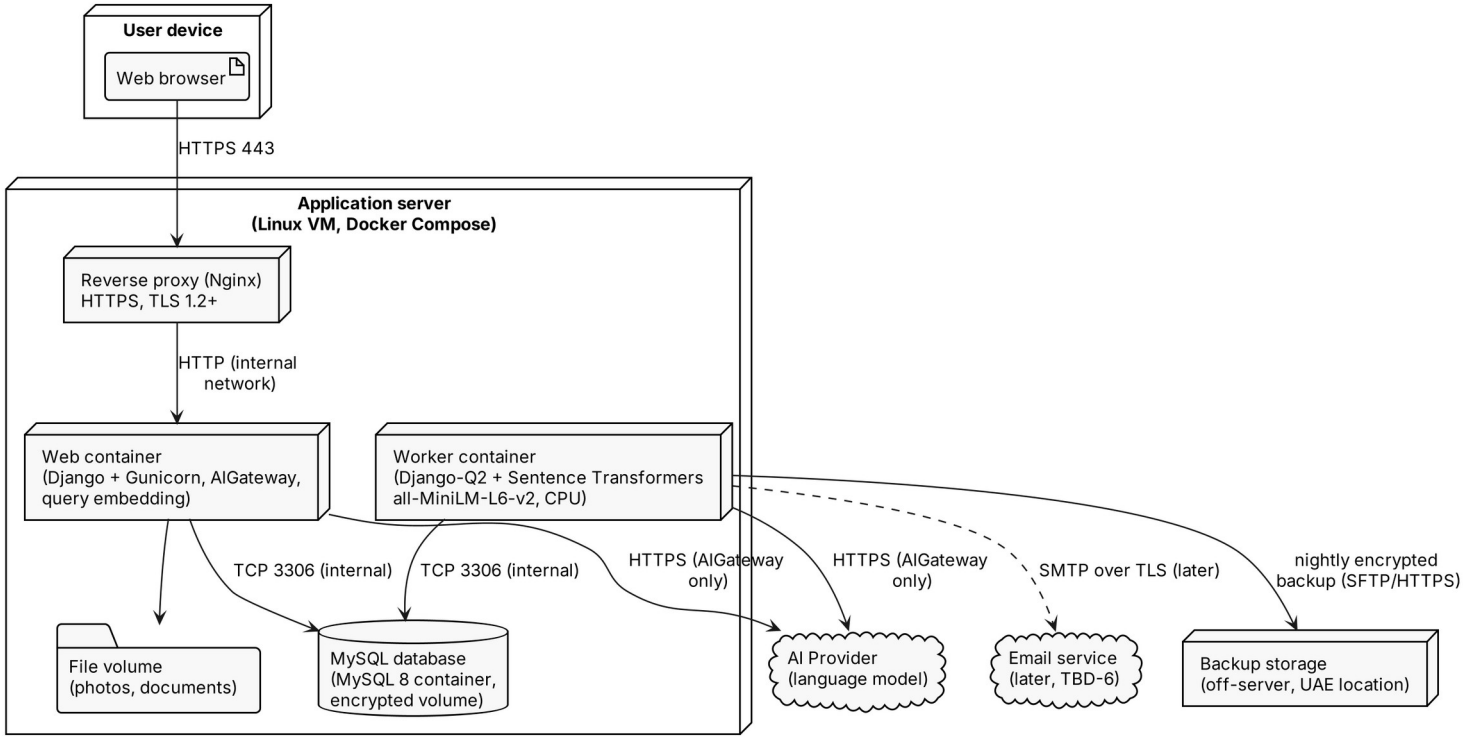


Figure 13-2: Deployment diagram

Table 13-1: Deployment nodes

Node	Runs	Notes
Reverse proxy	Nginx	Only public entry point, port 443; HTTP redirected; TLS certificate
Web container	Django, Gunicorn, AIGateway, query embedding	Serves pages and module services; restarts automatically
Worker container	Django-Q2, embedding model, scheduled jobs	Embeddings, retries, backups, monitoring
Database container	MySQL 8	Internal network only; encrypted volume
File volume	Photos and documents	Outside the web root; served after a permission check
Backup storage	Encrypted copies	UAE location; 14 daily and 8 weekly copies

14. AI Recommendation Workflow

Prepared by: Ahmed AlAbdouli

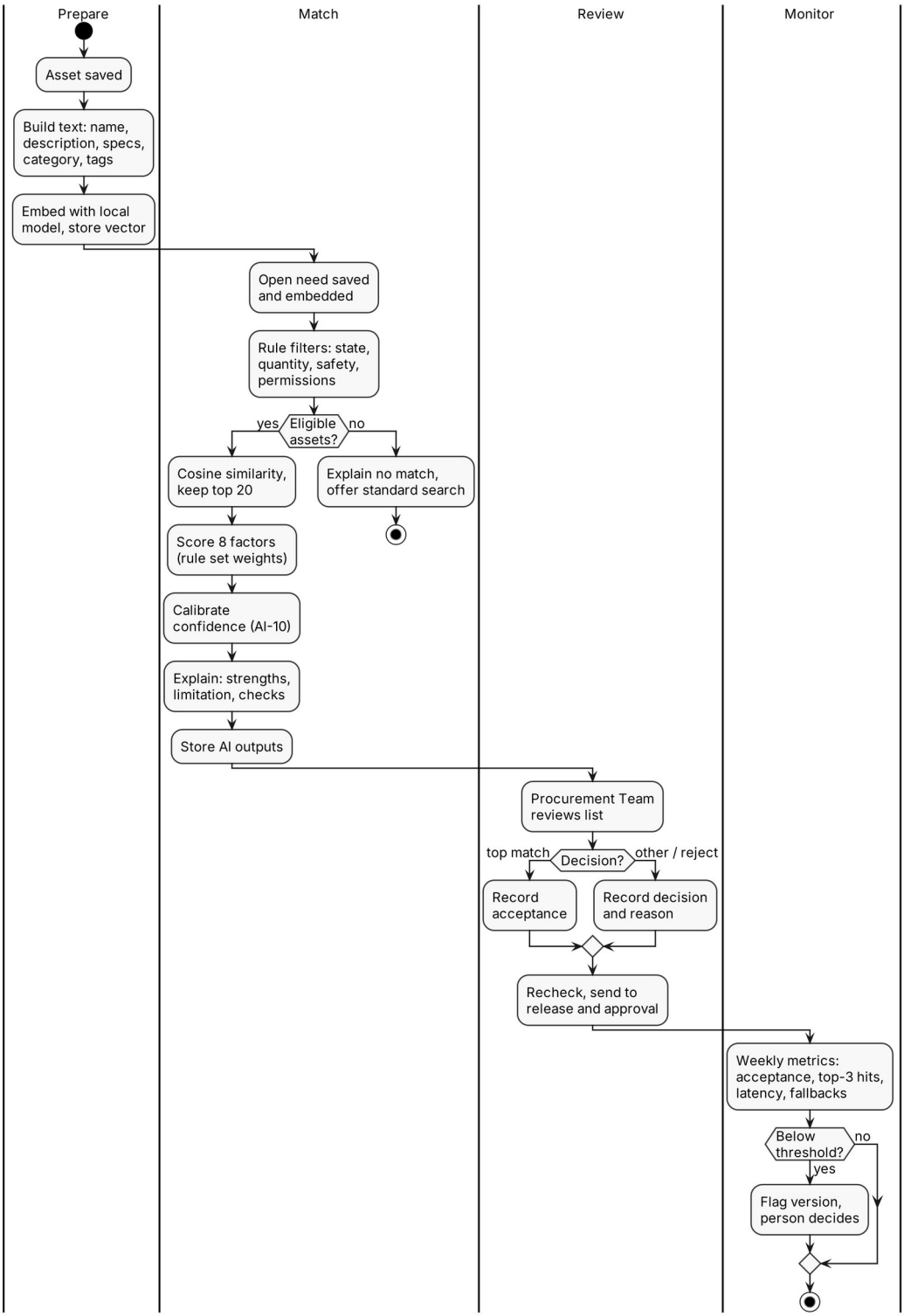


Figure 14-1: AI recommendation workflow

Figure 14-1 shows the full life of a match recommendation, from the asset being saved to the weekly monitoring of the results. It has four stages.

1. **Prepare (pattern B).** When an asset is saved, the worker builds its text (name, description, specifications, category, tags), embeds it and stores the vector with the text hash and model version.
2. **Match.** When a need is saved and embedded, rule filters remove assets that fail state, quantity, safety or permission checks (FR-8.2). Cosine similarity keeps the 20 closest, the ranking scores them on eight factors (Section 17), the confidence is calibrated (AI-10) and an explanation is written. Every result is stored before it is shown (AI-5). If nothing is eligible, the system says so and offers standard search (FR-8.9). If the request vector is not ready yet, the page shows the rule matches marked "not ranked" and refreshes when the job finishes.
3. **Review.** The Procurement Team member decides (H-2). A choice other than the top match, or a rejection, needs a reason (FR-8.7). The chosen asset is rechecked and sent to release and approval (FR-8.8).
4. **Monitor.** Each week the worker calculates the monitoring metrics of Section 19. A metric below its threshold flags the model version; a person decides what to do (H-13). Nothing is retrained automatically (FR-9.9).

15. Entity Relationship Diagram

Prepared by: **Mohammad Alsuwaidi**

The ERD has 32 tables and 61 foreign keys, drawn in five parts: accounts and catalogue (Figure 15-1); requests, approvals and transfers (Figure 15-2); maintenance, end of life and life-cycle events (Figure 15-3); AI records (Figure 15-4); and sustainability, notifications and audit (Figure 15-5). Every column is shown with its MySQL type and its keys (PK, FK, UQ). Every foreign key is drawn as its own line, labeled with the relationship name and the foreign key column, with the cardinality at both ends. A grey entity is a table drawn in full in the figure named under it; it is repeated so that each foreign key line can be drawn. Table 15-1 lists the same columns.

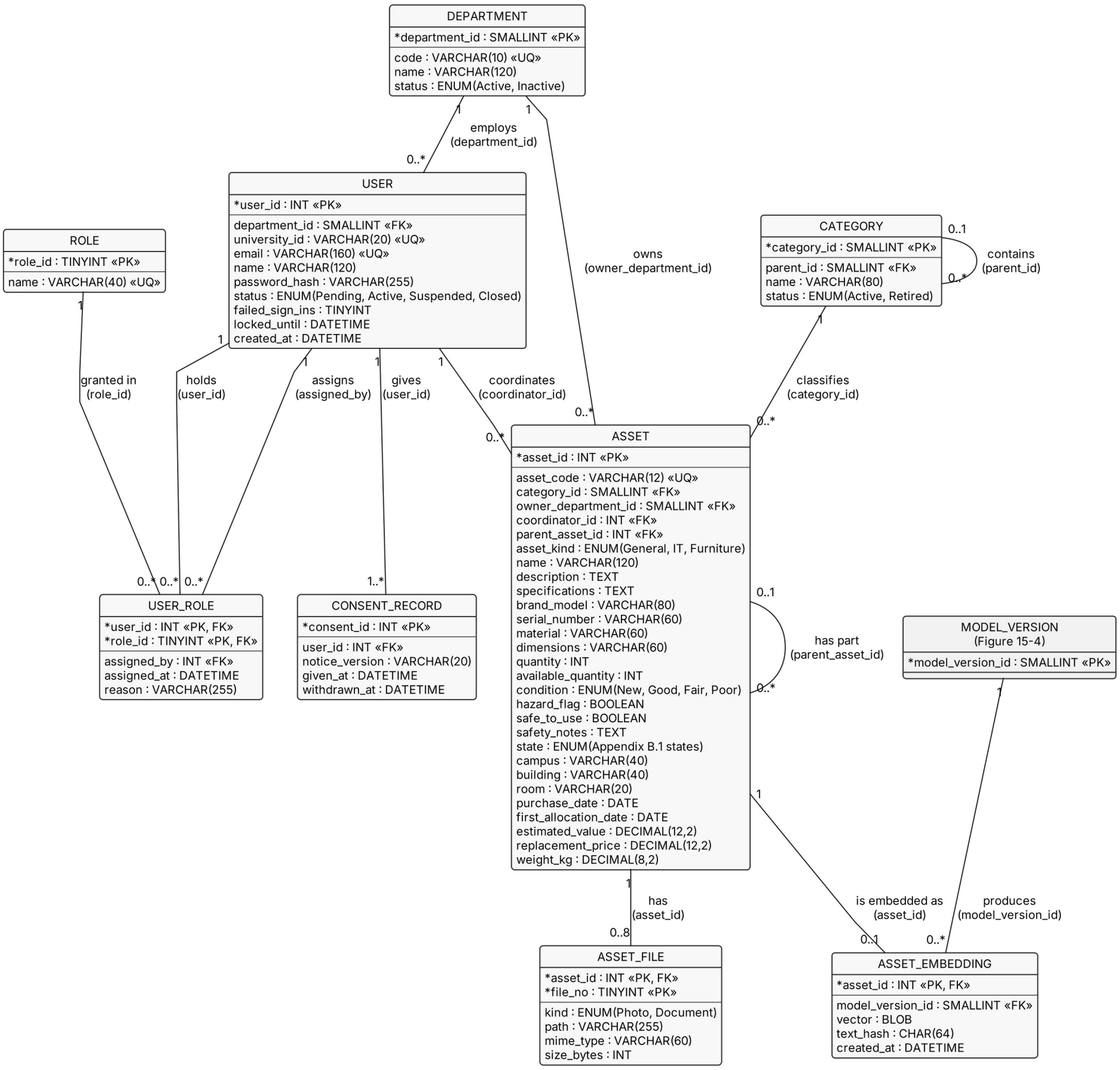


Figure 15-1: ERD part 1, accounts and catalogue

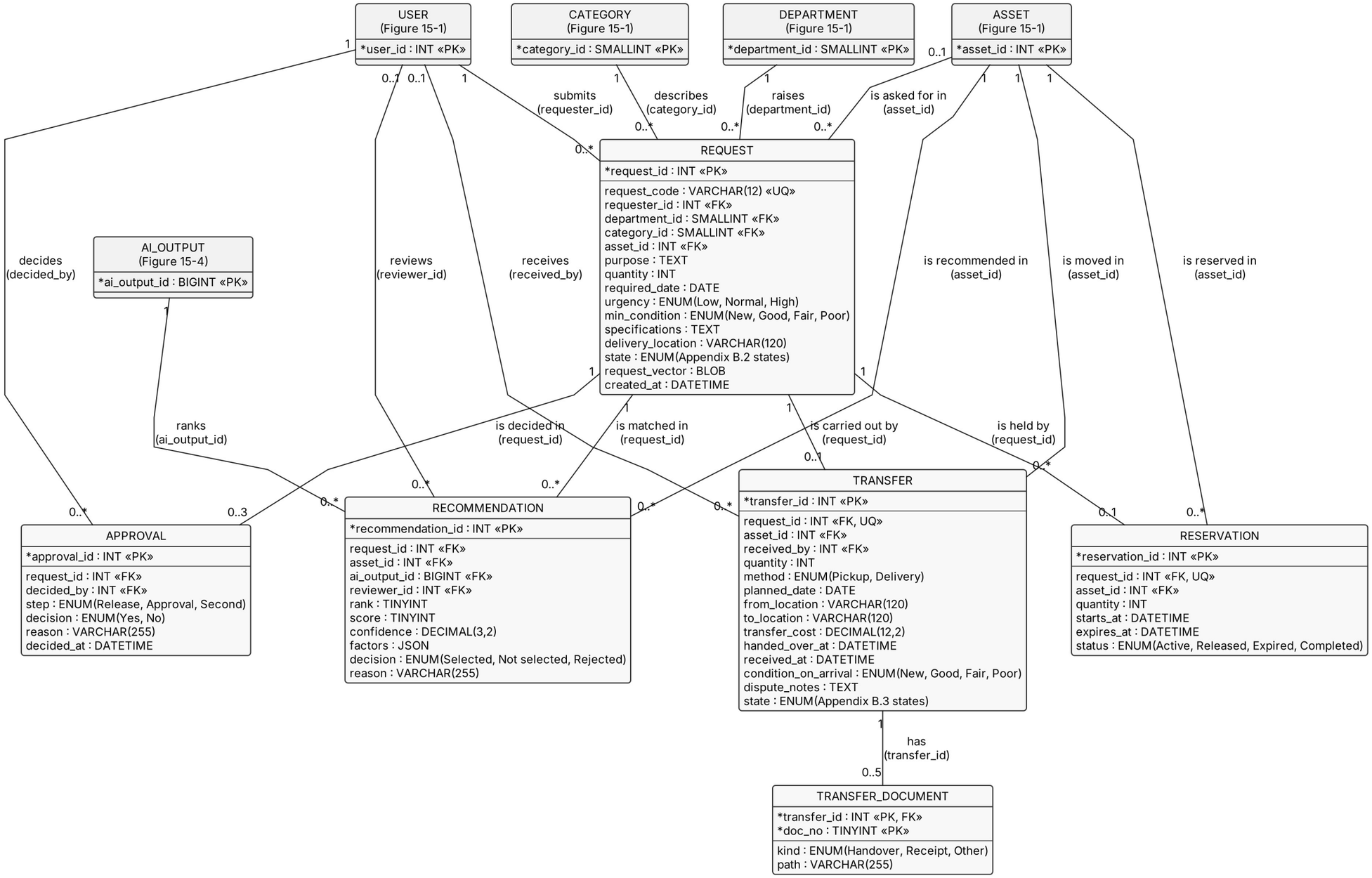


Figure 15-2: ERD part 2, requests, approvals and transfers

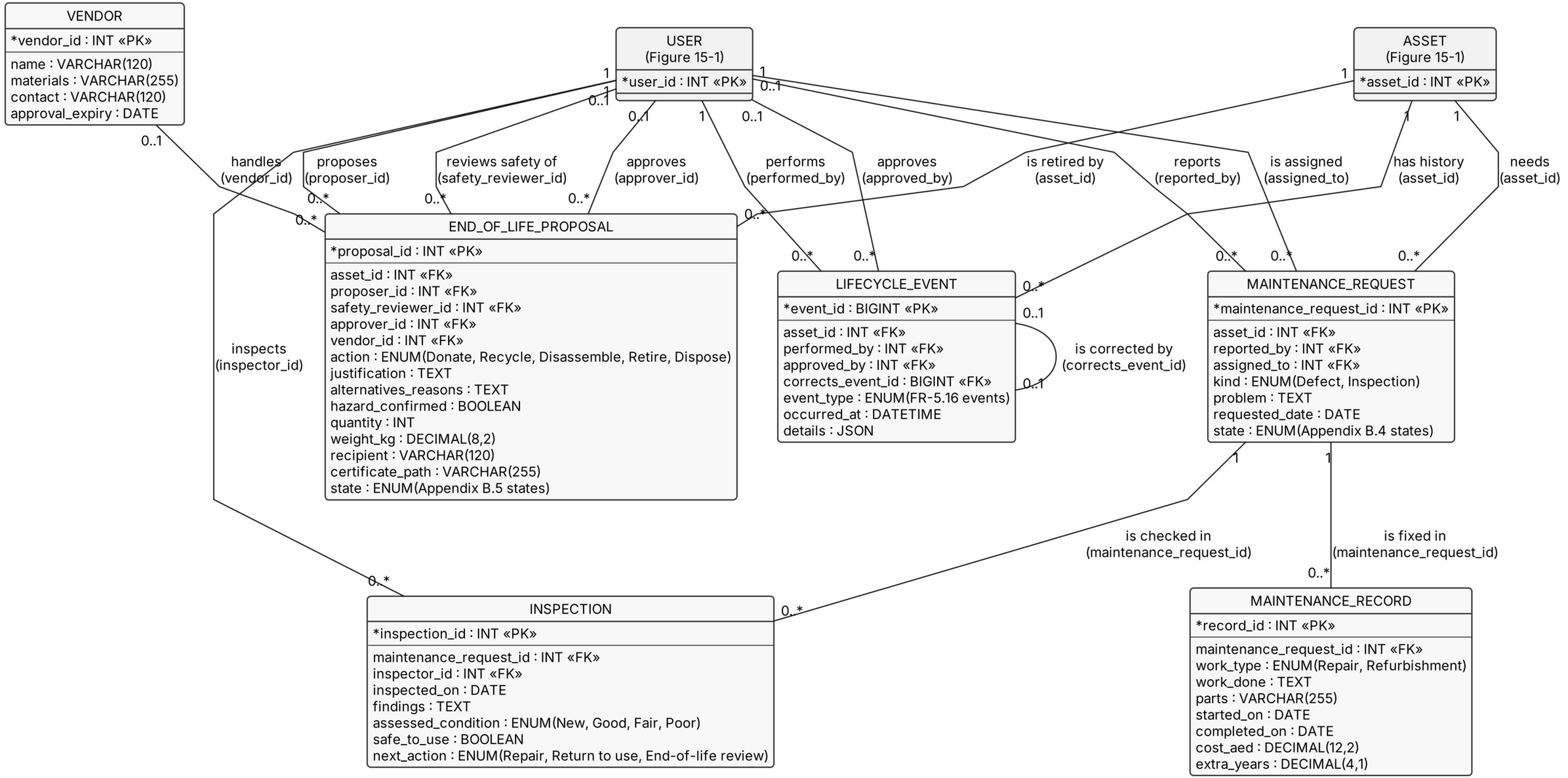


Figure 15-3: ERD part 3, maintenance, end of life and life-cycle events

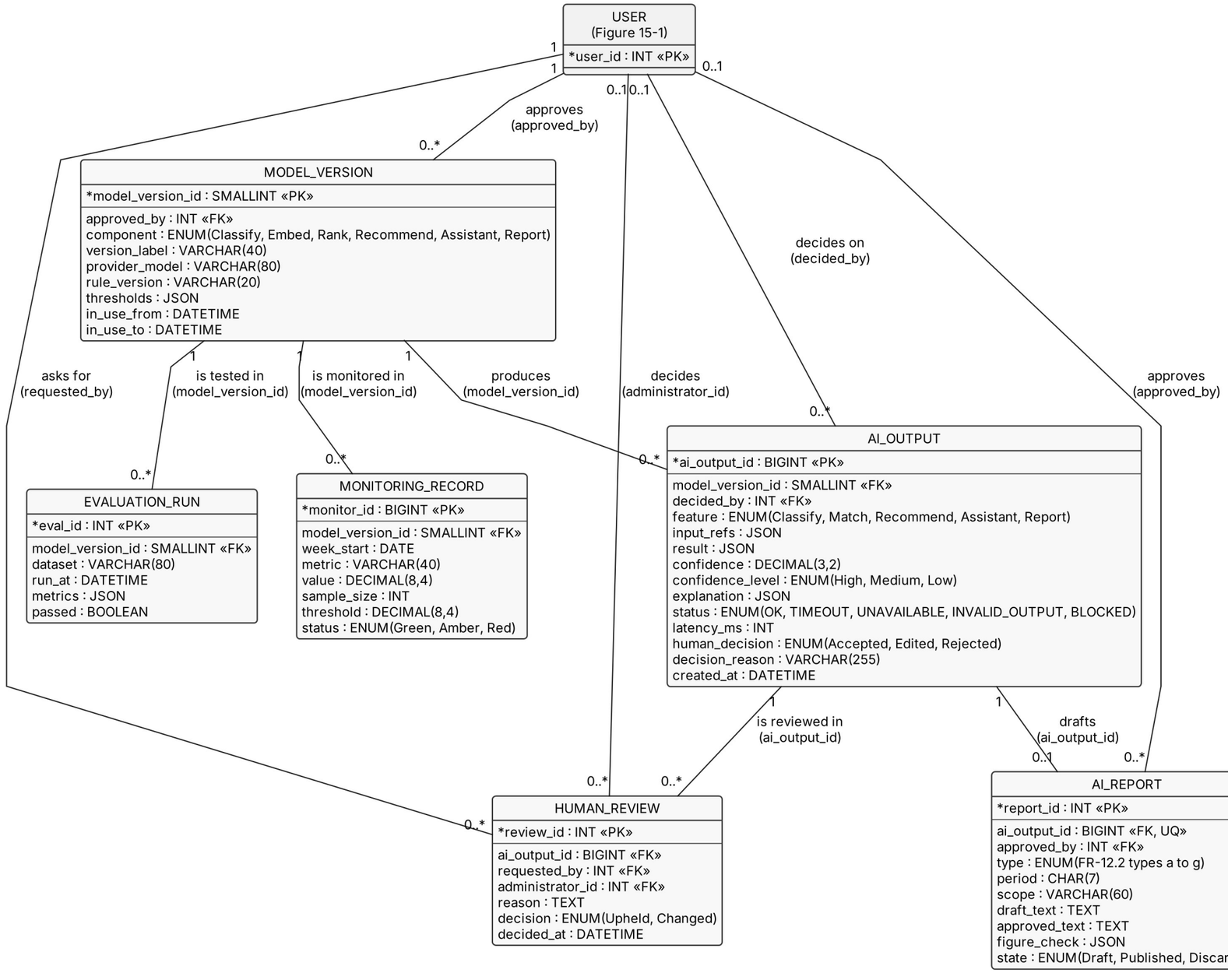


Figure 15-4: ERD part 4, AI records

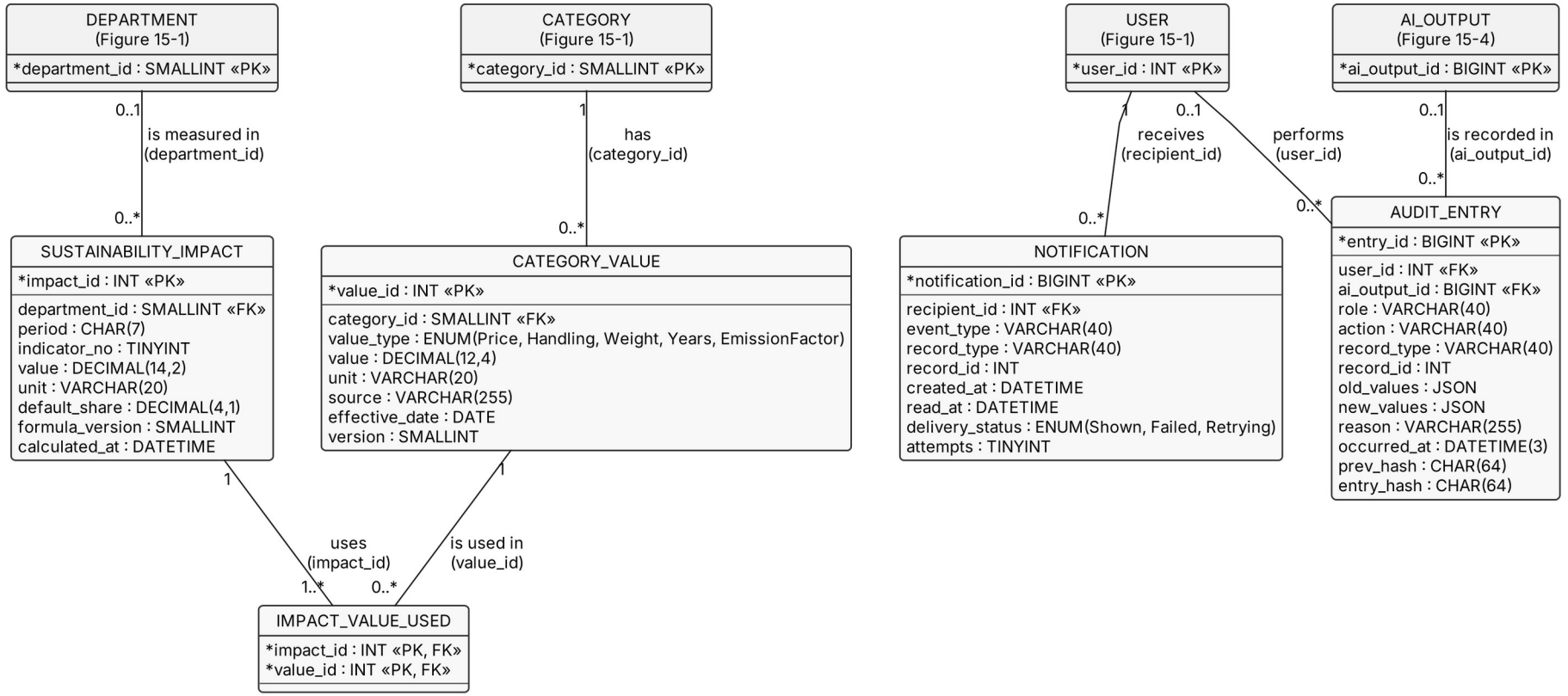


Figure 15-5: ERD part 5, sustainability, notifications and audit

15.1 Notation and Data Dictionary

Prepared by: Mohammad Alsuwaidi

- **Weak entities.** ASSET_FILE and TRANSFER_DOCUMENT use their owner key plus a partial key (file_no, doc_no).
- **Composite and derived attributes.** Location is stored as campus, building and room. Age is derived from purchase_date and today, so it is calculated, not stored (FR-5.19).
- **Many-to-many links.** USER_ROLE links users and roles; RECOMMENDATION links requests and assets and stores the score, explanation and model version, following [3]; IMPACT_VALUE_USED links each indicator result to the category value versions it used (FR-13.5).
- **Types.** ENUM columns list their values, or name the Phase 3 list they use (Appendix B states, FR-5.16 events, FR-12.2 report types). Money columns are DECIMAL(12,2) in AED and all dates are stored in UAE time.

Table 15-1: Tables and columns

Table	Holds	Keys (PK; FK)	Other columns	Requirements
DEPARTMENT	Campus departments	department_id	code, name, status	FR-6.9
USER	Accounts	user_id; department_id	university_id, email, name, password_hash, status, failed_sign_ins, locked_until, created_at	FR-6.1, SEC-6, SEC-7
ROLE	The nine roles	role_id	name	FR-6.8
USER_ROLE	Role assignments	user_id + role_id; assigned_by	assigned_at, reason	FR-6.8, BR-4
CONSENT_RECORD	Consent given and withdrawn	consent_id; user_id	notice_version, given_at, withdrawn_at	FR-6.2, SEC-8
CATEGORY	Approved category list	category_id; parent_id	name, status	FR-9.2, QUA-7
ASSET	Registered assets	asset_id; category_id, owner_department_id, coordinator_id, parent_asset_id	asset_code, asset_kind, name, description, specifications, brand_model, serial_number, material, dimensions, quantity, available_quantity, condition, hazard_flag, safe_to_use, safety_notes, state, campus, building, room, purchase_date, first_allocation_date, estimated_value, replacement_price, weight_kg	FR-1.2 to FR-1.10, FR-4.15
ASSET_FILE	Photos and documents (weak entity)	asset_id + file_no	kind, path, mime_type, size_bytes	FR-1.6, SEC-13
ASSET_EMBEDDING	Text vector of an asset	asset_id; model_version_id	vector, text_hash, created_at	FR-8.1
REQUEST	Requests and open needs	request_id; requester_id, department_id, category_id, asset_id	request_code, purpose, quantity, required_date, urgency, min_condition, specifications, delivery_location, state, request_vector, created_at	FR-3.1 to FR-3.15
RECOMMENDATION	One ranked match (links requests and assets)	recommendation_id; request_id, asset_id, ai_output_id, reviewer_id	rank, score, confidence, factors, decision, reason	FR-8.4, FR-8.7, QUA-12
APPROVAL	Release, approval and second approval	approval_id; request_id, decided_by	step, decision, reason, decided_at	FR-4.1 to FR-4.4
RESERVATION	Reserved units	reservation_id; request_id, asset_id	quantity, starts_at, expires_at, status	FR-3.8, FR-4.5
TRANSFER	Movement of an asset	transfer_id; request_id, asset_id, received_by	quantity, method, planned_date, from_location, to_location, transfer_cost, handed_over_at, received_at, condition_on_arrival, dispute_notes, state	FR-4.7 to FR-4.12
TRANSFER_DOCUMENT	Handover files (weak entity)	transfer_id + doc_no	kind, path	FR-4.8
MAINTENANCE_REQUEST	Defects and inspection requests	maintenance_request_id; asset_id, reported_by, assigned_to	kind, problem, requested_date, state	FR-5.1 to FR-5.7
INSPECTION	Inspection results	inspection_id; maintenance_request_id, inspector_id	inspected_on, findings, assessed_condition, safe_to_use, next_action	FR-5.5, SAF-5
MAINTENANCE_RECORD	Repair and refurbishment work	record_id; maintenance_request_id	work_type, work_done, parts, started_on, completed_on, cost_aed, extra_years	FR-5.6, FR-5.10
END_OF_LIFE_PROPOSAL	End-of-life proposal and completion	proposal_id; asset_id, proposer_id, safety_reviewer_id, approver_id, vendor_id	action, justification, alternatives_reasons, hazard_confirmed, quantity, weight_kg, recipient, certificate_path, state	FR-5.20 to FR-5.32, SAF-4
VENDOR	Approved recycling and disposal vendors	vendor_id	name, materials, contact, approval_expiry	FR-5.25
LIFECYCLE_EVENT	Life-cycle history (no updates)	event_id; asset_id, performed_by, approved_by, corrects_event_id	event_type, occurred_at, details	FR-5.16 to FR-5.18

Table	Holds	Keys (PK; FK)	Other columns	Requirements
CATEGORY_VALUE	Versioned prices, costs, weights, years and emission factors	value_id; category_id	value_type, value, unit, source, effective_date, version	FR-13.4, FR-13.5
SUSTAINABILITY_IMPACT	Calculated indicator values	impact_id; department_id	period, indicator_no, value, unit, default_share, formula_version, calculated_at	FR-13.1, FR-13.7
IMPACT_VALUE_USED	Category value versions used in each result	impact_id + value_id	-	FR-13.5
MODEL_VERSION	AI model, rule and threshold versions	model_version_id; approved_by	component, version_label, provider_model, rule_version, thresholds, in_use_from, in_use_to	AI-5, AI-10
AI_OUTPUT	Every AI result and its human decision	ai_output_id; model_version_id, decided_by	feature, input_refs, result, confidence, confidence_level, explanation, status, latency_ms, human_decision, decision_reason, created_at	FR-7.A4, AI-5
HUMAN_REVIEW	Requests for human review of an AI result	review_id; ai_output_id, requested_by, administrator_id	reason, decision, decided_at	SEC-10
AI_REPORT	AI report drafts and approved versions	report_id; ai_output_id, approved_by	type, period, scope, draft_text, approved_text, figure_check, state	FR-12.6 to FR-12.10
EVALUATION_RUN	Test-set evaluations of a model version	eval_id; model_version_id	dataset, run_at, metrics, passed	QUA-9, QUA-10
MONITORING_RECORD	Weekly monitoring metrics	monitor_id; model_version_id	week_start, metric, value, sample_size, threshold, status	QUA-12
NOTIFICATION	In-application notifications	notification_id; recipient_id	event_type, record_type, record_id, created_at, read_at, delivery_status, attempts	FR-7.N1 to FR-7.N5
AUDIT_ENTRY	Append-only audit trail	entry_id; user_id, ai_output_id	role, action, record_type, record_id, old_values, new_values, reason, occurred_at, prev_hash, entry_hash	FR-7.A1 to FR-7.A6

Django-Q2 adds its own job tables, and Django adds its session table. The data stores of Section 2.2 map to these tables as shown in Table 2-2.

16. Interface Contracts

Prepared by: Mohammad Alsuwaidi

Section 16.1 defines the operations that modules offer each other and the web pages; Section 16.2 defines the contracts between the modules and the AI services. All operations follow four common rules:

1. Every operation receives the acting user and checks access first (AccessCheck); a refusal raises `PERMISSION_DENIED` and is audited (FR-6.7, FR-7.A1).
2. Every operation that changes data runs in one database transaction with its audit entry; if the audit entry fails, nothing is saved (FR-7.A6).
3. Notifications are sent after the transaction commits; a notification failure never reverses the change (FR-7.N3).
4. Errors are returned as a code and a message that names the problem (for example `MISSING_FIELDS: location, condition`).

16.1 Module Operations

Prepared by: Mohammad Alsuwaidi

Table 16-1: Module service operations

Module	Operation	Input	Output	Errors	Requirements
Accounts and Access	apply, activate, refuse, assignRole, suspend, withdrawConsent	Applicant details and consent; user, role, department, reason	Account state	DUPLICATE, NOT_AUTHORIZED, CONSENT_MISSING	FR-6.1 to FR-6.12
Accounts and Access	allowed(user, action, record)	User, action, record	Yes or no	-	FR-6.7, SEC-5
Assets and Catalogue	saveDraft, publish, withdraw	Asset fields, files	asset_id, state	MISSING_FIELDS, INVALID_VALUE, SAFETY_REVIEW_NEEDED, FILE_REJECTED	FR-1, FR-2.1
Assets and Catalogue	search(criteria)	Text, filters, page	Results page with counts	INVALID_CRITERIA	FR-2.2 to FR-2.6
Assets and Catalogue	reserve, releaseReservation, completeTransfer	asset_id, quantity, request_id or transfer_id	Reservation or updated asset	NOT_ENOUGH_UNITS, RESTRICTED	FR-4.5, FR-4.11, FR-4.15
Requests and Reservations	submit, edit, cancel	Request fields	request_id, state	MISSING_FIELDS, NOT_OWN_DEPARTMENT, NOT_ENOUGH_UNITS, ASSET_RESTRICTED, WRONG_STATE	FR-3.1 to FR-3.15
Match Review	matches(request_id)	request_id	Ranked list or "not ranked" list	NO_ELIGIBLE_ASSETS	FR-8.1 to FR-8.9
Match Review	decide(request_id, asset_id, decision, reason)	Choice and reason	Request sent to release	REASON_REQUIRED, NO_LONGER_AVAILABLE	FR-8.7, FR-8.8
Approvals and Transfers	release, approve, secondApprove	request_id, decision, reason	Approval record; reservation on final approval	SAME_PERSON, REASON_REQUIRED, NOT_ENOUGH_UNITS	FR-4.1 to FR-4.6
Approvals and Transfers	schedule, handOver, confirmReceipt, reportProblem, resolveDispute	transfer details, condition, decision	Transfer state	NOT_APPROVED, UNSAFE, INSPECTION_TOO_OLD, NOT_RECEIVER	FR-4.7 to FR-4.16
Maintenance and End-of-Life	report, assign, startInspection, recordInspection, recordWork, complete, closeNotRepairable, cancel	Maintenance fields	Maintenance state	NOT_MAINTENANCE_ROLE, INVALID_DATES, MISSING_FIELDS	FR-5.1 to FR-5.15
Maintenance and End-of-Life	propose, safetyReview, decide, complete	Action, reasons, review, completion record	Proposal state	BLOCKED_BY_OPEN_WORK, SAME_PERSON, VENDOR_EXPIRED, CERTIFICATE_REQUIRED	FR-5.20 to FR-5.34
Reporting and Indicators	standardReport, indicators, requestAIReport, approveReport, discardReport	Report type, period, scope	Report, indicator set or AI report draft	NO_RECORDS, NOT_AUTHORIZED	FR-7.R, FR-12, FR-13
Notifications	notify(event, record, recipients)	Event and record	Notification IDs	Retried on failure	FR-7.N1 to FR-7.N5
Audit	write(user, action, record, before, after, reason, ai_output_id)	Entry fields	entry_id	AUDIT_FAILED (change rolled back)	FR-7.A1 to FR-7.A6

16.2 AI Service Contracts

Prepared by: Ahmed AlAbdouli

Modules, and the assistant screen, call AIGateway with a common envelope. The envelope stays inside ReSource; for the three features that use the AI Provider, only the fields in the "payload" go to the provider, after redaction. Table 16-2 lists the seven contracts, one for each operation of the interfaces in Figure 9-4. None of them contains a name, university ID or e-mail field (DR-4).

Table 16-2: AI service contracts

Operation	Request payload	Response	Leaves the server?
classify	asset text (name, description, specifications), photo references, approved category list	category, subcategory, tags, description, missing fields, confidence, reasons, conflicts	Text and photos to the AI Provider
embed	text (up to 256 word pieces)	384-number vector, model version	No
candidates	request_id, number of candidates (20)	eligible asset_ids with cosine similarity	No
rank	request_id, top-20 candidate asset_ids, urgency	ranked list with score, factors, confidence, explanation, rule checks	No
recommendAction	asset_id, condition, safety status, maintenance cost, open requests count	action options with score, confidence, estimates, explanation	No
chat	question, conversation so far, allowed read tools	answer, links, optional request draft, confidence	Question and tool results to the AI Provider
draftReport	report type, period, scope, standard figures	draft text, figure check, assumptions, action list with confidence	Figures and template to the AI Provider

Example request from Match Review to AIGateway for a ranking. AIGateway checks acting_user itself and never sends it to an AI service:

```
{
  "feature": "rank",
  "call_id": "AIQ-2026-31877",
  "acting_user": 1047,
  "time_limit_s": 20,
  "payload": {
    "request_id": "R-2026-0391",
    "urgency": "Medium",
    "candidates": ["A-2026-0412", "A-2026-0377", "..."]
  }
}
```

Example response, showing the first result only (Table 16-3 lists the other status values):

```
{
  "call_id": "AIQ-2026-31877",
  "status": "OK",
  "model_version": "embed-v1.0 / rank rules v3",
  "latency_ms": 640,
  "results": [{
    "asset_id": "A-2026-0412", "rank": 1, "score": 92,
    "factors": {"function": 0.96, "quantity": 1.00, "condition": 0.90,
                "location": 0.55, "ready_by": 1.00, "repair_cost": 1.00,
                "transfer_cost": 0.81, "waste_carbon": 1.00},
    "confidence": 0.84, "confidence_level": "High",
    "explanation": {
      "summary": "Same function and better than the 70 MHz asked for.",
      "strengths": ["function and specs", "quantity", "ready now"],
      "limitation": "On Main Campus: delivery to SAN costs about AED 120.",
      "rule_checks": ["available", "quantity", "safety", "permissions"]
    }
  }],
  "weak_matches": false,
  "ai_output_id": "AI-2026-08812"
}
```

Table 16-3: AI error statuses

Status	Meaning	Caller action
TIMEOUT	No usable result within 20 s	Show the fallback of Table 4-1; record the failed output
UNAVAILABLE	Service marked down after repeated failures	Show the fallback at once
INVALID_OUTPUT	Result failed validation (category not on the list, figure mismatch, bad format)	Treat as failed; show the fallback
BLOCKED	Request refused by an access or content check (AI-7, AI-9)	Tell the user what is not allowed; offer the standard function

17. Recommendation Score and Explanation Design

Prepared by: Ahmed AlAbdouli

17.1 Score

Prepared by: Ahmed AlAbdouli

Each eligible candidate gets a compatibility score from 0 to 100 (FR-8.4), the weighted sum of eight factor values, each between 0 and 1:

```
Score = round( 100 x (w1 F1 + w2 F2 + ... + w8 F8) ),  with w1 + ... + w8 = 1
```

The factors cover every criterion listed in FR-8.3 and FR-8.5. The weights belong to a versioned rule set (rule set v3), are visible to Administrators and can be changed only by creating a new version (LEG-9, QUA-7). The requesting department's name, ID and size are not inputs (LEG-9, QUA-13).

Table 17-1: Score factors (rule set v3)

Factor	Weight	Value between 0 and 1	Criterion
F1 Function and specifications	0.35	clamp((cosine - 0.30) / 0.55, 0, 1) x 0.5 to the power of the number of stated specifications not met	Category, purpose, specifications
F2 Quantity	0.15	min(available, requested) / requested	Quantity (FR-8.6)
F3 Condition	0.10	New 1.0, Good 0.9, Fair 0.6, Poor 0.3; at most 0.3 if below the requested minimum	Current and requested condition
F4 Location	0.10	Same building 1.0, same campus 0.85, other campus 0.55	Location
F5 Ready by the date	0.10	1.0 if available by the required date, 0.5 if up to 2 days late, else 0	Required date, urgency
F6 Repair cost	0.08	1 - (expected repair cost / replacement price), at least 0	Repair needs
F7 Transfer cost	0.07	1 - (transfer cost / (0.15 x replacement price)), at least 0	Transfer cost
F8 Waste and carbon saved	0.05	carbon avoided by this candidate / largest carbon avoided among the candidates	Waste and carbon reduction

Urgency changes the weight of F5: High urgency doubles it and Low urgency halves it, and all weights are then rescaled to add up to 1. If no candidate scores 60 or more, the list is shown with a "weak matches" warning and no top pick. A candidate that meets only part of the quantity is shown with that fact (FR-8.6).

17.2 Confidence

Prepared by: Ahmed AlAbdouli

The score ranks candidates; the confidence says how likely it is that the Procurement Team will find the candidate suitable. It is calculated with Platt scaling [14], a logistic curve fitted on the labeled evaluation set of QUA-10:

```
confidence = 1 / (1 + exp(-(a x S + b))),   S = score / 100
example fit: a = 4.60, b = -2.57 (refitted for each model version)
```

The level then follows Table 4-2. A candidate with missing required information (for example no condition or no location) is Low whatever its score (AI-10).

17.3 Explanation Structure

Prepared by: Ahmed AlAbdouli

Every recommendation has the same explanation structure, stored in AI_OUTPUT and shown on the screen (AI-2, QUA-14):

Table 17-2: Explanation fields

Field	How it is produced	Example (A-2026-0412)
summary	One sentence from a template using the two strongest factors	Same function and better than the 70 MHz you asked for, enough units.
strengths	The three factors with the largest weighted values (w x F)	Function and specs; quantity; ready now
limitation	The factor with the largest weighted gap, w x (1 - F)	On Main Campus: delivery to SAN costs about AED 120
rule_checks	Checks passed before scoring	Available, quantity, safety, permissions
confidence, level	Section 17.2	0.84, High
versions	Embedding model and rule set	embed-v1.0, rank rules v3

17.4 Worked Example

Prepared by: Ahmed AlAbdouli

For request R-2026-0391 (2 oscilloscopes, at least 70 MHz, minimum condition Good, SAN Campus, urgency Medium), Table 17-3 scores the two candidates shown in mockups 7.2 and 7.3.

Table 17-3: Worked scoring example

Factor (weight)	A-2026-0412: 100 MHz, Good, 3 units, Main Campus	A-2026-0377: 50 MHz, Fair, 1 unit, SAN Campus
F1 (0.35)	cosine 0.83, all specs met: 0.96	cosine 0.795, 1 spec not met: 0.90 x 0.5 = 0.45
F2 (0.15)	min(3, 2) / 2 = 1.00	1 / 2 = 0.50
F3 (0.10)	Good: 0.90	Fair below the Good minimum: 0.30
F4 (0.10)	Other campus: 0.55	Same campus: 0.85
F5 (0.10)	Ready now: 1.00	Ready now: 1.00
F6 (0.08)	No repair: 1.00	No repair: 1.00
F7 (0.07)	1 - 120 / (0.15 x 4,200) = 0.81	1 - 40 / (0.15 x 3,000) = 0.91
F8 (0.05)	119.2 kg CO2e, the largest: 1.00	68.8 / 119.2 = 0.58
Score	100 x 0.919 = 92	100 x 0.620 = 62
Confidence	0.84, High	0.57, Medium

Carbon avoided uses Section 18: 2 units x 2.6 kg x 22.92 kg CO2e per kg = 119.2 for A-2026-0412 and 1 unit x 3.0 kg x 22.92 = 68.8 for A-2026-0377. Transfer costs (AED 120 and AED 40) and the AED 3,000 replacement price of A-2026-0377 are sample values.

18. Sustainability Calculations

Prepared by: Rashid Alzaabi

The nine indicators are calculated with the fixed formulas of Phase 3, Table 2.13-1, never with AI (FR-13.2). This section designs how the values are found, stored and reproduced.

18.1 Calculation Pipeline

Prepared by: Rashid Alzaabi

1. **Trigger.** A completed transfer, repair, refurbishment, donation, recycling, disassembly, retirement or disposal (FR-13.3), a correction record, or a nightly run.
2. **Find each value.** For each record and value (replacement price, handling cost, weight, additional years, emission factor): use the actual value on the record; otherwise the category default in force on the action date; otherwise list the record under "Missing data" and leave it out (FR-13.6, FR-13.8).
3. **Calculate.** Apply the formulas per indicator, counting each asset unit once per period. A rate with a zero denominator is shown as "not available", not as 0.
4. **Store.** Save one SUSTAINABILITY_IMPACT row per period, scope and indicator with the share of default values, and link the CATEGORY_VALUE versions used in IMPACT_VALUE_USED (FR-13.5).
5. **Label.** Mark the indicator "Estimate" when any record used a default, and show the share (FR-13.7).

18.2 Values and Sources

Prepared by: Rashid Alzaabi

Table 18-1: Category values

Value	Unit	Maintained by	Source
Replacement price	AED per unit	Finance Officer	Recorded price, else category default (FR-13.4)
Storage and handling cost avoided	AED per unit	Finance Officer	Category default (TBD-8)
Weight	kg per unit	Sustainability Officer	Recorded weight, else category default (FR-5.32)
Additional years of use	years	Sustainability Officer	Maintenance Staff estimate, else category default
Emission factor	kg CO2e per kg	Sustainability Officer	EPA WARM version 16 source-reduction factors [15], Phase 3 Table 2.13-2

Every value has a source, an effective date and a version; changing a value creates a new version, so a past result can always be reproduced with the versions it used (FR-13.5).

18.3 Worked Example

Prepared by: **Rashid Alzaabi**

Two oscilloscopes of A-2026-0412 are transferred to Physics and receipt is confirmed. The recorded values are a replacement price of AED 4,200 and a weight of 2.6 kg per unit; the category defaults are an avoided storage and handling cost of AED 18 per unit (the Phase 2 assumption [5]) and 4 additional years of use per unit; the transfer cost is AED 120; the emission factor for "Other electronic equipment" (WARM Mixed Electronics) is 22.92 kg CO2e per kg.

Table 18-2: Worked indicator example

#	Indicator	Calculation	Result
1	Assets reused	2 units	2
3	Avoided purchases	2 x AED 4,200	AED 8,400
4	Financial savings (net)	8,400 + 2 x 18 - 120 - 0 maintenance	AED 8,316 (estimate)
5	Waste diverted	2 x 2.6 kg	5.2 kg
8	Asset-life extension	2 x 4 years (default)	8 years (estimate)
9	Estimated carbon reduction	5.2 kg x 22.92 kg CO2e per kg	119.2 kg CO2e

Indicators 4 and 8 use defaults and are labeled "Estimate". The rates (indicators 2 and 6) and the repair figures (indicator 7) are calculated over the whole period, not per transfer. All figures are estimates for internal planning and are not certified carbon or financial accounts (FR-13.12).

19. Model-Monitoring Records

Prepared by: Ahmed AlAbdouli

Two kinds of record show whether the AI keeps meeting its targets. An EVALUATION_RUN tests a model version on a fixed labeled data set before it is put into use and after any change. A MONITORING_RECORD stores one weekly metric from real use. Both point to the MODEL_VERSION they measure.

Table 19-1: Monitoring metrics and thresholds

Metric	How it is calculated	Source	Target or threshold	Requirements
Top-3 hit rate	Test requests with a known match whose match is in the first three	Evaluation run	At least 80%	QUA-10
Category accuracy	Test assets whose main category is suggested correctly	Evaluation run	At least 80%	QUA-9
First action acceptable	Test scenarios where the first recommended action is acceptable	Evaluation run	At least 80%	QUA-11
Department-name test	Rankings unchanged when only the department name or ID changes	Evaluation run	100% unchanged	QUA-13, LEG-9
Assistant correctness	Correct answers and correct refusals on the test questions	Evaluation run	At least 80% (TBD-12)	QUA-15
Calibration error	Average gap between confidence and actual acceptance, in 10 bins	Evaluation run	0.10 or less	AI-10
Match acceptance rate	Top matches accepted / match decisions, also by confidence level	Weekly	Amber below 60%, red below 40%	QUA-12
Override reasons	Count of each reason for choosing another match or rejecting	Weekly	Reviewed when one reason exceeds 30%	FR-8.7
Classification acceptance	Suggested fields accepted without edit / fields suggested	Weekly	Amber below 60%	FR-9.9
Response time	AI calls with a usable result within 15 s / all calls	Weekly	At least 90%	PER-4
Fallback rate	Calls ending in TIMEOUT, UNAVAILABLE or INVALID_OUTPUT	Weekly	Amber above 5%, red above 10%	AI-6
Report figure failures	AI report drafts rejected by the figure check	Weekly	Any failure is reviewed	FR-12.4

Table 19-2: Example monitoring records

week_start	model_version	metric	value	threshold	status
19/10/2026	rank rules v3	match_acceptance_rate	0.71	0.60	Green
19/10/2026	rank rules v3	response_time_within_15s	0.97	0.90	Green
19/10/2026	classify-v1.2	classification_acceptance	0.58	0.60	Amber
19/10/2026	classify-v1.2	fallback_rate	0.03	0.05	Green

An amber result notifies the System Administrator and the Administrator, who review the overrides and reasons. A red result, or a failed evaluation run, stops the version for new suggestions until a person decides to restore the previous version, publish a new rule set or thresholds, or accept the result with a reason (H-13). The AI is never retrained automatically (FR-9.9).

Appendix A: Requirement-to-Design Traceability

Prepared by: Mohamed Almarzooqi

Table A-1 traces each requirement group of the Phase 3 document [4] to the parts of this design that meet it.

Table A-1: Traceability from requirements to design

Requirements	DFD	Architecture and AI	Detailed design	Screens
FR-1 Asset registration	2.0, D2	Table 3-8	9.1, 10.3, 11.1, 12.1, 15, 16.1	7.1
FR-2 Publication and search	2.0, D2	Table 3-8	9.1, 10.3, 10.7, 12.1, 16.1	7.2
FR-3 Requests and reservations	3.0, D3, Level 2	Table 3-8	9.1, 10.6, 10.7, 11.1, 12.3, 16.1	7.2, 7.8
FR-4 Approvals and transfers	4.0, D4	Section 5 (H-4 to H-7)	9.2, 10.1, 10.2, 11.1, 11.2, 12.4, 15, 16.1	7.4
FR-5 Maintenance and life cycle	5.0, D5	Section 5 (H-8, H-9)	9.2, 10.9, 10.10, 11.3, 12.2, 12.5, 12.6, 15, 16.1	7.5
FR-6 Accounts and access	1.0, D1	Section 6	9.1, 10.8, 15, 16.1	-
FR-7 Notifications, reports, audit	6.0, D8, D9	Sections 3.3, 6.3	9.4, 10, 15, 16.1	7.6, 7.7
FR-8 Matching and ranking	3.0, Level 2, D7	Table 4-1	9.3, 10.4, 14, 15, 16.2, 17	7.2, 7.3
FR-9 Classification	2.0, D7	Table 4-1, Section 4.2	9.3, 10.3, 16.2	7.1
FR-10 Sustainable actions	5.0	Table 4-1	9.3, 10.10, 11.3, 16.2	7.5
FR-11 Assistant	7.0	Table 4-1	9.3, 10.6, 16.2	7.8
FR-12 AI reports	6.0, D9	Table 4-1, H-10	9.3, 10.5, 16.2	7.6, 7.7
FR-13 Indicators	6.0, D6, D9	Table 3-8	9.4, 15, 18	7.6
AI-1 to AI-10	D7	Sections 4, 5	9.3, 14, 16.2, 17, 19	All
PER-1 to PER-6	-	DR-3, DR-5, Table 4-1	13.2	AI-down states
SAF-1 to SAF-9	5.0	Section 5, Table 6-5	10.2, 10.10, 11.3, 12.2	7.5
SEC-1 to SEC-13	D1, D8	Section 6, DR-2, DR-4	13, 15	-
QUA-1 to QUA-15	-	DR-1, Section 6.5	17, 19	All
BR-1 to BR-12	4.0, 5.0	Section 5, Table 6-2	10, 11, 12	7.4
LEG-1 to LEG-9	-	DR-4, DR-6, Section 6.7	17.1	7.3
ORG-1 to ORG-4	-	DR-6, Section 6.7	-	Sample data only

ORG-4 is met outside the document: each member commits their own work to the course GitHub repository.

AI Contributions

Mohamed Almarzooqi

Prepared by: Mohamed Almarzooqi

I used an AI assistant (Claude, Anthropic) to help draw the context and data-flow diagrams, the architecture view (Section 3), the account sequence diagram (Section 10.8) and the component diagram (Section 13.1) from my design, and to check the flows, data stores and use case mapping against the Phase 2 and Phase 3 reports. I wrote the section content, chose the architecture options and reviewed every figure.

Mohammad Alsuwaidi

Prepared by: Mohammad Alsuwaidi

I used an AI assistant (ChatGPT astra6, OpenAI) to help draw the class diagrams in Sections 9.1 and 9.2, the sequence, activity and state diagrams in Sections 10.1, 10.2, 10.7, 11.1, 11.2, 12.1, 12.3 and 12.4, the ERD (Section 15) and the screen mockups in Sections 7.1, 7.2 and 7.4 from my design, and to check the states, entities and keys against the Phase 3 report. I wrote the section content and reviewed every figure.

Ahmed AlAbdouli

Prepared by: Ahmed AlAbdouli

I used an AI assistant (ChatGPT astra6, OpenAI) to help draw the AI class diagrams (Section 9.3), the sequence diagrams in Sections 10.3, 10.4 and 10.6, the recommendation workflow (Section 14) and the screen mockups in Sections 7.3 and 7.8 from my design, and to check the AI requirements, time limits and the worked example against the Phase 3 report. I wrote the section content and reviewed every figure.

Rashid Alzaabi

Prepared by: Rashid Alzaabi

I used an AI assistant (Claude, Anthropic) to help draw the notification, audit and sustainability class diagram (Section 9.4), the sequence diagrams in Sections 10.5, 10.9 and 10.10, the maintenance activity and state diagrams (Sections 11.3, 12.2, 12.5 and 12.6), the deployment diagram (Section 13.2) and the screen mockups in Sections 7.5 to 7.7 from my design, and to check the permissions and sustainability figures against the Phase 2 and Phase 3 reports. I wrote the section content and reviewed every figure.

Integration Review

We used an AI assistant (Claude, Anthropic) to check the combined document against the earlier phase reports (requirement IDs, role names and sample data) and to point out inconsistencies. Each of us reviewed and corrected our own sections.

References

[1] University, "Intelligent, AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System," COSC 336 Introduction to Software Engineering project description, University, Abu Dhabi, UAE, Fall 2026.

[2] University, "Phase 4: Architecture Design," COSC 336 lab slides, University, Abu Dhabi, UAE, Fall 2026.

[3] University, "Phase 4 & 5: Design Document, Sequence, ER and Class Diagrams," COSC 336 lab slides, University, Abu Dhabi, UAE, Fall 2026.

[4] R. Alzaabi, M. Alsuwaidi, A. AlAbdouli and M. Almarzooqi, "ReSource Phase III: Requirements Document," COSC 336, University, Abu Dhabi, UAE, Oct. 2026.

[5] R. Alzaabi, M. Alsuwaidi, A. AlAbdouli and M. Almarzooqi, "ReSource Phase II Feasibility Study," COSC 336, University, Abu Dhabi, UAE, Sep. 2026.

[6] Django Software Foundation, "User authentication in Django," Django documentation. [Online]. Available: https://docs.djangoproject.com/en/stable/topics/auth/

[7] Anthropic, "Models overview," Claude Platform Docs, 2026. [Online]. Available: https://platform.claude.com/docs/en/about-claude/models/overview

[8] Django Q2 project, "Brokers," Django Q2 documentation. [Online]. Available: https://django-q2.readthedocs.io/en/stable/brokers.html

[9] World Wide Web Consortium, "Web Content Accessibility Guidelines (WCAG) 2.2," W3C Recommendation, Dec. 2024. [Online]. Available: https://www.w3.org/TR/WCAG22/

[10] Sentence Transformers, "all-MiniLM-L6-v2," model card, Hugging Face. [Online]. Available: https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2

[11] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in Proc. 2019 Conf. Empirical Methods in Natural Language Processing (EMNLP-IJCNLP), Hong Kong, 2019, pp. 3982-3992.

[12] OWASP Foundation, "OWASP Top 10:2025," 2025. [Online]. Available: https://owasp.org/Top10/2025/

[13] United Arab Emirates, Federal Decree-Law No. 45 of 2021 Concerning the Protection of Personal Data, 2021. [Online]. Available: https://uaelegislation.gov.ae/en/legislations/1972

[14] J. Platt, "Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods," in Advances in Large Margin Classifiers, A. Smola et al., Eds. Cambridge, MA, USA: MIT Press, 1999, pp. 61-74.

[15] U.S. Environmental Protection Agency, "Documentation chapters for greenhouse gas emission, energy and economic factors used in the Waste Reduction Model (WARM)," WARM version 16, Dec. 2023. [Online]. Available: https://www.epa.gov/warm/documentation-chapters-greenhouse-gas-emission-energy-and-economic-factors-used-waste